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Lecture Notes in Modern Electrodynamics

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Chapter 1

Introduction to Spacetime

This chapter gives an introduction to spacetime and the algebra of the vectors that make up Spacetime.

1.1. Spacetime and Relativity

What is time? According to Newton time was absolute; meaning that all observers always agree on what time is. That doesn't tell us what it is, and infact this is a hard question. Obviously we use time to record *change*, such as the change in position of a particle, or the change in the chemical composition of some material. We measure time by choosing one such process to represent a standard and compare other changes to that. Hence, it is an intrinsically relative concept. As such we should expect that the more fundamental our investigations become, the more cumbersome our concept of time will be. In modern physics we look at time slightly differently; we simply put forward the assertion that time is one of the four dimensions of *spacetime*. So we say that time and space are, though not identical, then at least similar things.

We measure *durations* in terms of units such as seconds. Lengths on the other hand have magnitudes that we measure in units such as meters. If the time and space are to be different dimensions of a single spacetime they should be measurable in the same units! Furthermore all observers should agree on the conversion of units of time into units of length. So the assumption that space and time form a single entity *spacetime* requires the existence of a conversion-factor which should have units like *meters per second*; i.e. there must exist a *fundamental* speed which is the same for all observers. We will call this fundamental speed c and use it to denote an arbitrary *event* in *Bob's* Spacetime with the vector

$$\mathbf{x} = x^0 \mathbf{e}_0 + x^1 \mathbf{e}_1 + x^2 \mathbf{e}_2 + x^3 \mathbf{e}_3, \quad x^0 = ct, \quad (1.1.1)$$

assuming that Bob can choose his basis-vectors $\mathbf{e}_\mu$ to be perpendicular of of *unit*-length. In order to make progress we need to know exactly what the words *perpendicular* and *length* mean in Spacetime. To establish that we will make use of the principles of relativity.

Inertial Observers

Let us first consider so-called *inertial observers* i.e. observers that move at relative constant velocities. The theory of *Special Relativity* then starts from two laws;

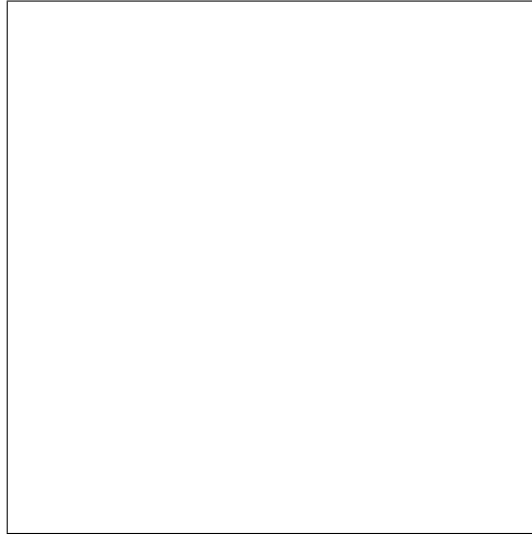


fig. 1: An event, its vector and its coordinates

- **A:** The Laws of Nature are the same for all inertial observers.
- **B:** The *fundamental speed*, c , is a constant of nature.

The first principle was already known to Galileo, the second was put in place by Albert Einstein in 1905 when electromagnetic waves (light) were the only known fundamental waves. Nowadays we trace the origin of the fundamental speed c back to the geometrical structure of spacetime. Traditional treatments replace the statement that space and time form a single entity known as *Spacetime* by the postulate of the absoluteness of the speed of light in vacuum. This reflects the chronological rather than the cognitive order. Once the spacetime geometry is fixed, we will see later, light has no other choice but to travel at the fundamental speed.

Spacetime is an *absolute* set of points called *events*. Observers in relative motion use different bases in spacetime and thus we transformation equations that allow us to calculate the change in basis between two observers in relative motion: these are the *Lorentz transformations*. The Lorentz transformations do not *change* anything in Spacetime, they only express how bases of different inertial observers are related. We say they are *passive* transformations. Let us establish the geometric properties of Spacetime that comply with the notion of relativity.

1.2. Worldlines and Lightcones

Lightsignals travel in the vacuum with precisely the fundamental speed c . We will see later in these notes why this is so. But because of this they make for great anchors in drawings of spacetime processes. Consider an observer Bob and draw his time-axis, ct vertically and one of his space-axes, x , horizontally. Because we have chosen the units ct on the time axis a lightsignal travelling through spacetime will be represented by a diagonal line at exactly $\frac{\pi}{4}$ radians, i.e. 45 degrees. If you draw all the lightsignals emitted from the origin of Bob's coordinates you obtain the *future lightcone*. If you draw all the lightsignals that Bob receives at $ct = 0$

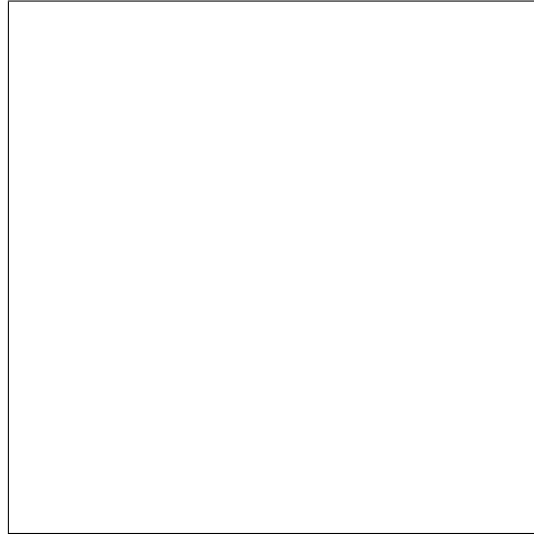


fig. 2: The lightcones of the origin in Bob's spacetime

and $x = 0$ you draw a *past lightcone*. When you look up to the stars at night you are looking into you past lightcone.

Problem 1: Mother of all diagrams: Radar-Bob

- a: Draw Bob's spacetime diagram according to the specifications in the text.
 b: Why is Bob not "at rest" in his own spacetime diagram but moving through it along the ct -axis? This line is now called the *worldline* of Bob.
 c: Draw lines through all the events that are connected to the origin $\mathcal{O}$ by a light signal. (This is called the *lightcone* of the event $\mathcal{O}$).
 d: Bob emits a radar signal at $t = t_0$ and receives the echo of some event $\mathbf{x}$ at $t = t_1$. Draw the situation and show, by drawing or algebra, that the event has the cartesian coordinates $t = \frac{t_1+t_0}{2}$ and $x = \pm c\frac{t_1-t_0}{2}$. What determines the sign?

Consider that Bob, who is convinced that he is at rest, uses a basis $\mathbf{e}_\mu$, with $\mu = 0, 1, 2, 3$, in which $\mathbf{e}_0$ is the time-direction. According to Bob it is possible to move in the time-direction *without* moving in any of the space-directions. From that he concludes that $\mathbf{e}_0$ can be taken to be perpendicular to any of the other $\mathbf{e}_j$, with $j = 1, 2, 3$ *. Any sensible innerproduct on spacetime should allow for the following algebraic statement of perpendicularity to be true

$$\mathbf{e}_0 \cdot \mathbf{e}_j = 0 . \quad (1.2.1)$$

If an object such as an elementary particle moves through spacetime its position $\mathbf{x}$ is a function of the time on its own clock. This we call the *proper* time of the particle and we denote it by τ . We will see later in this chapter that the proper time has a very definite *geometric* interpretation.

*From now on Greek indices will always run from 0 through 3, and latin indices run from 1 through 3.

If Bob has set up a time- and space-axis after choosing an orthonormal basis $\mathbf{e}_\mu$ he can denote the spacetime position of the particle by

$$\mathbf{x}(\tau) = ct(\tau)\mathbf{e}_0 + x^1(\tau)\mathbf{e}_1 + x^2(\tau)\mathbf{e}_2 + x^3(\tau)\mathbf{e}_3 .$$

We call this function $\mathbf{x}(\tau)$ the *worldline* of the particle. So if we do physics of particles and objects in spacetime, then we study their worldlines. Extended



fig. 3: Worldlines and worldsheet

objects, such as strings, do not sweep out worldlines in spacetime but, in the case of string for example, *worldsheets*.

1.3. The Lorentz transformations on basis-vectors

Now consider a second observer Alice. According to Bob she moves with respect to Bob with a constant speed v in the $\mathbf{e}_1$ -direction.

Problem 2: Alice's Worldline according to Bob

- a: Draw Bob's spacetime diagram again.
- b: Why is Alice moving through the same diagram along a line given by $\mathbf{x}_A(t) = ct\mathbf{e}_0 + \frac{v_A}{c}ct\mathbf{e}_1$?
- c: If ct' is the time in Alice's clock and $\mathbf{e}'_0$ is her time-direction, then why will she describe her motion through spacetime with $\mathbf{x}_A(t') = ct'\mathbf{e}'_0$?

Alice uses a different set of basisvectors $\mathbf{e}'_\mu$. According to the principles of relativity Alice may just as well argue that *she* is actually at rest whereas it is Bob that moving with respect to her. There is no physical experiment any one of them can do that will differentiate between these two points of view. As a result, Alice can take her time-direction $\mathbf{e}'_0$ to be the direction in which she moves. When we look at her motion from Bob's frame we can easily read of that the direction of her $\mathbf{e}'_0$ is given by

$$\mathbf{e}'_0 = \gamma\left\{\mathbf{e}_0 + \frac{v}{c}\mathbf{e}_1\right\} , \quad (1.3.1)$$

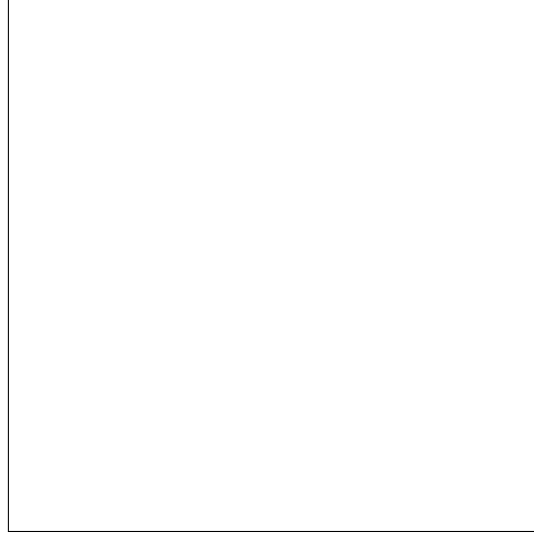


fig. 4: Alice in Bob's spacetime

where γ is some unknown number. Now for every observer *lightcones* are sliced exactly in between space and time directions. As a result, Alice's space-direction $\mathbf{e}'_1$ must be on the exact opposite side of the lightcone from $\mathbf{e}'_0$. As a result it must be given by

$$\mathbf{e}'_1 = \gamma \left\{ \mathbf{e}_1 + \frac{v}{c} \mathbf{e}_0 \right\} . \quad (1.3.2)$$

The remaining two directions of Bob and Alice can be identical

$$\begin{aligned} \mathbf{e}'_2 &= \mathbf{e}_2 , \\ \mathbf{e}'_3 &= \mathbf{e}_3 , \end{aligned}$$

as there is no relative motion between Alice and Bob in these directions. The equations we have derived here are called the *Lorentz transformation on basisvectors* for a *boost* in the $\mathbf{e}_1$ -direction. These equations translate between the basis of inertial observers moving relative to one another.

1.4. A conceptual Space-time split

One of these consequences is that the split between space and time is *relative* i.e. observer dependent.

Problem 3: Bob and Alice disagree about Time and Space

- a: Why must in general Alice's worldline be the same thing as her ct' -axis?
- b: Why must Alice's x'_1 -axis be on the exactly opposite side of the light signals when compared to her ct' -axis?
- c: Use this fact to draw Alice's x'_1 -axis in Bob's Spacetime Diagram.

The notion of *simultaneity* is connected to the split you make as an observer between Space and Time.

Problem 4: Observed Simultaneity and Retardation

The definition of simultaneity is not the same as saying that you *observe* events at the same time. This diagram will show you why.

- a: Draw Bob's Spacetime diagram.
- b: Draw the worldlines of light signals that reach Bob exactly at $t = 0$.
- c: Can you draw at least two events that *do not* occur simultaneously but that are visible to Bob at the same time $t = 0$?
- d: Draw two events that are simultaneous but that *are not* visible at the same time to Bob.

Because we want laws of nature to be absolute (i.e. the same for all inertial observers) the concept of simultaneity cannot play a role in the formulation of such laws.

Tachyons

Signals that move at speeds larger than the speed of light simply make no sense. If we study the transmission of such a *tachyonic* signal from the point of view of a fast co-moving observer then the order of transmission and reception of the signal turns out to be *relative*!

Problem 5: Tachyons: faster than light signals

Hypothetically we could wonder what happens when two observers communicate with one another using so-called *tachyons*: faster-than-light particles.

- a: Draw a spacetime diagram of two observers, Bob and Boba, both at rest. Draw a tachyon signal with a speed $v = 5c$ from Bob to Boba.
- b: Do Bob and Boba both agree that Bob sent the signal before Boba received it?
- c: Now Alice passes both Bob and Boba on her way to infinity. Is it possible for Alice to pick a speed such that for her both the emission by Bob as well as the reception of the signal by Boba are *simultaneous*?
- d: Draw another observer for which the reception of the signal in fact happens prior to the emission by Bob.
- e: Could anyone communicate with such signals?

Rigidity

Another issue with simultaneity that affects our physical intuition concerns rigidity; in classical mechanics we like to think about rigid bodies as those bodies for which all parts of the body are in the same state of motion at the same time. A rigid body does not deform under accelerations, ensuring that all parts of the body undergo the same accelerations at the same time. In relativity such rigid bodies cannot exist as their rigidity appeals to a sense of simultaneity. Still, space and time can be split up and often it is useful to do so when studying abstract equations in spacetime from the point of view of some particular observer.

1.5. Problems

This section contains a number of useful and relevant problems.

Problem 6: Strings

Consider a piece of string moving through spacetime.

- a: Draw a spacetime diagram and a straight bit of string on the x -axis at time $t = 0$.
- b: If this string moves with a constant speed; sketch its motion in the spacetime diagram.

Problem 7: Relativistic International Politics (RIP)

Two fractions, A and B, of the People's Liberation Front of the Previous Past (PLFPP) are on a collision course in Spacetime. They are destined to collide at the location of the UN office in Geneva at some time in the future. Both fractions have a representative at the UN.

- a: Draw a Spacetime diagram with the UN "at rest".
- b: Draw the worldlines of the incoming fractions.
- c: Draw how it is possible that the representative at the UN of fraction B signs the peace-treaty *simultaneously* (according to fraction A) with the leader of fraction A, while according to the leader of fraction B his representative signs *after* the leader of A has signed?
- d: Show in a drawing that vice versa is true as well; and explore for yourself how these options open up novel prospects for relativistic international politics!

Problem 8: Bob's rigid ruler

Suppose Bob measures distances in his frame with a rigid ruler at rest until $t = 0$. As Alice comes along at $t = 0$ Bob instantaneously accelerates his rigid ruler to the same velocity as Alice.

- a: Draw a spacetime diagram of the situation from Bob's frame.
- b: Draw the same situation but now from Alice's frame.
- c: Is the ruler rigid in Alice's frame?

Problem 9: Retardation and Length contraction

Consider a train that in its own frame has a length L , driving at the hilarious speed of $0.5c$ relative to Bob. At $t = 0$ the front end passes Bob at $x = 0$.

- a: Draw Bob's Spacetime diagram of the moving train by indicating the motion of the front and the rear of the train.
- b: What is the length the train has on an axis parallel to Bob's x_1 -axis?
- c: Draw the light signals that Bob receives at $t = -1$ and at $t = +1$ from the front and the rear of the train.
- d: What can you say about the visual appearance of the train according to Bob: Is it contracted?

In the previous problem you'll have seen that retardation is by far more important in determining the visual appearance of a moving object than the Lorentz contraction.

Chapter 2

Spacetime Geometry

In this chapter we will discuss the fundamentals of the geometry of flat spacetime and the algebraic relations that it induces among spacetime vectors.

2.1. Vector algebra

Standard vector algebra deals with the addition of vectors and scalar-multiplication that are ingrained in the definition. For our uses it is possible and desirable to add a little more structure to the algebra of vectors in the form of the *geometric product*. We define it by specifying a few of its properties as axioms.

Given two, vectors **a** and **b** from modelspace we will denote their *geometric product* as **ab**. Before you ask what the product means, geometrically, let us first fix its mathematical properties. First of all we will assume that the product between vectors is *associative*:

$$(\mathbf{a}\mathbf{b})\mathbf{c} = \mathbf{a}(\mathbf{b}\mathbf{c}) . \quad (2.1.1)$$

Secondly we will also assume that the product is *distributive* with respect to vector addition,

$$\mathbf{a}(\mathbf{b} + \mathbf{c}) = \mathbf{a}\mathbf{b} + \mathbf{a}\mathbf{c} , \quad (2.1.2)$$

from the left as well as from the right,

$$(\mathbf{b} + \mathbf{c})\mathbf{a} = \mathbf{b}\mathbf{a} + \mathbf{c}\mathbf{a} . \quad (2.1.3)$$

Now we arrive at the two essential axioms. First of all we will assume commutativity *only with respect to scalar multiplication* ! So for any real number k and vector **a** we assume that the following holds;

$$k\mathbf{a} = \mathbf{a}k . \quad (2.1.4)$$

Finally we will assume that the geometric product of a vector with itself yields a real number. So we assume that

$$\mathbf{a}^2 \text{ is a number} . \quad (2.1.5)$$

This concludes the assumptions we make about the algebraic nature of vectors. One immediate consequence is that the so-called dot-product

$$\mathbf{a} \cdot \mathbf{b} \equiv \frac{1}{2}(\mathbf{ab} + \mathbf{ba}) , \quad (2.1.6)$$

yields a number as well. This number is important because we associate to it the notion of perpendicularity. In other words, we say that vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular when $\mathbf{a} \cdot \mathbf{b} = 0$. Everything else follows from the principle of relativity.

Problem 10: The symmetrised product of two vectors is a scalar

a: Consider a vector $\mathbf{a}$ as the sum of two others $\mathbf{a} = \mathbf{b} + \mathbf{c}$. Then show from the axioms $\mathbf{bc} + \mathbf{cb}$ is indeed scalar.

b: Show that if $\mathbf{a} \cdot \mathbf{b} = 0$ then $\mathbf{ab} = -\mathbf{ba}$.

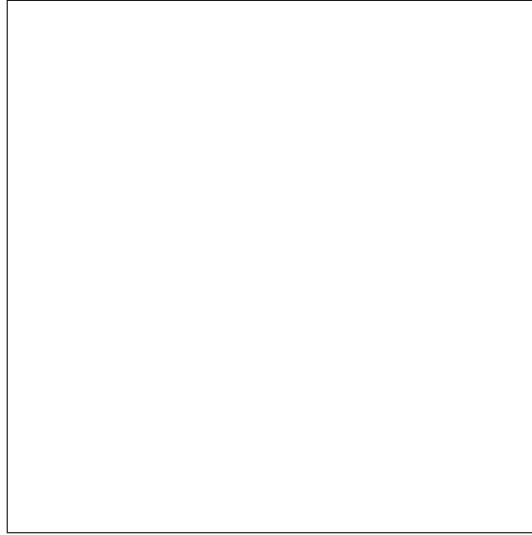


fig. 5: Graphic impression of the dotproduct $\mathbf{a} \cdot \mathbf{b}$

2.1.1. A Spacetime dot-product

In spacetime the notions of perpendicularity are different from those in ordinary space. So we expect the dot-product to be different as well.

Alice is physically but also geometrically equivalent to Bob. So in any experiment she does she is allowed to take her $\mathbf{e}'_0$ and $\mathbf{e}'_1$ as perpendicular as well! This implies that a sensible, *relativistic* dotproduct on spacetime should allow for

$$\mathbf{e}'_0 \cdot \mathbf{e}'_1 = 0 .$$

If we substitute the expressions we found above we find

$$\mathbf{e}'_0 \cdot \mathbf{e}'_1 = \gamma^2 \frac{v}{c} \{ \mathbf{e}_0 \cdot \mathbf{e}_0 + \mathbf{e}_1 \cdot \mathbf{e}_1 \} = 0 .$$

This can only be true when

$$\mathbf{e}_0 \cdot \mathbf{e}_0 = -\mathbf{e}_1 \cdot \mathbf{e}_1 .$$

The principle of relativity *forces* us to consider dotproduct with a negative square of a vector. The choice we make here is called the *west-coast* metric: for an orthonormal basis we take

$$\mathbf{e}_0 \cdot \mathbf{e}_0 = +1 , \quad (2.1.7)$$

$$\mathbf{e}_j \cdot \mathbf{e}_j = -1 . \quad (2.1.8)$$

This choice is typically favoured by workers in General Relativity. The alternative is to swap the signs. This is a choice often used by people in the particle physics community. As far as we know now the choice is purely a matter of convention.

Problem 11: Using the west-coast metric

With the above choice of inner-product show that the vector

$$\mathbf{a} = a^0 \mathbf{e}_0 + a^1 \mathbf{e}_1 + a^2 \mathbf{e}_2 + a^3 \mathbf{e}_3 ,$$

satisfies

$$\mathbf{a}^2 = (a^0)^2 - (a^1)^2 - (a^2)^2 - (a^3)^2 . \quad (2.1.9)$$

We use the dot-product to define what we mean by an *orthonormal basis*. If we have a set of linearly independent basis vectors $\mathbf{e}_\mu$ that satisfy

$$\mathbf{e}_\mu \cdot \mathbf{e}_\nu = 0 , \text{ for } \mu \neq \nu , \quad (2.1.10)$$

then we say these basis vectors are *orthogonal*. If furthermore we have that

$$\mathbf{e}_\mu^2 = \pm 1 , \quad (2.1.11)$$

we say they are *orthonormal*.

If Bob's basisvectors are orthonormal vectors, then so should Alice's. The lorentz transformations cannot yield a change in their lengths as that would have physical consequences. This requirement fixes the unknown constant γ , for we find

$$\begin{aligned} 1 &= \mathbf{e}'_0 \cdot \mathbf{e}'_0 , \\ &= \gamma^2 \left\{ 1 - \frac{v^2}{c^2} \right\} . \end{aligned}$$

As a result we find that

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} . \quad (2.1.12)$$

This gamma-factor is a very typical hallmark of relativistic physics. What you see here is that its origin is deeply geometric.

Problem 12: Training dotting

This is just a formal little exercise. Assume that we have an orthonormal basis $\mathbf{e}_j^2 = +1$ for $j = 1, 2, 3$. Given

$$\mathbf{a} = \mathbf{e}_1 + 3\mathbf{e}_2, \mathbf{b} = 2\mathbf{e}_1 + 3\mathbf{e}_3, \mathbf{c} = -\mathbf{e}_3.$$

a: Compute $\mathbf{a}^2$, $\mathbf{b}^2$, and $\mathbf{c}^2$.

b: Is any of these three vectors perpendicular to one of the other two? Check this by calculating their dot-products.

The Reciprocal basis and Components of vectors

A very usefull concept is that of the *reciprocal frame* $\mathbf{e}^\mu$ defined by

$$\mathbf{e}_\mu \cdot \mathbf{e}^\nu = 0, \mu \neq \nu, \quad (2.1.13)$$

$$\mathbf{e}_\mu \cdot \mathbf{e}^\mu = +1. \quad (2.1.14)$$

Note that for a standard orthonormal spacetime frame $\mathbf{e}^\mu$ this means that

$$\begin{aligned} \mathbf{e}^0 &= \mathbf{e}_0, \\ \mathbf{e}^j &= -\mathbf{e}_j. \end{aligned}$$

The use of this frame, as well as of our way of handling the index up-or-down, becomes evident as soon as we want to compute components of spacetime vectors. Consider an arbitrary vector $\mathbf{a}$ in Spacetime as viewed by Bob. Bob will expand this vector as

$$\mathbf{a} = \sum_{\mu=0}^3 a^\mu \mathbf{e}_\mu$$

The components a^μ are numbers and using the reciprocal basis we can give an easy algebraic expression for them

$$a^\nu = \mathbf{e}^\nu \cdot \mathbf{a}. \quad (2.1.15)$$

Due to the fact that the square of a vector can be either positive, null or negative we come to the following classification of spacetime vectors:

- A vector $\mathbf{b}$ is called *spacelike* if $\mathbf{b}^2 < 0$.
- A vector $\mathbf{b}$ is called *timelike* if $\mathbf{b}^2 > 0$.
- A vector $\mathbf{b}$ is called *lightlike* or *null* if $\mathbf{b}^2 = 0$.

Problem 13: Visualizing the classification

Ignore the dimensions $\mathbf{e}_2$ and $\mathbf{e}_3$ for the moment and consider only those vectors $\mathbf{b} = b^0 \mathbf{e}_0 + b^1 \mathbf{e}_1$.

a: Draw a spacetime diagram and indicate the $\mathbf{e}_0$ and the $\mathbf{e}_1$ directions.

b: A vector $\mathbf{b}$ can now be represented by points in this spacetime plane. Indicate the areas where we find timelike vectors $\mathbf{b}$.

c: Indicate the areas where we find spacelike vectors $\mathbf{b}$.

d: Finally indicate where the lightlike vectors can be found.

Although Lorentz transformations may seem odd to you they do preserve the geometry of spacetime.

Problem 14: Class conservation

- a: Show that if $\mathbf{e}_0$ is a timelike vector then so is $\mathbf{e}'_0$.
 b: Show that if $\mathbf{e}_1$ is a spacelike vector then so is $\mathbf{e}'_1$.

Quadratic equations in Spacetime

It is of great practical interest, as we will see a little later, to study the solutions of equations like

$$\mathbf{x}^2 = c . \quad (2.1.16)$$

In ordinary spaces the solutions to this equation would lie on circles or spheres around the origin. In spacetime this is a little more complicated.

Problem 15: Spacelike Shells

Consider the events described by $\mathbf{x} = ct\mathbf{e}_0 + x\mathbf{e}_1$.

- a: Draw a spacetime diagram with the ct - and the x -axes.
 b: Write down and solve the equation $\mathbf{x}^2 = -1$.
 c: Sketch the solutions in your diagram.

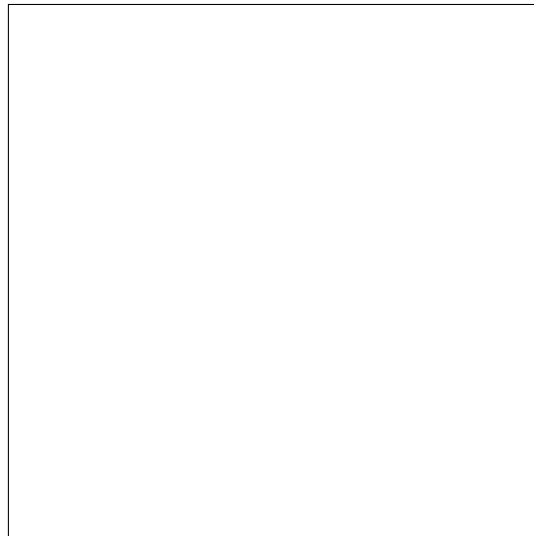


fig. 6: A time-like shell: the solution to $\mathbf{x}^2 = +1$

If $c = 0$ then the events satisfying this equation form the lightcone through the origin. But if $c \neq 0$ then you will find hyperbolic surfaces that are either timelike if $c > 0$ or spacelike if $c < 0$.

2.1.2. Bivectors

Given two vectors $\mathbf{a}$ and $\mathbf{b}$ we define

$$\mathbf{a} \wedge \mathbf{b} \equiv \frac{1}{2}(\mathbf{a}\mathbf{b} - \mathbf{b}\mathbf{a}) , \quad (2.1.17)$$

the so-called *wedge*- or outer-product.

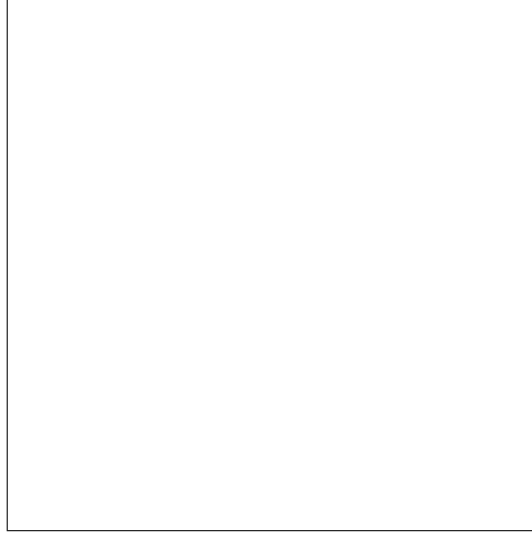


fig. 7: Wedge-product and surface areas

Problem 16: Wedge product

We have defined the *wedge product* between $\mathbf{a}$ and $\mathbf{b}$ as

$$\mathbf{a} \wedge \mathbf{b} \equiv \frac{1}{2}(\mathbf{a}\mathbf{b} - \mathbf{b}\mathbf{a}) . \quad (2.1.18)$$

a: Show that if the vector $\mathbf{b}$ is parallel to $\mathbf{a}$ then $\mathbf{a} \wedge \mathbf{b} = 0$.

b: Show from the definition of dot and wedge product that

$$\mathbf{a}\mathbf{b} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b} . \quad (2.1.19)$$

c: Let $\hat{\mathbf{a}}$ be a vector for which $\hat{\mathbf{a}}^2 = 1$. Let $\mathbf{b}$ be any other vector. Show that

$$\mathbf{b} = (\mathbf{b} \cdot \hat{\mathbf{a}})\hat{\mathbf{a}} + (\mathbf{b} \wedge \hat{\mathbf{a}})\hat{\mathbf{a}} . \quad (2.1.20)$$

d: Why is $(\mathbf{b} \cdot \hat{\mathbf{a}})\hat{\mathbf{a}}$ the part of $\mathbf{b}$ parallel to $\hat{\mathbf{a}}$?

e: So what *must* $(\mathbf{b} \wedge \hat{\mathbf{a}})\hat{\mathbf{a}}$ be?

The wedgeproduct of two vectors introduces a *new* geometric object. It represents a plane-element. For example for the basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ we may think of $\mathbf{e}_1 \wedge \mathbf{e}_2$ as a square with sides of length 1 in the 1 – 2-plane. Because $\mathbf{e}_1 \cdot \mathbf{e}_2 = 0$

we can write $\mathbf{e}_1 \wedge \mathbf{e}_2 = \mathbf{e}_1 \mathbf{e}_2$.

Problem 17: Training wedging

Assume that we have an orthonormal basis $\mathbf{e}_j^2 = +1$ for $j = 1, 2, 3$. Given

$$\mathbf{a} = \mathbf{e}_1 + 3\mathbf{e}_2, \mathbf{b} = 2\mathbf{e}_1 + 3\mathbf{e}_3, \mathbf{c} = \mathbf{e}_1 - \mathbf{e}_3 .$$

a: Compute the wedges $\mathbf{a} \wedge \mathbf{b}$ and $\mathbf{b} \wedge \mathbf{c}$.

b: Draw the vectors $\mathbf{a}$ and $\mathbf{c}$ in a single plane, compute the wedge $\mathbf{b} \wedge \mathbf{c}$ and compare the answer to the surface area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{c}$.

The wedgeproduct between two vectors is not a vector. The wedge product maps two vectors into a *bivector* representing the plane-element that they span. If we start out from a orthonormal basis $\mathbf{e}_\mu$ we can define the following set of *six* different bivectors

$$\begin{aligned} K_j &\equiv \mathbf{e}_j \mathbf{e}_0 , \\ J_1 &\equiv \mathbf{e}_2 \mathbf{e}_3 , \quad J_2 \equiv \mathbf{e}_3 \mathbf{e}_1 , \\ J_3 &\equiv \mathbf{e}_1 \mathbf{e}_2 . \end{aligned} \tag{2.1.21}$$

Together they span the space of *all* bivectors. Bivectors will play a very important role so try get used to them. They have many fascinating properties, some of which you will prove in the next problem.

Problem 18: Squaring Bivectors

a: Show that for any of the bivectors K_j we have $K_j^2 = +1$.

b: Show that for any of the bivectors J_k we have $J_k^2 = -1$.

The different classes of spacetime vectors, *spacelike*, *timelike* and *null*-vectors, can also be found in the set of bivectors. The bivectors K_n defined above are called timelike bivectors, and similarly the J_n are called spacelike bivectors. Null-bivectors also exist, they square to 0 and are combinations of timelike and spacelike bivectors.

Problem 19: A Null Bivector

Show that the bivector $B = (\mathbf{e}_0 + \mathbf{e}_1)\mathbf{e}_2$ is a null-bivector, i.e. $B^2 = 0$.

Bivectors also play a very important role in *generating* rotations and Lorentz transformations as we will see in the next chapter. Here the distinction between spacelike and timelike is the key to the distinction between ordinary rotations and lorentzboosts.

Problem 20: Spacelike plane-elements

Consider two spacelike vectors $\mathbf{a}$ and $\mathbf{b}$.

a: Why should we modify the "old" dotproduct formula into $\mathbf{a} \cdot \mathbf{b} = -ab \cos(\theta)$ for spacelike vectors?

b: Choose an easy basis to calculate $\mathbf{a} \wedge \mathbf{b}$ (hint: Choose $\mathbf{e}_1$ parallel to $\mathbf{b}$ or $\mathbf{a}$).

c: Draw the situation and show that $\mathbf{a} \wedge \mathbf{b}$ is proportional to $ab \sin(\theta)$ and derive from the drawing that this is the area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$.

2.1.3. Relative spacetime vectors

Consider the timelike basis vector $\mathbf{e}_0$ and some arbitrary spacetime vector $\mathbf{a}$. Using the fact that $\mathbf{e}_0^2 = 1$ we can write

$$\mathbf{a} = \mathbf{a} \mathbf{e}_0^2 = (\mathbf{a} \cdot \mathbf{e}_0) \mathbf{e}_0 + (\mathbf{a} \wedge \mathbf{e}_0) \mathbf{e}_0 . \quad (2.1.22)$$

The first term in this expression clearly represent the part of $\mathbf{a}$ that points in the time-like direction of $\mathbf{e}_0$. The second term we can then intepret as the *remaining space-like part*. We will, from now on, use the standard vector-notation

$$\vec{a} \equiv \mathbf{a} \wedge \mathbf{e}_0 , \quad (2.1.23)$$

for the *bivector* representing such a *relative* spacelike part.

2.2. Time-dilation and Lorentz transformations

A basic result from relativity is the occurence of time-dilation and length contraction. Here we give a rather standard argument.

Consider a clock whose "ticks" are produced by a Big Fat Photon travelling between two mirrors. If Alice carries along such a "light-clock" and keeps it perpendicular in space to the $\mathbf{e}_1$ -direction direction, then Bob can easily calculate what the time is that he records between two ticks.

Alice's clock consists of two mirrors separated by a distance l_0 . As we just saw the Lorentz transformations don't do anything in directions perpendicular to the velocity, so this length will not be affected by it. In the time between two Ticks the Big Fat Photon covers a distance $2l_0$, and it takes a time $\delta t_0 = \frac{2l_0}{c}$ to do so.

As seen from Bob's frame the clock moves. The Big Fat Photon thus seems to travel a larger distance l during a time-interval δt . The time-interval δt can be computed through the Pythagorean theorem,

$$\begin{aligned} \delta t^2 &= \frac{4l^2}{c^2} , \\ &= \frac{4l_0^2 + v^2 \delta t^2}{c^2} \end{aligned}$$

Re-ordering the terms involving δt gives

$$\begin{aligned} \left(1 - \frac{v^2}{c^2}\right) \delta t^2 &= \frac{4l_0^2}{c^2} = \delta t_0^2 , \\ \delta t &= \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \delta t_0 . \end{aligned}$$

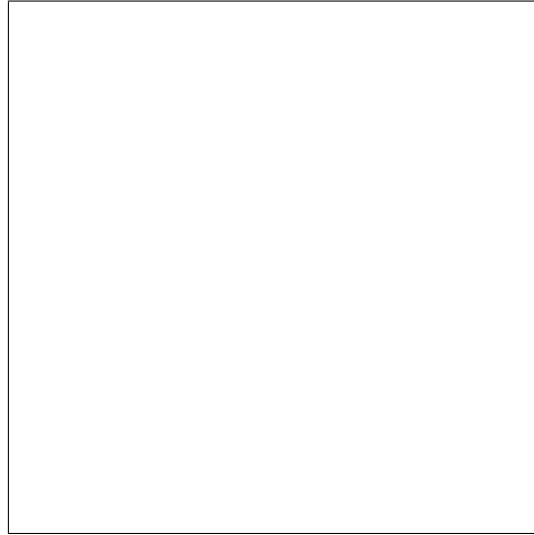


fig. 8: The Lightclock

The factor in front of δt_0 is,

$$\gamma(v) = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} . \quad (2.2.1)$$

This factor is responsible for the relativistic effect called *time-dilation*.

Problem 21: Time-dilation and Lorentz transformations

Draw a spacetime diagram with Bob and Alice's worldlines. Let Bob and Alice's clocks be at $t = 0$ and $t' = 0$ when they pass each other in the origin. Alice's clock tocks at $t' = 0$ and again at $t' = \delta t_0$.

- a: Draw the events on Alice's worldline at which the clock tocks.
- b: Read off the components of these two events in Alice's frame.
- c: Use the Lorentz transformation to determine the time t in Bob's frame at which the second tock is produced by Alice's clock.
- d: Compare your result to the result above.

Example:

Consider an astronaut travelling aboard a star ship towards the nearest star that is approximately 4 ly away. As she travels with, say 75 percent of the speed of light relative to the Earth, the time-dilation between her and the Earth amounts to $\gamma = 2\sqrt{\frac{2}{3}}$. In other words the clocks of the astronaut are a factor $\frac{1}{\gamma}$ slower than those on the Earth. As a result when, according to Earth clocks, her trip takes about 5.3 years, on her own clocks only ... years pass.

Problem 22: Muons fall

Due to the irradiation by cosmic rays a strange type of "heavy electrons", called muons, are produced high in the atmosphere. These muons are instable particles and typically decay after about 10^{-6} seconds. Suppose such a muon is created at an altitude of 80 km above ground level. What is the minimal speed with which this particle must travel so that it reaches the ground before it decays?

There is something very odd about the previous time-dilation example. The astronaut knows that she's travelling at a speed of 75 percent of c . So as she reads of her clock and measures that it takes her only ... years to complete the trip she will estimate the distance between Earth and her destination cannot have been 4 ly but much less! In fact she will record a distance which is exactly a compensating factor γ *smaller*! This phenomenon is called *length-* or *Lorentz-contraction*.

Suppose Alice measures the length of a rod, extending in the $\mathbf{e}'_1$ -direction to be λ_A . As she is moving relative to Bob in the $\mathbf{e}'_1$ -direction with a speed v , Bob's length measurement on the moving rod will yield a value

$$\lambda_B = \frac{\lambda_A}{\gamma} . \quad (2.2.2)$$

Note that this is mutual, just like time dilation! Time dilation and Length-contraction are two sides of the same medal. They mutually assure that no undesirable inconsistencies arise. At the same time however they also represent purely observer-dependent effects. There is no actual mechanical contraction that stores elastic energy. Similarly time dilation does not generate additional time. Unfortunately most *graphical* illustrations of length contraction completely ignore other important effects such as *retardation*. This is discussed further in one of the revision problems.

Now the above derivation may seem very fancy and *deep*. But if you take a closer look at where the γ -factor comes from then notice that if we have two observers one using the time-like basis vector $\mathbf{e}_0$ and another in constant relative motion using $\mathbf{e}'_0$ then their relative γ -factor is given by

$$\gamma = \mathbf{e}_0 \cdot \mathbf{e}'_0 . \quad (2.2.3)$$

We may use this to derive the lorentz transformation equations for *components* rather than for vectors. Let some spacetime point $\mathbf{x}$ be given in the frame of Alice as

$$\mathbf{x} = ct'\mathbf{e}'_0 + x'\mathbf{e}'_1 ,$$

while for Bob we have *for the same* spacetime point

$$\mathbf{x} = ct\mathbf{e}_0 + x\mathbf{e}_1 .$$

If we want to compute the components that Alice observes in terms of those that Bob observes we simply need to calculate

$$\begin{aligned} x'^\mu &= \mathbf{e}'^\mu \cdot \mathbf{x} , \\ &= ct\mathbf{e}'^\mu \cdot \hat{\mathbf{e}}_0 + x\mathbf{e}'^\mu \cdot \hat{\mathbf{e}}_1 . \end{aligned}$$

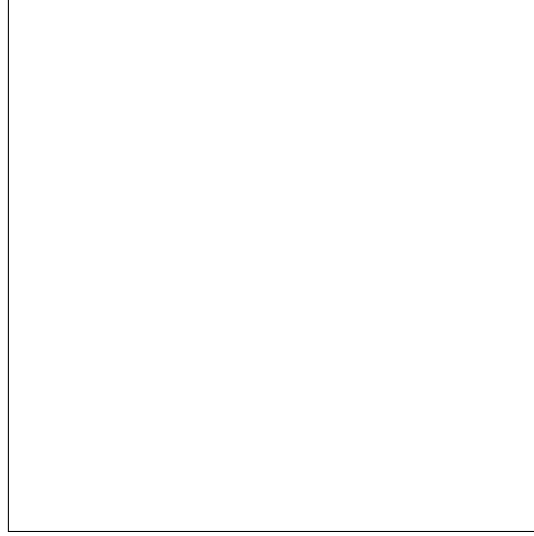


fig. 9: One event, two observers, two sets of coordinates

If we use the expression derived earlier for the $\mathbf{e}'_\mu$ in terms of the $\mathbf{e}_\mu$ we find the following result

$$x'^0 = ct' = \gamma \left\{ ct - \frac{v}{c} x \right\} , \quad (2.2.4)$$

$$x'^1 = x' = \gamma \left\{ x - \frac{v}{c} ct \right\} . \quad (2.2.5)$$

This is the *component form* of the lorentz transformation equation for the boost in the $\mathbf{e}_1$ -direction.

Lorentz transformations in their component-form reveal how the Lorentz transformations relate the coordinates of two inertial observers in relative motion. Yet there are other coordinate systems that can be very useful although they cannot be made using Lorentz transformations.

Problem 23: Null basis in Spacetime

Given an orthonormal basis vectors $\mathbf{e}_\mu$. One way obtaining a new basis is through Lorentz transformations. But there are other possibilities as well.

a: Draw a spacetime diagram and sketch $\mathbf{e}_0$ and $\mathbf{e}_1$ as well as the vectors

$$\mathbf{k}_\pm = \frac{1}{\sqrt{2}}(\mathbf{e}_0 \pm \mathbf{e}_1) \quad (2.2.6)$$

in a single drawing.

b: Prove that the two vectors $\mathbf{k}_\pm$ are *linearly independent*.

c: Note that the $\mathbf{k}_\pm$ seem to be perpendicular in the drawing, but compute $\mathbf{k}_+ \cdot \mathbf{k}_-$. Does it vanish?

d: The basis vectors $\mathbf{k}_\pm$ are sometimes called *lightcone vectors*. Now let

$$\mathbf{x} = u\mathbf{k}_+ + w\mathbf{k}_- + y\mathbf{e}_2 + z\mathbf{e}_3 . \quad (2.2.7)$$

Determine u and w in terms of x and ct . The coordinates u and w are called *lightcone coordinates*.

The coordinates u and w computed in the previous problem are called *lightcone coordinates*.



fig. 10: Lightcone Coordinates

Lorentz transformation matrices

Two observers, Alice and Bob, use different sets of orthonormal basis-vectors $\mathbf{e}_\mu$ and $\mathbf{e}'_\nu$ due to their constant relative motion. As a result when they observe *one and the same* vector $\mathbf{a}$ they assign *different* components a^μ and a'^ν to it. These different components can be related by

$$\begin{aligned} a'^\nu &= \mathbf{e}'^\nu \cdot \mathbf{a} \\ &= \sum_{\mu=0}^3 a^\mu \mathbf{e}'^\nu \cdot \mathbf{e}_\mu , \\ &= \sum_{\mu=0}^3 \Lambda_\mu^\nu a^\mu , \end{aligned} \tag{2.2.8}$$

where we introduced a matrix Λ with the matrix elements Λ_μ^ν given by

$$\Lambda_\mu^\nu = \mathbf{e}'^\nu \cdot \mathbf{e}_\mu . \tag{2.2.9}$$

If we represent vectors as arrays of components, then the lorentz transformations can be represented as *matrix*-multiplications. This is the standard approach in many textbooks, despite the fact that it describes a coordinate-free theory in terms of coordinates.

Problem 24: Transforming arbitrary vectors

Consider an arbitrary vector $\mathbf{a}$ observed by Alice and *the same* vector observed by Bob.

- a: Write $\mathbf{a}$ out in terms of the basis-vectors $\mathbf{e}'_\nu$ of Alice and her components $a^{\nu'}$ (ignore the $\mathbf{e}'_2$ and $\mathbf{e}'_3$ directions for simplicity).
- b: Determine the components a^μ that Bob observes by taking the dot-products $\mathbf{e}_\mu \cdot \mathbf{a}$.
- c: Show that $(a^0)^2 - (a^1)^2 = (a^{0'})^2 - (a^{1'})^2$.
- d: Why does this mean that Lorentz transformations preserve the Spacetime length of vectors?

The last exercise clearly shows that the spacetime length of a vector is something that all observers will agree on. Such things are called *lorentz invariants*. The close relationship between Lorentz transformations and ordinary rotations can be revealed more explicitly by working with the so-called *hyperbolic* sines and cosines.

Problem 25: Hyperbolic functions

Now that we know what γ is we take a closer look at the transformation rule

$$\mathbf{e}'_0 = \gamma \mathbf{e}_0 + \gamma \frac{v}{c} \mathbf{e}_1 .$$

- a: Show that $\gamma^2 - (\gamma \frac{v}{c})^2 = 1$.
- b: Define the functions

$$\begin{aligned} \cosh(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{2} , \\ \sinh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{2} , \\ \tanh(\alpha) &= \frac{\sinh(\alpha)}{\cosh(\alpha)} . \end{aligned} \tag{2.2.10}$$

- c: Show that $\cosh(\alpha)^2 - \sinh(\alpha)^2 = 1$.
- d: Show that if we set $\cosh(\alpha) = \gamma$ then $\sinh(\alpha) = \gamma \frac{v}{c}$.
- e: Show that $\tanh(\alpha) = \frac{v}{c}$.

They allow us to write the transformation equations for the vectors as, for example,

$$\begin{aligned} \mathbf{e}'_0 &= \cosh(\alpha) \mathbf{e}_0 + \sinh(\alpha) \mathbf{e}_1 , \\ \mathbf{e}'_1 &= \cosh(\alpha) \mathbf{e}_1 + \sinh(\alpha) \mathbf{e}_0 . \end{aligned} \tag{2.2.11}$$

If you plot the last expression in a spacetime diagram for all values of α you will find that the vectors parametrise an entire future timelike shell. Spacelike shells can be obtained in a similar way by considering the hyperbolic form of the transformation on $\mathbf{e}_1$. I leave that to an exercise.

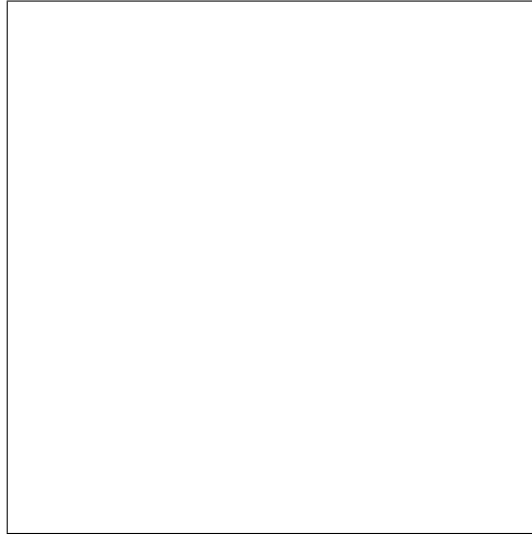


fig. 11: Hyperbolic functions and timelike vectors

Problem 26: Shells and dotproducts

Consider a vector $\mathbf{b}$ given by

$$\mathbf{b} = \sinh(\alpha)\mathbf{e}_0 + \cosh(\alpha)\mathbf{e}_1 .$$

- a: Show that this vector is spacelike.
- b: Draw the line that this vector parametrizes for α between $-\infty$ and $+\infty$.
- c: Suppose you are give two spacelike vectors $\mathbf{c}$ and $\mathbf{d}$, choose a suitable basis and argue that we can find a β such that $\mathbf{c} \cdot \mathbf{d} = -\cosh(\beta)$.
- d: Suppose you are given two timelike vectors $\mathbf{m}$ and $\mathbf{n}$, choose a suitable basis and show that there is an α such that $\mathbf{m} \cdot \mathbf{n} = +\cosh(\alpha)$.
- e: Finally, consider one timelike vector $\mathbf{a}$ and one spacelike vector $\mathbf{b}$ and show that there is a σ such that $\mathbf{a} \cdot \mathbf{b} = \sinh(\sigma)$.

In the next chapter we will see that the correspondence between rotations and Lorentz transformations is even closer than we have seen so far.

2.3. A little more Algebra

The use of the algebra among vectors is so vast that we can only scratch a few of its applications in this text. Consider, for example, the two timelike vectors $\mathbf{a} = a^0\mathbf{e}_0$ and $\mathbf{b} = b^0\mathbf{e}_0 + b^1\mathbf{e}_1$. It is easy to check that

$$\mathbf{a}\mathbf{b} = a^0b^0 - a^0b^1K_1 .$$

Now note that squaring a sum of a bivector and a scalar is an easy caculation gives

$$(c - dK_1)^2 = c^2 + d^2 - 2cdK_1 .$$

Combining the two results above we can give a proper meaning to an expression like $\sqrt{\mathbf{a}\mathbf{b}}$. If we can find two number c and d such that

$$\begin{aligned} c^2 + d^2 &= a^0b^0 , \\ 2cd &= a^0b^1 , \end{aligned}$$

then we can write

$$\sqrt{\mathbf{a}\mathbf{b}} = c - dK_1 . \quad (2.3.1)$$

We will have use for this type of a result in the next chapter.

Problem 27: Determining c and d

- a: Use the two equations above to solve for d and c in terms of a_0 , b_0 and b_1 .
b: Why does the result show that the square root of the product of *any* pair of timelike vectors is well-defined?
c: Can you find two vectors $\mathbf{a}$ and $\mathbf{b}$ for which $\sqrt{\mathbf{a}\mathbf{b}} = \mathbf{a}\mathbf{b}$? (Hint: consider the above expressions for $c = d = \frac{1}{2}$).

2.3.1. Trivectors and the pseudo-scalar

Multiplying together three vectors gives us further new possibilities. For example we can define things called *trivectors*

$$\begin{aligned} T_0 &= \mathbf{e}_1\mathbf{e}_2\mathbf{e}_3 , \\ T_n &= \mathbf{e}_0J_n . \end{aligned} \quad (2.3.2)$$

The trivector T_0 represents a small cube with sides of unit-length in the three space-directions. Such a *volume-element* has a magnitude but also an orientation. However what is at rest with respect to Bob will seem to flow with respect to Alice. As a result T_0 will be transformed into the other trivectors T_n under Lorentz transformations.

Problem 28: Multiplying J 's and K 's.

- a: Show that for any choice of $n = 1, 2, 3$ we find that $K_nJ_n = \mathbf{e}_0\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$

The products among bivectors gave use something new. We call it the pseudo-scalar

$$i = \mathbf{e}_0\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3 . \quad (2.3.3)$$

It represents a small element of spacetime, i.e. a small 4-dimensional cube. It allows us to write down a few useful relationships between all the objects we have found so far.

Problem 29: Dualities

- a: Show that the following is true $K_n = iJ_n$
b: Show that $T_\mu = \mathbf{e}_\mu i$.
c: Finally show that $i^2 = -1$.

You will get accustomed to all these different objects in due time. For us, most important of all will be the bivectors.

2.4. Problems

This section contains a number of useful and relevant problems.

Problem 30: Exercising the geometric product

Assume we have chosen an orthonormal basis in Spacetime, $\mathbf{e}_\mu$. Compute the geometric product $\mathbf{a}\mathbf{b}$ in the following cases.

a: $\mathbf{a} = \mathbf{e}_1 + \mathbf{e}_2$ and $\mathbf{b} = \mathbf{e}_1 - \mathbf{e}_2$.

b: $\mathbf{a} = 2\mathbf{e}_0 + \mathbf{e}_2$ and $\mathbf{b} = \mathbf{e}_0 - 2\mathbf{e}_2$.

c: $\mathbf{a} = \mathbf{e}_1 + 3\mathbf{e}_2$ and $\mathbf{b} = 3\mathbf{e}_0 - 2\mathbf{e}_3$.

Problem 31: Bubbles in the Early Universe

In modern physics some phase transitions are expected to occur in the early universe. They often happen as a result of a symmetry in the underlying field theories breaking down when the system cools down into a new phase. The solutions to the field equations that describe the formation of a bubble of a new phase are commonly called *Bounce solutions*. The wall of such a bubble is usually a fixed *spacetime distance* away from the initial centre of the bubble.

a: Suppose that at $t = 0$ a bubble of new phase appears centered on $x = 0$ with a radius R . Draw the centre and the bubble walls at $t = 0$ in a spacetime diagram.

b: The spacetime distance between the centre and the bubble walls remains constant. If you assume that the bubble centre does not move, sketch the bubble walls for all time $t > 0$.

c: Find an expression for the speed v with which the bubble walls appear to move outward, relative to the centre, after t seconds.

Problem 32: Lightcone components

Reconsider the lightcone basis from a previous problem.

a: What is the definition of the lightcone basis vectors $\mathbf{k}_\pm$?

b: Show that $\mathbf{k}_\pm \cdot \mathbf{e}_2 = \mathbf{k}_\pm \cdot \mathbf{e}_3 = 0$.

c: Show that the reciprocal basis vectors satisfy $\mathbf{k}^\pm = \mathbf{k}_\mp$. d: Explain why this means that a vector $\mathbf{a}$ can be expanded in the lightcone basis like

$$\mathbf{a} = a^+ \mathbf{k}_+ + a^- \mathbf{k}_- + a^2 \mathbf{e}_2 + a^3 \mathbf{e}_3 .$$

Problem 33: Time and Spacelike lines

Consider the unit vectors $\hat{\mathbf{a}}$, being timelike, and $\hat{\mathbf{b}}$ being spacelike.

a: Sketch a spacetime diagram with the lines represented by $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ and perform a double reflection of some arbitrarily chosen vector in them.

b: Compute $\hat{\mathbf{a}} \cdot \hat{\mathbf{b}}$ and $\hat{\mathbf{a}} \wedge \hat{\mathbf{b}}$ in a conveniently chosen basis in terms of hyperbolic functions.

c: Is the transformation $\mathbf{n}' = \hat{\mathbf{a}}\hat{\mathbf{b}}\mathbf{n}\hat{\mathbf{b}}\hat{\mathbf{a}}$ a Lorentz transformation?

Problem 34: Basis Blades and Commutations

This exercise is a lot of tedious work, but grinding through all these expressions will make you feel very at home in the machinery of geometric algebra. The questions are simple.

a: Establish the following relations among the spacelike bivectors

$$\begin{aligned}\frac{1}{2}(J_1 J_2 - J_2 J_1) &= J_3 , \\ \frac{1}{2}(J_2 J_3 - J_3 J_2) &= J_1 , \\ \frac{1}{2}(J_3 J_1 - J_1 J_3) &= J_2 .\end{aligned}\tag{2.4.1}$$

These algebraic relations are known as the *commutation relations of the lie group $SO(3)$* and they contain all the properties of *rotations*. The relations are called *closed* because any commutator between two of this set of bivectors always returns another bivector from the same set.

b: Show that if we multiply the timelike bivectors among eachother we get

$$\begin{aligned}\frac{1}{2}(K_1 K_2 - K_2 K_1) &= -J_3 , \\ \frac{1}{2}(K_2 K_3 - K_3 K_2) &= -J_1 , \\ \frac{1}{2}(K_3 K_1 - K_1 K_3) &= -J_2 .\end{aligned}\tag{2.4.2}$$

The algebra of these bivectors does not close and we need to include the space-like bivectors as well.

c: Show that if we multiply these among eachother we get

$$\frac{1}{2}(K_i J_i - J_i K_i) = 0 ,\tag{2.4.3}$$

for all i .

d: Don't think this was it! Also show that

$$\begin{aligned}\frac{1}{2}(K_1 J_2 - J_2 K_1) &= -K_3 , \\ \frac{1}{2}(K_2 J_3 - J_3 K_2) &= -K_1 , \\ \frac{1}{2}(K_3 J_1 - J_1 K_3) &= -K_2 ,\end{aligned}\tag{2.4.4}$$

and last but not least

$$\begin{aligned}\frac{1}{2}(K_1 J_3 - J_3 K_1) &= -K_2 , \\ \frac{1}{2}(K_2 J_1 - J_1 K_2) &= -K_3 , \\ \frac{1}{2}(K_3 J_2 - J_2 K_3) &= -K_1 ,\end{aligned}\tag{2.4.5}$$

All these algebraic relations are easy to compute, it just takes a little time.

e: Is the algebra of the spacetime bivectors closed?

It is called the commutator-algebra of the $SO(3,1)$ Lie-group.

Problem 35: Commuting trivectors

- a: Compute the commutators $[T_\mu, T_\nu]$ and show they yield bivectors or vanish.
 b: Show that the set of bivector *and* trivectors taken together are a closed set under commutation.

These 10 generators generate a transformation group known as $SO(4, 1)$.

Problem 36: Linear Maps and Matrices

A *linear map* $G(\mathbf{b})$ on vectors $\mathbf{b}$ is defined as a map satisfying the following two axioms

$$G(\mathbf{b} + \mathbf{a}) = G(\mathbf{b}) + G(\mathbf{a}) , \quad (2.4.6)$$

$$G(a\mathbf{b}) = aG(\mathbf{b}) , \quad (2.4.7)$$

where a is some scalar. Now assume that we have been given some orthonormal basis $\mathbf{e}_\nu$.

- a: Show that the *Lorentz transformation* is a linear map.

Given a linear map G that maps *vectors into vectors* and define $\mathbf{b}' = G(\mathbf{b})$.

- b: Explain why the action of G is completely fixed once we know all the $G(\mathbf{e}_\nu)$.

- c: Define

$$G_\nu^\mu \equiv \mathbf{e}^\mu \cdot G(\mathbf{e}_\nu) , \quad (2.4.8)$$

Show that the components of $\mathbf{b}'$ are given by

$$b'^\mu = \sum_{\nu=0}^3 G_\nu^\mu b^\nu . \quad (2.4.9)$$

Problem 37: Products of Linear Maps

Consider two linear maps G and H both mapping vectors into vectors. Let $G_{\mu\nu}$ and $H_{\mu\nu}$ be the matrices corresponding to these maps. Now we define a new map $K(\mathbf{b}) = G(H(\mathbf{b}))$.

- a: Show that K is also a linear map.
 b: Compute the matrix of K in terms of the matrices of G and H .

Problem 38: Time dilation

Consider a species of particles that have an average lifetime δ . Now a sample of these is observed deep in the hot core of a star and they turn out to decay, on average, after a time 5δ . At what speed do these particle move in the stellar core?

Chapter 3

Kinematics and Lorentz transformations

In the previous chapter you have encountered Lorentz transformations in the form of transformation rules on basis vectors and transformation rules on components of vectors. The use of matrices is of great practical use there, but it too presupposes the choice of a basis. Our vectoralgebra allows a different, basis-free and very powerful method of describing Lorentz transformations. As an added advantage it also is very geometric in nature.

You may recall (see the appendix A if you do not) that rotations can be represented as suitable double reflections in ordinary geometry. The same holds true for Lorentz transformations. Now reflections are easily described using the algebra among the spacetime vectors, and so are Lorentz transformations.

Problem 39: Unit and inverse vectors

Given some vector $\mathbf{a}$ and consider the vector $\hat{\mathbf{a}}$ defined by,

$$\hat{\mathbf{a}} \equiv \frac{\mathbf{a}}{\sqrt{|\mathbf{a}^2|}} . \quad (3.0.1)$$

a: Explain under which conditions it is well-defined, and show it has length ± 1 .
The inverse of a vector $\mathbf{a}$ defined as

$$\mathbf{a}^{-1} \equiv \frac{\mathbf{a}}{\mathbf{a}^2} . \quad (3.0.2)$$

b: Show that $\mathbf{a}\mathbf{a}^{-1} = 1$.

c: Explain for which $\mathbf{a}$ an inverse does *not* exist.

d: Compute the unit-vectors and inverses of the vectors $\mathbf{a}$ from the previous problem.

3.1. Reflections and Parity

Given a vector $\mathbf{b}$ and a non-parallel unit vector $\hat{\mathbf{a}}$, $\hat{\mathbf{a}}^2 = +1$, we can write

$$\begin{aligned} \mathbf{b} &= (\mathbf{b} \cdot \hat{\mathbf{a}})\hat{\mathbf{a}} + (\mathbf{b} \wedge \hat{\mathbf{a}})\hat{\mathbf{a}} , \\ &\equiv \mathbf{b}_{//} + \mathbf{b}_{\perp} . \end{aligned}$$

Now suppose that we would make a new vector $\mathbf{b}'$ from the old one by changing

the sign of the perpendicular part, i.e.

$$\begin{aligned}\mathbf{b}' &= \mathbf{b}_{//} - \mathbf{b}_{\perp} , \\ &= (\mathbf{b} \cdot \hat{\mathbf{a}})\hat{\mathbf{a}} + (\hat{\mathbf{a}} \wedge \mathbf{b})\hat{\mathbf{a}} , \\ &= \hat{\mathbf{a}}\mathbf{b}\hat{\mathbf{a}} .\end{aligned}\tag{3.1.1}$$

This *sandwich* expression is the reflection of $\mathbf{b}$ in the $\hat{\mathbf{a}}$ -axis.

Problem 40: Another reflection

Suppose that we would have used a unit-vector $\hat{\mathbf{c}}$, $\hat{\mathbf{c}}^2 = -1$. Give a *sandwich* expression for the reflection of $\mathbf{b}$ in the $\hat{\mathbf{c}}$ -axis.

So we see that simple transformations can be performed directly by taking the geometric product.

Parity

One particularly important reflection is the one in the time-axis. This operation is called a *parity* transformation and it takes a multivector $\mathbf{a}$ into a vector $\mathbf{a}_P$ given by

$$\mathbf{a}_P = \mathbf{e}_0 \mathbf{a} \mathbf{e}_0 .\tag{3.1.2}$$

The parity transformation is very important in elementary particle physics.

3.2. Double reflections and Rotations

Now suppose that we reflect the vector $\mathbf{b}$ first in the $\hat{\mathbf{c}}$ -axis and subsequently in the $\hat{\mathbf{a}}$ -axis. The resulting vector $\mathbf{b}'$ can be computed through

$$\mathbf{b}' = \hat{\mathbf{a}}\hat{\mathbf{c}}\mathbf{b}\hat{\mathbf{c}}\hat{\mathbf{a}} .\tag{3.2.1}$$

So if we define a so-called *rotor* $R = \hat{\mathbf{a}}\hat{\mathbf{c}}$ and use the notation $\tilde{R}$ for the *reverse*, i.e. the product with the order of all factors reversed, we get

$$\begin{aligned}\mathbf{c}' &= R\mathbf{b}\tilde{R} , \\ R\tilde{R} &= 1 .\end{aligned}\tag{3.2.2}$$

These rotors can contain ordinary rotations, but also Lorentz transformations. To see this consider the case in which $\hat{\mathbf{a}}$ and $\hat{\mathbf{c}}$ are spacelike vectors in the $\mathbf{e}_1\mathbf{e}_2$ -plane. In that case we have

$$\begin{aligned}R &= \hat{\mathbf{a}}\hat{\mathbf{c}} = \hat{\mathbf{a}} \cdot \hat{\mathbf{c}} + \hat{\mathbf{a}} \wedge \hat{\mathbf{c}} , \\ &= -\cos(\theta) + \sin(\theta)\mathbf{e}_1\mathbf{e}_2 .\end{aligned}$$

The calculation of the reversed is easy now and yields

$$\tilde{R} = \hat{\mathbf{c}}\hat{\mathbf{a}} = -\cos(\theta) - \sin(\theta)\mathbf{e}_1\mathbf{e}_2 .$$

So the final result would be that if we apply this rotor to a vector $\mathbf{b}$ we get

$$\mathbf{b}' = (\cos(\theta) - \sin(\theta)\mathbf{e}_1\mathbf{e}_2)\mathbf{b}(\cos(\theta) + \sin(\theta)\mathbf{e}_1\mathbf{e}_2) .$$

Writing this out entirely indeed shows that $\mathbf{b}'$ is simply $\mathbf{b}$ rotated by an angle of 2θ in the $\mathbf{e}_1\mathbf{e}_2$ -plane.

Problem 41: Rotating $\mathbf{e}_1$

Use the last expression found above to compute $\mathbf{b}'$ if we choose $\mathbf{b} = \mathbf{e}_1$.

There is an easier method to compute this than by straightforward expansion of all products. But before we can treat that we need to know a little more about how these rotations are related to the bivectors. First we treat the Lorentz transformations, then we return to this problem.

3.3. The Rotor form of Lorentz transformations

When the lines in the double reflection are timelike the resulting transformation is not a rotation but a Lorentz transformation. We find, a basis where $\hat{\mathbf{c}} = \mathbf{e}_0$,

$$\begin{aligned} R &= \hat{\mathbf{a}}\hat{\mathbf{c}} = \hat{\mathbf{a}} \cdot \hat{\mathbf{c}} + \hat{\mathbf{a}} \wedge \hat{\mathbf{c}} , \\ &= \cosh(\alpha) + \sinh(\alpha)\mathbf{e}_1\mathbf{e}_0 . \end{aligned}$$

So if we apply this rotor to a vector $\mathbf{b}$ we get

$$\mathbf{b}' = (\cosh(\alpha) + \sinh(\alpha)\mathbf{e}_1\mathbf{e}_0)\mathbf{b}(\cosh(\alpha) - \sinh(\alpha)\mathbf{e}_1\mathbf{e}_0) .$$

Consider the exponential function $\exp(x)$. In your math courses you will have encountered its Taylor expansion

$$\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} . \quad (3.3.1)$$

Now if you consider an expression like $\exp(\alpha K_1)$ and use that $K_1^2 = 1$ you will find that

$$\begin{aligned} \exp(\alpha K_1) &= \sum_{n=0}^{\infty} \frac{\alpha^n K_1^n}{n!} , \\ &= \sum_{n=\text{even}} \frac{\alpha^n}{n!} + \sum_{n=\text{odd}} \frac{\alpha^n}{n!} K_1 , \\ &= \cosh(\alpha) + \sinh(\alpha)K_1 . \end{aligned}$$

This last expression is remarkable because it is a rotor representing a Lorentz transformation in the $\mathbf{e}_1$ -direction. So a *boost* L_j in the ϕe_j direction can be written as

$$L_j(\alpha) = \exp\left(\frac{\alpha}{2}K_j\right) . \quad (3.3.2)$$

Problem 42: Do the Boost

a: Compute $\mathbf{u} = L_1\mathbf{e}_0\tilde{L}_1$.

b: Show that the vector $\mathbf{u}$ represents a proper velocity for a particle moving with ordinary speed $v = c \tanh(\alpha)$ along the $\mathbf{e}_1$ axis.

Similarly the bivectors J_n can be used to generate rotations in the planes represented by these bivectors. A rotation in the J_n plane about an angle θ is represented by the rotor $R_n(\theta)$ given by

$$R_n(\theta) = \exp\left(\frac{\theta}{2}J_n\right). \quad (3.3.3)$$

Problem 43: Check the powerseries

Expand the formula for $R_n(\theta)$ in powers and check that you indeed get

$$R_n(\theta) = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)J_n. \quad (3.3.4)$$

It is extremely usefull to work with these exponential forms of rotations and Lorentz transformations. If you obtain skill in manipulating them this allows you to short cut many algebraic derivations. For example the tedious multiplications necessary to calculate the action of a rotor on a vector become simpler. I will give an example.

Example

Consider the vector $\mathbf{e}_1$ and a rotation in the J_1 plane. First note that

$$\mathbf{e}_1 J_1 = J_1 \mathbf{e}_1 ,$$

i.e. $[\mathbf{e}_1, J_1] = 0$, we say that they *commute*. As a result we find that

$$\mathbf{e}_1 \exp(-\alpha J_1) = \exp(-\alpha J_1) \mathbf{e}_1 .$$

Now it is very easy to see that

$$\begin{aligned} \exp(\alpha J_1) \mathbf{e}_1 \exp(-\alpha J_1) &= \exp(\alpha J_1) \exp(-\alpha J_1) \mathbf{e}_1 , \\ &= \exp(\alpha J_1 - \alpha J_1) \mathbf{e}_1 = \mathbf{e}_1 . \end{aligned}$$

Geometric insight would have said: $\mathbf{e}_1$ is perpendicular to the plane J_1 and so rotating in that plane does not change $\mathbf{e}_1$. Note that in the last step we used $\exp(A)\exp(B) = \exp(A+B)$; this is *only allowed when* $[A, B] = 0$, i.e. when the two terms commute. If they do not commute this simplification cannot be done and a much more involved formulae exists, the so-called Campbell-Baker-Hausdorff formula. It says that if

$$\exp(A)\exp(B) = \exp(C) ,$$

then C is given by an infinite series

$$C = A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] - \frac{1}{12}[B, [A, B]] = \dots .$$

Problem 44: CBH-formula

- a: Look up the entire Campbell-Baker-Hausdorff relation and quote the next two terms in the series.
- b: If $A = \alpha J_1$ and $B = \alpha J_2$, compute C up to order α^3 .
- c: Suppose that $D = [A, B]$ gives a D that commutes with both A and B . How does this simplify the CBH-relation?

But now let's consider $\mathbf{e}_2$ and the same rotation. This time we have

$$\mathbf{e}_2 J_1 = -J_1 \mathbf{e}_2 ,$$

and so

$$\mathbf{e}_2 \exp(-\alpha J_1) = \exp(+\alpha J_1) \mathbf{e}_2 .$$

So we find

$$\begin{aligned} \exp(\alpha J_1) \mathbf{e}_2 \exp(-\alpha J_1) &= \exp(\alpha J_1) \exp(+\alpha J_1) \mathbf{e}_2 , \\ &= \exp(2\alpha J_1) \mathbf{e}_2 = (\cos(2\alpha) + \sin(2\alpha) J_1) \mathbf{e}_2 . \\ &= \cos(2\alpha) \mathbf{e}_2 + \sin(2\alpha) \mathbf{e}_3 . \end{aligned}$$

From the above example we see that the key role in the geometry of transformations is played by the elements that *generate* the transformations, i.e. the bivectors, and the *commutation* relations with the other objects in the algebra. For a vector $\mathbf{b}$ and a bivector B we can say in general: if $\mathbf{b}$ is *invariant* under transformations generated by B , then

$$\begin{aligned} [B, \mathbf{b}] &= 0 , \text{ i.e.} \\ B\mathbf{b} &= \mathbf{b}B . \end{aligned} \tag{3.3.5}$$

A bivector B that *does* affect a vector $\mathbf{b}$ satisfies

$$B\mathbf{b} = -\mathbf{b}B .$$

3.3.1. Reconstructing a Lorentz-transformation

Suppose that you would be given two vectors $\mathbf{a}'$ and $\mathbf{a}$. Can you find a Lorentz-transformation R for which $\mathbf{a}' = R\mathbf{a}\tilde{R}$? The answer is yes, and again the algebra will be straightforward once you use geometric insight.

As we only want to find *a* Lorentztransformation we may want to choose a transformation that *does not contain any transformations that leave $\mathbf{a}$ invariant*. In other words, we choose R to be generated by a bivector B such that $B\mathbf{a} = -\mathbf{a}B$.

Problem 45: Squaring the rotor

Show that if for a bivector B and a vector $\mathbf{b}$, then the rotor $R = \exp(B)$ satisfies

$$R\mathbf{b}\tilde{R} = R^2\mathbf{b} .$$

In that case we can rewrite the transformation equation as

$$bfa' = Ra\tilde{R} = R^2a .$$

This equation can be easily solved for R by multiplication with a^{-1} from the right/ This yields

$$R = \sqrt{a'a^{-1}} . \quad (3.3.6)$$

Problem 46: Necessary condition

Show that for this procedure for finding such a rotor R to work properly we must have $(a')^2 = a^2$.

3.4. Proper time and Proper velocity

Inertial observers move along straight worldlines, but particles that they study need not do so. What is the correct way of handling velocities in Spacetime? And if these particles carry internal clocks, then how would they behave if the particle itself is not an inertial observer?

3.4.1. Proper time and proper velocity

Consider Alice moving with some speed $v(\tau)$ as she moves past Bob. For simplicity assume that both their clocks indicate $t = 0$ for Bob and $\tau = 0$ for Alice when they pass each other, their origin's coinciding. According to herself Alice is at rest, and so a very small time $d\tau$ later (on her clock!) her position in spacetime is

$$\mathbf{x}_A(\tau) = cd\tau\mathbf{e}'_0 .$$

For Bob Alice is moving with a speed $v(0)$ along the $\mathbf{e}_1$ -direction, and so for him after a small time dt Alice is at

$$\mathbf{x}_A(dt) = (c\mathbf{e}_0 + v(0)\mathbf{e}_1)dt .$$

Now we know from the Lorentztransformations that

$$\mathbf{e}'_0 = \gamma\{\mathbf{e}_0 + \frac{v(0)}{c}\mathbf{e}_1\} ,$$

and during the time-interval dt on Bob's clock a time $d\tau$ has passed on ALice's clock as she moved from her origin to

$$\mathbf{x}_A(d\tau) = \mathbf{e}'_0 d\tau . \quad (3.4.1)$$

If we compare the two expressions for Alice's position in Spacetime, we find that they are equal if $d\tau\gamma = dt$, i.e. when

$$d\tau = \sqrt{1 - \frac{v(t)^2}{c^2}} dt .$$

The effect that Alice's clock seems to run slower than Bob's clock is what we

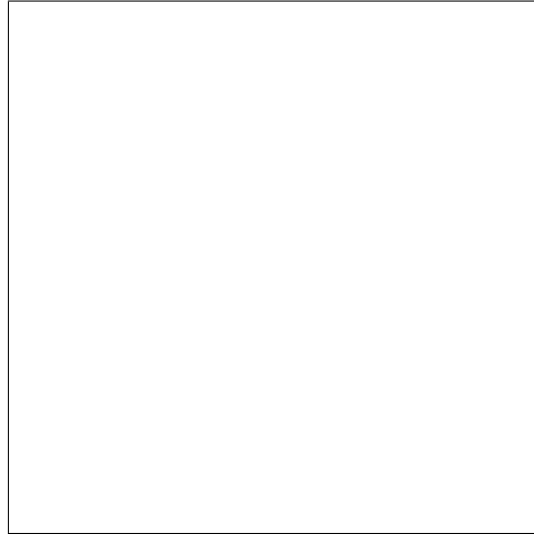


fig. 12: A worldline and proper velocity

called *time-dilation*. The derivation here is a more precise alternative to the one given previously. Now there is something peculiar about the square-root; have you noticed it?

Problem 47: Time-dilation and inner product

a: Show that for the spacetime velocity $\mathbf{v}(t)$ of the particle according to Bob we could also write

$$\sqrt{1 - \frac{v(t)^2}{c^2}} = \sqrt{\frac{\mathbf{v}(t)^2}{c^2}}.$$

b: Show that $\mathbf{v}(t)^2 > 0$ for $v(t) < c$.

The relation between the two intervals is actually

$$d\tau = \sqrt{\frac{\mathbf{v}(t)^2}{c^2}} dt. \quad (3.4.2)$$

So if we "sum up" all the little intervals dt between Bob's times t_0 and t_1 we find the integral

$$\tau_1 - \tau_0 = \int_{t_0}^{t_1} \sqrt{\frac{\mathbf{v}(t)^2}{c^2}} dt, \quad (3.4.3)$$

representing the proper time that has passed on a clock in the restframe of the particle. We call the time that passes in the rest-frame of a moving object its *proper time*, and we will denote it by the Greek letter τ .

Problem 48: Half-Life

Consider a very small box of unstable particles at $x = 0$. The box explodes, in Bob's frame at $t = 0$, and the unstable particles fly apart with very different speeds in all directions. In their rest frame these particles are known to have a half-life time of $t_{\frac{1}{2}} = 1.0\text{s}$. Draw a spacetime diagram and sketch the lines where one, two and three decay times have passed.

In the expression for the proper time we integrate the spacetime length of the tangent-vector $\mathbf{v}(t)$ along the worldline. The result is ofcourse the spacetime length of the worldline.

Problem 49: Twins that Travel

Consider a twin Boba and Alex. Alex never does anything and can always be considered at rest.

a: Draw a spacetime frame according to Alex.

Boba is a bounty-hunter and so must remain on the move. At $t = 0$ Boba leaves Alex and travels with a speed $v = 0.5c$ into the $\mathbf{e}_1$ -direction. After one year (on Alex's clock) Boba turns around instantaneously and travels back at the same speed.

b: How much time has passed on Alex's clock when Boba returns?

c: How much time has passed on Boba's clock when he returns to Alex? Are Boba and Alex still equally old?

d: Which distance has Boba covered according to Alex?

e: Which distance has Boba covered according to himself?

Proper time is also a relevant concept in the study of large amounts of relativistic decaying matter. In such cases proper time describes the local rate at which time passes. An example is the evolution of a quark-gluon plasme originating from energetic collissions of heavy ions.

Problem 50: A moving Bubble

Consider a heavy ion collision in which two large atomic nuclei are smashed together, at $t = 0$ and $x = 0$ in the lab frame, forming a so-called *quark-gluon* plasma for a very short period. The plasma expands in all directions for about 10^{-21} s.

a: Draw a spacetime diagram of this process and draw the line at which all parts of the expanding plasma reach a proper time of 10^{-21} s.

After this time the plasma *hadronizes* into ordinary particles such as protons, neutrons and lots of mesons. Suppose this transition from the plasma to the hadronic phase occurs due to the expansion of a single bubble whose centre is initially moving with a speed of $v = 0.5c$ in the lab-frame in the positive x -direction.

b: Indicate the centre of the bubble on the hadronization line and draw a space-axis and a time axis through it.

c: In the rest frame of the bubble the bubble walls are initially one Fermi away from the centre. Draw them on the space-axis of the initial bubble.

d: Each point on each of the two walls has the same constant spacetime distance from the centre of the initial bubble. Sketch the time evolution of the bubble walls.

e: Is this evolution of the bubble walls Lorentz covariant?

Proper velocity

We define the *proper velocity* $\mathbf{u}(\tau)$ of an object following the worldline $\mathbf{x}(\tau)$ as

$$\mathbf{u}(\tau) = \frac{d}{d\tau} \mathbf{x}(\tau) . \quad (3.4.4)$$

A naive definition of spacetime velocity for a particle reads

$$\mathbf{v}(t) = \frac{d}{dt} \mathbf{x}(t) = c\mathbf{e}_0 + v(t)\mathbf{e}_1 .$$

If we consider the observer's time t to be a function of the particle's proper time τ , and use

$$d\tau = \sqrt{1 - \frac{v(t)^2}{c^2}} dt ,$$

we find from the chain rule

$$\begin{aligned} \frac{d}{d\tau} \mathbf{x}(\tau) &= \frac{d}{dt} \mathbf{x}(t(\tau)) \frac{d}{d\tau} t(\tau) , \\ &= \frac{1}{\sqrt{1 - \frac{v(t)^2}{c^2}}} \mathbf{v}(t) . \end{aligned}$$

In the case of an object, moving in the $\mathbf{e}_1$ -direction with respect to Bob, this means

$$\mathbf{u}(\tau) = \gamma \{ c\mathbf{e}_0 + v(\tau)\mathbf{e}_1 \} . \quad (3.4.5)$$

This proper velocity has the very special property that $\mathbf{u}(\tau)^2 = c^2$.

Problem 51: Proper velocity can be easy

Take another look at all your Alice and Bob pictures and comment on the following statements.

- a: Bob's proper velocity is $\mathbf{u}_{Bob} = c\mathbf{e}_0$.
- b: Alice's proper velocity is $\mathbf{u}_{Alice} = c\mathbf{e}'_0$.
- c: The proper velocity of a particle is just c times it's own time-direction.
- d: The time-dilation factor γ between Bob and Alice is just $\gamma = \frac{\mathbf{u}_{Alice} \cdot \mathbf{u}_{Bob}}{c^2}$.

3.4.2. Relative velocity and velocity addition

The proper velocity of the particle $\mathbf{u}(\tau)$ observed by the observer using the basis $\mathbf{e}_\mu$ contains a time-like part proportional to $\mathbf{e}_0$ and a remainder, the spacelike part. Now if we recall that $c\mathbf{e}_0$ is just the proper velocity of the observer and denote it $\mathbf{u}_O$, we can write

$$\vec{v}(\tau) = \frac{\mathbf{u}(\tau) \wedge \mathbf{u}_O}{\mathbf{u}(\tau) \cdot \mathbf{u}_O} . \quad (3.4.6)$$

The bivector represents the spacelike part of the relative velocity between the observer and the object. This algebraic extraction of a relative velocity from the two

proper velocities will turn out to be a very usefull equation in many cases.

Problem 52: Relative velocity

Consider a particle with a proper velocity $\mathbf{v}(\tau)$ and an inertial observer with a proper velocity $\mathbf{u}_O$.

a: Expand both proper velocities in an orthonormal basis for the observer O .

b: Compute $\vec{v}(\tau)$ according to the equation above and show that it has the general form

$$\vec{v}(\tau) = \sum_{j=1}^3 v_j(\tau) \mathbf{e}_j \mathbf{e}_0 .$$

c: Show that $\vec{v}(\tau)^2 = \sum_{j=1}^3 v_j(\tau)^2$.

Let Bob assign a proper velocity $\mathbf{u}$ to Alice, as well as a proper velocity $\mathbf{v}$ to a moving particle. The relative velocity between Alice and the particle can be expressed as a bivector

$$\vec{v}_A = \frac{\mathbf{v} \wedge \mathbf{u}}{\mathbf{v} \cdot \mathbf{u}} c . \quad (3.4.7)$$

Velocity Addition

Velocity addition is an odd thing. Velocities belong to different particles and there is no evidence that they can or should be added. What is meant is something else. If, in non-relativistic physics, two observers A and B move at a relative velocity $\mathbf{v}$ and each observes a particle and assigns to it a velocity $\mathbf{v}_A$ for A and $\mathbf{v}_B$ for B, then

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v} .$$

However this equation is only approximately right, as we will see. Furthermore we are not adding velocities here (mathematically ofcourse we are) but we are expressing the velocity observed by Alice, $\mathbf{v}_A$, in terms of the velocity observed by Bob $\mathbf{v}_B$. We are comparing observations of two inertial observers in relative motion. So we are performing a Lorentz transformation and nothing else! Many undergraduate textbooks do not discuss it that way, but we do because that is what it is. Our *translation* problem now requires the following: Express the components of $\vec{v}_A$ in the basis used by Alice, in terms of the components measured by Bob.

We have a frame-free expression for the relative velocity, but to evaluate it we may ofcourse choose a suitable frame. So, let Bob choose a basis in which the proper velocities take the following form

$$\begin{aligned} \mathbf{u} &= \gamma_u \{ c \mathbf{e}_0 + u \mathbf{e}_1 \} , \\ \mathbf{v} &= \gamma_v \{ c \mathbf{e}_0 + v \cos(\theta) \mathbf{e}_1 + v \sin(\theta) \mathbf{e}_2 \} . \end{aligned}$$

Bob has chosen the $\mathbf{e}_1$ -direction to be in the direction in which Alice moves, and he has chosen the $\mathbf{e}_2$ -direction in such a way that the particle moves in the 1–2 plane. This is allowed and represents no assumption on any physics involved. If we use these two equations and plug them into the equation for $\vec{v}_A$ we find $\vec{v}_A$ in terms of the components that Bob assigns to $\mathbf{u}$ and $\mathbf{v}$. We find from a little bit of algebra

$$\vec{v}_A = \frac{[v \cos(\theta) - u] \mathbf{e}_1 \mathbf{e}_0 + v \sin(\theta) \mathbf{e}_2 \mathbf{e}_0 - v \sin(\theta) \frac{u}{c} \mathbf{e}_1 \mathbf{e}_2}{1 - \frac{uv}{c^2} \cos(\theta)} .$$

Problem 53: Check it

Don't just take the above equation on faith; Check it.

Now let's the components of this *same* bivector in the basis of Alice. We could take dot-products with the basis-bivectors of Alice, but simply reading them of is easier.

The basis Alice uses, expressed in terms of Bob's basis vectors, is

$$\begin{aligned}\mathbf{e}'_0 &= \frac{\mathbf{u}}{c} = \gamma_u \{c\mathbf{e}_0 + u\mathbf{e}_1\} , \\ \mathbf{e}'_1 &= \gamma_u \left\{ \mathbf{e}_1 + \frac{u}{c}\mathbf{e}_0 \right\} , \\ \mathbf{e}'_2 &= \mathbf{e}_2 .\end{aligned}$$

If Alice expands the bivector $\vec{v}_A$ in terms of her basis and subsequently inserts the expressions above, this yields

$$\begin{aligned}\vec{v}_A &= v_{//}\mathbf{e}'_1 \wedge \mathbf{e}'_0 + v_{\perp}\mathbf{e}'_2 \wedge \mathbf{e}'_0 , \\ &= v_{//}\gamma_u^2\left(1 - \frac{u^2}{c^2}\right)\mathbf{e}_1\mathbf{e}_0 + v_{\perp}\gamma_u\mathbf{e}_2\mathbf{e}_0 - v_{\perp}\gamma_u\frac{u}{c}\mathbf{e}_1\mathbf{e}_2 .\end{aligned}$$

Comparing the two expression for $\vec{v}_A$, and using that $\gamma_u^2(1 - \frac{u^2}{c^2}) = 1$, we can read off the following identifications

$$v_{//} = \frac{v \cos(\theta) - u}{1 - \frac{uv}{c^2} \cos(\theta)} , \quad (3.4.8)$$

$$v_{\perp} = \frac{1}{\gamma_u} \frac{v \sin(\theta)}{1 - \frac{uv}{c^2} \cos(\theta)} . \quad (3.4.9)$$

The first of these equations is usually given in introductory texts for colinear velocities. i.e. for the special case where $\theta = 0$, then it reads

$$v_{//} = \frac{v - u}{1 - \frac{uv}{c^2}} .$$

Problem 54: Receding Galaxies

We observe light emitted by two Galaxies, B and C, receding from our Galaxy, A, with a relative speed of $0.8c$ in opposite directions.

a: Sketch the situation in a spacetime diagram and determine the increase per second of the distance between B and C according to us?

b: What is the increase per second in distance between B and C according to an observer co-moving with C?

The second one shows that perpendicular velocities are suppressed by a factor γ_u . This equation is given in fewer introductory textx and then mostly only for the case that $\theta = \frac{\pi}{2}$ in which case we get

$$v_{\perp} = \frac{1}{\gamma_u} v .$$

These two equations represent the general result for translating a velocity observed by one observer to that observed by another. It was a tricky translation exercise rather than a physics problem!

3.5. Proper acceleration and the Eigen-rotor

The proper velocity of a particle has a fixed spacetime length. So what *could* a change in the *magnitude* of proper velocity mean? It would require a change in the rate at which all clocks tick that an observer may carry with him. But if all clocks change, then how does an observer notice this? We can also give a geometric argument with the same content.

Problem 55: Proper time is the time

Comment on the correctness of the following statements.

- a: A change in the magnitude of proper velocity is like a change in the rate at which proper-time passes.
- b: If proper time "passes faster" the particle moves along the worldline "more quickly".
- c: But as the worldline and it's length, stay the same, physically everything stays the same when proper time speeds up or slows down.
- d: Changes in rate of proper time or in the magnitude of proper velocity are physically meaningless.

The fact that proper-velocity has a constant magnitude has several interesting consequences. One is, for example, that proper velocities are always related by active Lorentz transformations.

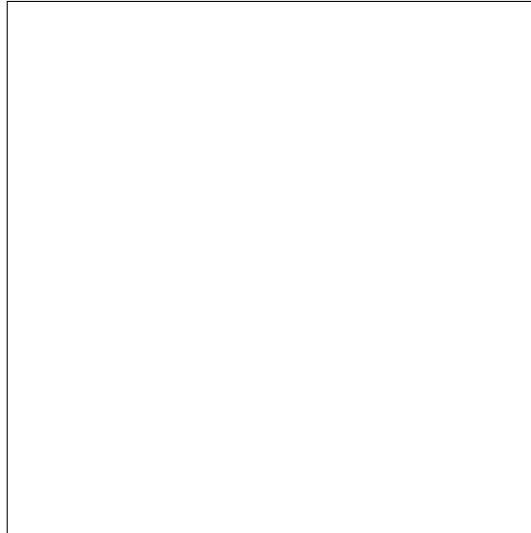


fig. 13: Proper velocity and a moving frame

Problem 56: Lorentz transformations and proper velocity

Consider a particle that moves with a proper velocity $\mathbf{u}(\tau)$. We compare the velocity at two different proper times τ_1 and τ_2 . Explain why we can always find an *active* Lorentz transformation that maps $\mathbf{u}(\tau_1)$ onto $\mathbf{u}(\tau_2)$.

Furthermore the *proper acceleration* $\mathbf{a}(\tau)$, defined as

$$\mathbf{a}(\tau) = \frac{d}{d\tau} \mathbf{u}(\tau) , \quad (3.5.1)$$

satisfies

$$\mathbf{a}(\tau) \cdot \mathbf{u}(\tau) = 0 , \quad (3.5.2)$$

due to the constant length of the proper velocity. Proper velocity and proper acceleration are always perpendicular!

Problem 57: Showing it

It may be a surprising result that proper velocity and proper acceleration are always perpendicular. But the result is understandable.

- a: Differentiate (formally) $\mathbf{u}(\tau) \cdot \mathbf{u}(\tau) = c^2$ by using the product rule and derive the result.
- b: Now consider a spacetime diagram of Bob, who is at rest. Draw the unit-vectors he uses. Suppose a force, Alice's anger, acts on Bob in which direction will Bob's proper velocity change?
- c: If a change in the magnitude of spacetime velocity is physically meaningless, why then will a physically relevant acceleration be perpendicular to this velocity?

The proper acceleration of a particle has a simple interpretation: it is the acceleration as measured in the rest-frame of the particle. The fact that the

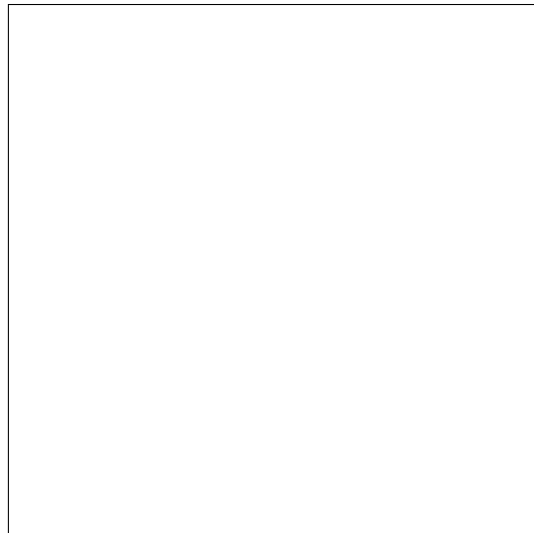


fig. 14: Proper velocity and acceleration

proper velocity has a constant magnitude and that it is physically meaningless to

allow forces to change that length results in an alternative way of describing the motion of particles. The proper velocity $\mathbf{u}(\tau)$ of a particle can always be connected to the initial value $\mathbf{u}(0)$ by means of a Lorentz transformation.

Problem 58: The *minimal boost rotor*

In the previous section we have derived an expression for a rotor connecting two vectors via a Lorentz transformation.

a: Explain why the rotor $R(\tau) = \sqrt{\mathbf{u}(\tau)\mathbf{u}(0)^{-1}}$ is a *pure boost* from the $\tau = 0$ restframe of the particle to its frame at τ ?

b: Let $\mathbf{u}(0) = c\mathbf{e}_0$ and $\mathbf{u}(\tau) = \gamma\{c\mathbf{e}_0 + v(\tau)\mathbf{e}_1\}$. Compute $R(t)$.

Let the rotor $\Lambda(\tau)$ represent a general Lorentz transformation connecting $\mathbf{u}(\tau)$ and $\mathbf{u}(0)$. Then we have

$$\mathbf{u}(\tau) = c\Lambda(\tau)\mathbf{u}(0)\tilde{\Lambda}(\tau) . \quad (3.5.3)$$

This $\Lambda(\tau)$ is called the *eigen-rotor* of the particle. All rotors, including eigen-rotors, satisfy $\Lambda(\tau)\tilde{\Lambda}(\tau) = 1$ so we find

$$\begin{aligned} \frac{d}{d\tau}\Lambda(\tau)\tilde{\Lambda}(\tau) &= 0 , \\ \left\{\frac{d}{d\tau}\Lambda(\tau)\right\}\tilde{\Lambda}(\tau) &= -\Lambda(\tau)\left\{\frac{d}{d\tau}\tilde{\Lambda}(\tau)\right\} . \end{aligned}$$

So we define the *acceleration bivector* Ω as

$$\Omega(\tau) \equiv 2\left\{\frac{d}{d\tau}\Lambda(\tau)\right\}\tilde{\Lambda}(\tau) , \quad (3.5.4)$$

and using it we can write for the acceleration

$$\begin{aligned} \mathbf{a}(\tau) &= \frac{1}{2}\{\Omega(\tau)\mathbf{u}(\tau) - \mathbf{u}(\tau)\Omega(\tau)\} , \\ &\equiv [\Omega(\tau), \mathbf{u}(\tau)] . \end{aligned}$$

3.6. Problems

This section contains a number of useful and relevant problems.

Problem 59: The Fizeau Experiment

Consider a tube filled with some gas flowing into the positive x -direction with a speed v . Through this gas a beam of light travels against the stream with the in-medium light speed $c_n = \frac{c}{n}$. At which speed does the light signal travel in the lab frame and find an approximate expression if $v \ll c$.

Problem 60: The *Centre-of-velocity* frame

Given two proper velocities $\mathbf{u}_A$ and $\mathbf{u}_B$ and their relative time-dilation

$$\gamma_{AB} = \frac{\mathbf{u}_A \cdot \mathbf{u}_B}{c^2} .$$

We can use them to construct a frame in which the proper velocities have the same time-like component but opposite spacelike components. One can call this the *centre of velocity* frame.

a: We define the vectors

$$\mathbf{z}_{\pm} = \frac{1}{\sqrt{2c^2}} \{ \mathbf{u}_A \pm \mathbf{u}_B \} ,$$

show that $\mathbf{z}_+ \cdot \mathbf{z}_- = 0$.

b: Compute $\mathbf{z}_{\pm}^2$ and show that we can always identify a timelike and a spacelike vector.

c: Use the vectors $\mathbf{z}_{\pm}$ to define appropriate unit-vectors $\mathbf{e}_{\pm}$.

d: Now show that in terms of these unit vectors we indeed have

$$\mathbf{u}_A = a_+ \mathbf{e}_+ + a_- \mathbf{e}_- , \quad (3.6.1)$$

$$\mathbf{u}_B = a_+ \mathbf{e}_+ - a_- \mathbf{e}_- , \quad (3.6.2)$$

and determine the constants $a_{\pm}$.

Problem 61: Rapidities and Dilations

An observer observes two particles moving with proper velocities $\mathbf{u}_1$ and $\mathbf{u}_2$. The observer can write for the proper velocities of the particles,

$$\mathbf{u}_1 = \cosh(\alpha_1) \mathbf{e}_0 + \sinh(\alpha_1) \mathbf{e}_1 ,$$

$$\mathbf{u}_2 = \cosh(\alpha_2) \mathbf{e}_0 + \sinh(\alpha_2) \{ \cos(\phi) \mathbf{e}_1 + \sin(\phi) \mathbf{e}_2 \} .$$

a: Compute the time-dilation of particle 1 relative to particle 2 in terms of the rapidities α_1 and α_2 of the particles and the angle ϕ between the direction of their velocities in space.

b: Suppose $\phi = 0$. Show that the relative time dilation is *not* just the product of the time-dilations between the observer and the particles.

c: Show that the relative time-dilation at $\phi = 0$ takes the simple form

$$\gamma_{12} = \cosh(\alpha_1 - \alpha_2) . \quad (3.6.3)$$

d: In which situation is the relative time-dilation just the product of the time-dilations?

Problem 62: Commutators and Infinitesimal Rotations

a: Consider an arbitrary vector $\mathbf{a}$ and the bivector K_1 defined in the text. Compute the commutator $[\mathbf{a}, K_1]$.

b: Consider some unit-bivector F , i.e. $F^2 = \pm 1$, then we know that the rotor

$$R = \exp\left(-\frac{\alpha}{2}F\right),$$

generates a rotation or a Lorentzboost. Now show that if $\alpha \approx 0$ then we can write for any vector $\mathbf{a}$

$$\mathbf{a}' \equiv R\mathbf{a}\tilde{R} \approx \mathbf{a} + \alpha[\mathbf{a}, F],$$

when we ignore terms quadratic or higher powers in α .

c: Consider the following two rotors; $L_1 = \exp\left(-\frac{\alpha_1}{2}K_1\right)$ and $L_2 = \exp\left(-\frac{\alpha_2}{2}K_2\right)$. Now we investigate the equation

$$L_1 L_2 = \exp\left(-\frac{\alpha_1}{2}K_1 - \frac{\alpha_2}{2}K_2\right).$$

Analyse all terms including quadratic terms in α_1, α_2 and terms proportional to $\alpha_1\alpha_2$. What do you conclude about the meaning of the commutator $[K_1, K_2]$?

Problem 63: Commutation relations are Real

This problem has a manual component best executed with a fellow student.

a: Show that a rotation in the J_k plane about an angle of π is represented by the rotor $R = J_k$.

b: Now pick up a book, put it on a table. Let the flat table be the plane represented by J_3 and choose the two other planes J_1 and J_2 .

c: Perform a rotation of the book in the J_1 plane about π radians, followed by a rotation in the J_2 plane also about π radians and carefully memorize the orientation of the book. Why is the rotor describing this double rotation $J_2 J_1$?

d: Put the book back in its original orientation. Perform a rotation of the book in the J_2 plane about π radians, followed by a rotation in the J_1 plane also about π radians and carefully memorize the orientation of the book. Why is the rotor describing this double rotation $J_1 J_2$?

e: What is the difference in orientation between the result under c: and the result under d:? (If done correctly the difference in results will be a rotation in the J_3 plane about π radians. Exactly what the commutator $[J_1, J_2] = J_3$ says!).

Problem 64: Passive and Active

If a physical system described by a vector $\mathbf{b}$ is observed by two inertial observers the measured properties the two observers find are related through

$$\begin{aligned} b'^{\mu} &= \mathbf{e}^{\mu'} \cdot \mathbf{b} , \\ b^{\mu} &= \mathbf{e}^{\mu} \cdot \mathbf{b} . \end{aligned}$$

- a: Justify why we could call the transformation $b'^{\mu} \rightarrow b^{\nu}$ a *passive* transformation.
b: There is ofcourse a rotor R describing the Lorentz transformation between the two observers such that $\mathbf{e}'_{\mu} = R\mathbf{e}_{\mu}\tilde{R}$. Show that

$$\mathbf{b} \cdot (R\mathbf{a}\tilde{R}) = (\tilde{R}\mathbf{b}R) \cdot \mathbf{a} . \quad (3.6.4)$$

- c: Show that $b'^{\mu} = \mathbf{e}^{\mu} \cdot (\tilde{R}\mathbf{b}R)$.
d: Why can we call the representation of the same Lorentz transformation as before but now via $\mathbf{b} \rightarrow (\tilde{R}\mathbf{b}R)$ an *active* Lorentz transformation.

Problem 65: Yet another group

In an earlier exercise we have studied the commutation relations of all the trivectors and bivector among eachother. In this exercise we will look at an interesting subgroup.

- a: Show, using the previous results, that the set of the three spacelike bivectors J_j and the spacelike trivectors T_j form a closed commutator algebra.
b: This commutator algebra is called $SO(4)$. It represents the algebra of rotations in a 4 dimensional euclidean space! Define the multi-vectors

$$H_j^{\pm} \equiv \frac{1}{2}\{1 \pm \mathbf{e}_0\}J_j . \quad (3.6.5)$$

Why do you expect them to generate the same group as the J_j and the T_j ?

- c: Show that $[H_j^+, H_k^-] = 0$ and argue that this means that $SO(4)$ can be decomposed into a product of two subgroups.
d: Show that both H_j^+ as well as the H_k^- form an $SO(3)$ commutator algebra.
e: Is $\frac{1}{2}(1 \pm \mathbf{e}_0)$ a projector?

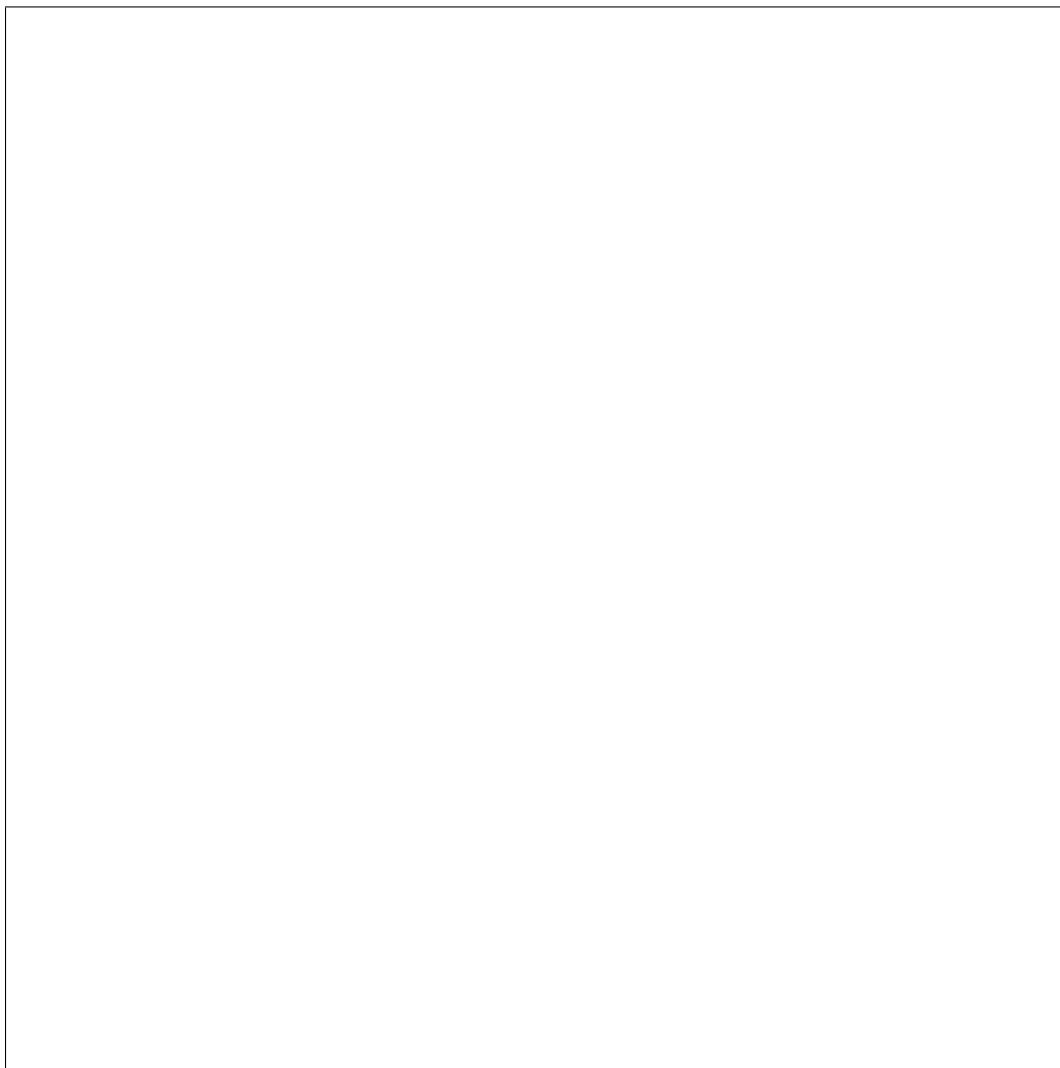


fig. 15: Pictorial commutation relations

Chapter 4

Elementary physics in Spacetime

In the previous chapter we have studied some of the geometric properties of flat spacetime. In this chapter we will focus on the description of motion by massive and massless particles, as well as on what Lorentz transformations tell us about the way different inertial observers measure these kinematical quantities. We start with the two most wellknown effects in spacetime: time-dilation and length contraction.

4.1. Momentum and Energy

Consider a particle moving through spacetime. Now as you know from standard mechanics we associate to this motion a *momentum*, depending on the mass m of the particle. We also do that in relativity, however we use the proper velocity instead of the ordinary velocity. So we set

$$\mathbf{p} \equiv m\mathbf{v} . \quad (4.1.1)$$

If you pick an arbitrary spacelike component, for example p_1 you will notice that it reads

$$p^1 = m\gamma(v)v^1 . \quad (4.1.2)$$

Problem 66: Momentum-Mass relation

Consider the relativistic momentum $\mathbf{p}$ of a particle with restmass m .

a: Show that $\mathbf{p}^2 = m^2c^2$.

b: Use this relation to express p^0 as a function of momentum.

Many textbooks call the expression

$$m_{rel} = m\gamma(v) ,$$

the *relativistic* mass. Whenever we talk about mass, we mean the *restmass* m and not the relativistic mass.

Problem 67: Non-relativistic approximations

Use the Taylor-series expansion to show that $\gamma \approx 1 + \frac{1}{2} \frac{v^2}{c^2}$ if $\frac{v}{c} \ll 1$.

What is obvious from our definition of momentum is that its increase with v

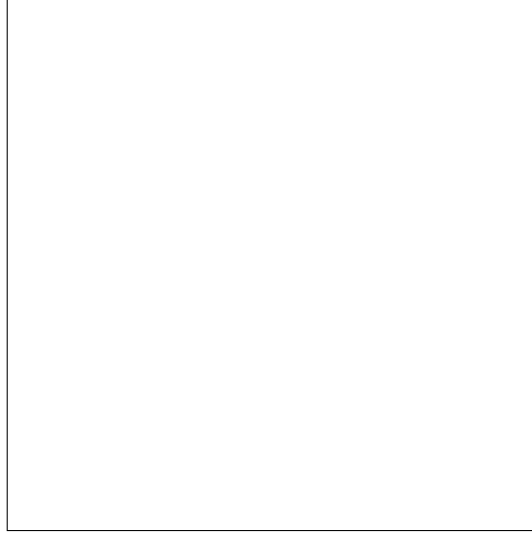


fig. 16: Relativistic momentum as a function of speed

is stronger than that of non-relativistic mechanics. What is not obvious is what the meaning is of the time-like component of proper momentum. It is easy to find though, just consider what it looks like for low velocities,

$$\begin{aligned} p^0 &= mc\gamma(v) , \\ &\approx mc + \frac{1}{2}m\frac{v^2}{c} , \\ &= \frac{mc^2 + \frac{1}{2}mv^2}{c} , \end{aligned}$$

The second term in the numerator we identify as the non-relativistic *kinetic energy*. So we are dealing with an energy, we define

$$p^0 \equiv \frac{E}{c} . \quad (4.1.3)$$

Notice that if the particle is at rest this definition entails

$$E = mc^2 , \quad (4.1.4)$$

Einstein's famous T-Shirt equation. If we observe a particle with proper momentum $\mathbf{p}$ we will expand it as

$$\mathbf{p} = \frac{E}{c}\mathbf{e}_0 + p^1\mathbf{e}_1 + p^2\mathbf{e}_2 + p^3\mathbf{e}_3 . \quad (4.1.5)$$

The effect of forces is to change the momentum of the particle as well as its kinetic energy.

Problem 68: Energy-Momentum relation

- a: Use the formula you found for p^0 as a function of momentum and mass to derive a relation for the energy E as a function of momentum and mass.
- b: What is the non-relativistic limit of your answer under a:?

Finally we can use the algebra of vectors to write down a frame-free relationship between the proper velocity $\mathbf{u}_O$ of an observer and the energy-momentum vector $\mathbf{p}$ of a moving particle.

Problem 69: Frame-free Energy

The energy of a particle, according to an observer, is c times its time-component.

- a: Why will observers generally disagree about the energy they assign to a particle?
- b: Why is the energy of the particle with proper momentum $\mathbf{p}$ according to Bob equal to $E_B = c\mathbf{e}_0 \cdot \mathbf{p}$?
- c: Alice has a proper velocity $\mathbf{u}_{Alice}$ why is the energy according to her given by $E_{Alice} = \mathbf{u}_{Alice} \cdot \mathbf{p}$?

An observer with a proper velocity $\mathbf{u}$ will measure an energy given by

$$E = \mathbf{u} \cdot \mathbf{p} . \quad (4.1.6)$$

This is a nice equation not referring to any specific frame, but only to the properties of a particle, $\mathbf{p}$, and of an observer $\mathbf{u}$.

Problem 70: Energy in the Rest frame

Show that for a particle with a proper velocity $\mathbf{u}$ and a mass m its energy in its own rest-frame is always $E = mc^2$.

4.2. Waves

Waves have two properties that are related to the way they propagate through spacetime. First of all we have the frequency, usually denoted by f . It signifies the *number of cycles per second* the wave makes when passing through some point in space. Said differently, $\frac{c}{f}$ is the distance in the time-direction between two wavecrests. We will usually use the so-called *angular frequency* defined as

$$\omega \equiv 2\pi f . \quad (4.2.1)$$

It has the SI-unit *radians per second*. Next to this a wave is characterised by the distance *in space* between two maxima or minima, the so-called wavelength which is usually denoted by λ . Note that ω is measured per second whereas λ is in meters. We could ofcourse also count k , the *number of cycles per meter* in space instead of the length of one cycle. we define a spacetime vector that contains these properties for waves,

$$\mathbf{k} \equiv \frac{\omega}{c}\mathbf{e}_0 + k_1\mathbf{e}_1 + k_2\mathbf{e}_2 + k_3\mathbf{e}_3 . \quad (4.2.2)$$

Using it we can also define a *relative* vector, called *the wave vector*

$$\vec{k} = \mathbf{k} \wedge \mathbf{e}_0 , \quad (4.2.3)$$

as the vector pointing in the direction a wave propagates and satisfying

$$|\vec{k}| = \frac{2\pi}{\lambda} . \quad (4.2.4)$$

This vector measures the number of cycles in radians per meter! But the relative vectors are, as always, observer dependent.

Now waves usually display a relationship between the frequency ω and the wavevector $\vec{k}$, this is called a *dispersion relation*. In secondary school you will have encountered the relation between frequency f , wavelenght λ and speed of propagation c of a lightwave,

$$\begin{aligned} \frac{f}{c} = \frac{1}{\lambda} & \rightarrow \frac{\omega}{c} = |\vec{k}| , \text{ so } , \\ \frac{\omega^2}{c^2} - \vec{k}^2 & = 0 \end{aligned}$$

In terms of spacetime vectors this simply reduces to

$$\mathbf{k} \cdot \mathbf{k} = 0 . \quad (4.2.5)$$

Different inertial observers will disagree on what the exact frequency and wave number of the wave are, depending on their choice of frame.

Problem 71: Frame-free frequency

Consider a general wave described by a wave-vector $\mathbf{k}$. Why will an observer that travels with a proper velocity $\mathbf{u}$ measure a frequency ω given by $\omega = \mathbf{u} \cdot \mathbf{k}$.

4.2.1. Lightwaves: the Doppler Effect

Assume that Alice and Bob are both observing a lightwave described by a wave-vector $\mathbf{k}$. Assume that Bob assigns a proper velocity $\mathbf{u}_A$ to Alice and he is using a standard orthonormal frame $\mathbf{e}_\nu$. Alice and Bob assign to the lightwave a frequency $\omega_{A/B}$ given by

$$\omega_{A/B} = \frac{1}{c} \mathbf{u}_{A/B} \cdot \mathbf{k} . \quad (4.2.6)$$

Similarly the *direction of propagation* that the observers assign to the lightwave can be expressed by means of the bivector $\hat{k}_{A/B}$

$$\hat{k}_{A/B} = \frac{\mathbf{k} \wedge \mathbf{u}_{A/B}}{\mathbf{k} \cdot \mathbf{u}_{A/B}} . \quad (4.2.7)$$

This construction is entirely analogous to the one we used to discuss the velocity addition theorem. So we could once again ask ourselves a question like: What are the components that Alice assigns to a lightwave-vector after it has been expressed in terms of the components that Bob assigns to it.

So, let Bob choose a basis in which $\mathbf{u}_B = c\mathbf{e}_0$, and Alice's proper velocity and the wavevector of the lightwave take the following form

$$\begin{aligned}\mathbf{u}_A &= \gamma_u \{c\mathbf{e}_0 + u\mathbf{e}_1\} , \\ \mathbf{k} &= \frac{\omega_B}{c} \mathbf{e}_0 + k_B \cos(\theta) \mathbf{e}_1 + k_B \sin(\theta) \mathbf{e}_2 .\end{aligned}$$

Bob has chosen the $\mathbf{e}_1$ -direction to be in the direction in which Alice moves, and he has chosen the $\mathbf{e}_2$ -direction in such a way that the lightwave moves in the 1 – 2plane. This represents no assumption on any physics involved. If we use these two equations and plug them into the equation for $\vec{k}_A$ we find $\vec{k}_A$ in terms of the components that Bob assigns to $\mathbf{u}$ and $\mathbf{k}$. If we compute the dot-product first we get

$$\mathbf{k} \cdot \mathbf{u}_A = \gamma_u \{\omega_B - uk_B \cos(\theta)\} .$$

This result is called the *Doppler effect*. The frequency that Alice observes is shifted with respect to the one Bob observes. If we use that for this light wave we have $k_B = \frac{\omega_B}{c}$ we get

$$\omega_A = \gamma_u \omega_B \left\{ 1 - \frac{u \cos(\theta)}{c} \right\} . \quad (4.2.8)$$

In case the lightwave and Alice move in the same direction with respect to Bob, i.e. $\theta = 0$ then we find the result that is usually quoted in introductory texts

$$\omega_A = \omega_B \sqrt{\frac{1 - \frac{u}{c}}{1 + \frac{u}{c}}} .$$

This special case of the Doppler effect sometimes goes under the name *longitudinal* Doppler effect. It is used on a day to day basis in the so-called Doppler radars that are used to monitor air-traffic and to enforce speed limits.

Problem 72: Mars shuttles

In 2020 the United Nations operate a Shuttle service between Earth and Mars. Every shuttle has front- and backlights that emit light at the same wavelength λ . The cruise speed v of the shuttles is such that, as seen from the Earth, outgoing shuttle have red backlights (600 nm) and incoming shuttles have green front lights (500 nm). Determine the wavelength λ and the cruise speed v .

But even in the case that $\theta = \frac{\pi}{2}$ there is a Doppler-effect as we find $\omega_A = \gamma_u \omega_B$. This effect traditionally goes under the name *transversal* Doppler effect. It is purely relativistic and can be explained from the fact that there is a time-dilation between the two observers. The longitudinal effects were known before special relativity. If we make a Taylor-expansion of the relativistic formulae for small velocities, i.e. $\frac{u}{c} \ll 1$, the non-relativistic results receive corrections that are

quadratic $\frac{u}{c}$. At the time of the advent of special relativity these "second-order" $\frac{u^2}{c^2}$ -effects were not measurable due to limitations in experimental accuracy.

For the $\vec{k}_A$ we finally find from a bit of algebra

$$\begin{aligned}\vec{k}_A &= \frac{[k_B \cos(\theta) - \frac{\omega_B u}{c}] \mathbf{e}_1 \mathbf{e}_0 + k_B \sin(\theta) \mathbf{e}_2 \mathbf{e}_0 - k_B \sin(\theta) \frac{u}{c} \mathbf{e}_1 \mathbf{e}_2}{\frac{\omega_B}{c} - \frac{u}{c} k_B \cos(\theta)} , \\ &= \frac{[\cos(\theta) - \frac{u}{c}] \mathbf{e}_1 \mathbf{e}_0 + \sin(\theta) \mathbf{e}_2 \mathbf{e}_0 - \sin(\theta) \frac{u}{c} \mathbf{e}_1 \mathbf{e}_2}{1 - \frac{u}{c} \cos(\theta)} ,\end{aligned}$$

where we use $k_B = \frac{\omega_B}{c}$ in the last step.

Problem 73: Check it

Don't just take the above equation on faith; Check it.

Now let's calculate the components of this *same* bivector in the basis of Alice. We could take dot-products with the basis-bivectors of Alice, but again simply reading them off is easier. The basis Alice uses, expressed in terms of Bob's basis vectors, is again

$$\begin{aligned}\mathbf{e}'_0 &= \frac{\mathbf{u}}{c} = \gamma_u \{c \mathbf{e}_0 + u \mathbf{e}_1\} , \\ \mathbf{e}'_1 &= \gamma_u \{ \mathbf{e}_1 + \frac{u}{c} \mathbf{e}_0 \} , \\ \mathbf{e}'_2 &= \mathbf{e}_2 .\end{aligned}$$

So if Alice expands the bivector $\vec{k}_A$ in terms of her basisvector, the wavenumber k_A and frequency ω_A and angle θ_A she observes this yields

$$\vec{k}_A = c \frac{k_A \cos(\theta_A) \mathbf{e}'_1 \wedge \mathbf{e}'_0 + k_A \sin(\theta_A) \mathbf{e}'_2 \wedge \mathbf{e}'_0}{\omega_A} .$$

Ofcourse for Alice we also have $k_A = \frac{\omega_A}{c}$, using this and the expression for the basis vectors yields

$$\vec{k}_A = \cos(\theta_A) \mathbf{e}_1 \mathbf{e}_0 + \sin(\theta_A) \gamma_u \mathbf{e}_2 \mathbf{e}_0 - \sin(\theta_A) \gamma_u \frac{u}{c} \mathbf{e}_1 \mathbf{e}_2 .$$

Comparing the two expression for $\vec{k}_A$ we can read off

$$\cos(\theta_A) = \frac{\cos(\theta) - \frac{u}{c}}{1 - \frac{u}{c} \cos(\theta)} . \quad (4.2.9)$$

This formula provides a relationship between the angle θ_A observed by Alice and the angle θ_B observed by Bob. This effect is called the relativistic *abberation* of lightwaves.

The Headlights effect

Consider the following very specific application of the above formulae. Suppose a particle (Alice) moves with high-speed along the $\mathbf{e}_1$ direction and emits a lightwave

that is along the $\mathbf{e}_2$ -direction in the particle's restframe. In the above equation this would mean that $\theta_A = \frac{\pi}{2}$, and thus that we must have

$$0 = \cos(\theta) - \frac{u}{c} .$$

So the angle that Bob observes is $\theta < \frac{\pi}{2}$ and to him it seems that the particle has emitted the lightwave partially in the direction of motion. In fact if $\frac{u}{c} \approx 1$ then $\theta \approx 0$. So the faster the particle moves, the more it seems to Bob that the particle emits its light in the forward direction of motion. This effect is called the *headlights effect* and is important in understanding the properties of *synchrotron radiation*. In many astrophysical systems, such as for example neutron stars,

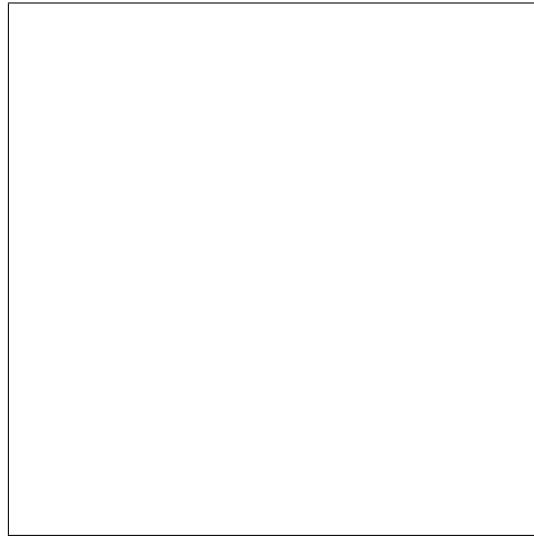


fig. 17: Synchrotron radiation

highly energetic electrons travel through strong magnetic fields. As a result their trajectories are strongly curved, they are gyrating around the magnetic fieldlines with a typical frequency determined by the strength of the magnetic field. Due to these accelerations they emit radiation and because of the headlights effect this radiation is emitted in narrow bundles in the direction of motion. As a result these electrons appear as a large number of small "lighthouses" whose bundles sweep through spacetime with a particular frequency. When observed from the Earth, this frequency can be measured and thus information can be obtained about the strength of the local magnetic fields.

4.2.2. Quantum waves and Massless particles

In quantum physics particles are also described by waves. The frequencies and wavelengths of those waves are associated with energies and momenta of the particles. It was Louis de Broglie, a physicist well acquainted with special relativity, who suggested the following relations

$$\begin{aligned} E &= \hbar\omega , \\ p &= \hbar k , \end{aligned}$$

where $\hbar$ is Planck's constant divided by 2π . Now we can give a much nicer equation as we have recognised them as components of spacetime vectors,

$$\mathbf{p} = \hbar \mathbf{k} . \quad (4.2.10)$$

If we consider photons, i.e. the elementary excitations of the electromagnetic field, or "light particles", we know that they satisfy $\mathbf{k}^2 = 0$. As a result we have for photons

$$m^2 c^2 = \mathbf{p}^2 = 0 , \quad (4.2.11)$$

i.e. photons have no restmass. Because of this we call them *massless* particles.

Problem 74: Energy-momentum relation if $m = 0$

Consider a massless particle. Show that the relationship between its energy and its momentum is $E = \pm c \sqrt{p_1^2 + p_2^2 + p_3^2}$.

If we denote the magnitude of spacelike momentum by $p = \sqrt{p_1^2 + p_2^2 + p_3^2}$, then for photons and other massless particles we have a simple relationship between energy and momentum

$$E = cp . \quad (4.2.12)$$

Problem 75: Momentum and speed

Consider a particle moving with a relativistic momentum of magnitude p and with energy E . Check that for massless as well as for massive particle the speed v at which the particle moves satisfies

$$\frac{v}{c} = \frac{pc}{E} .$$

If we compute the speed of these massless particle we find that $v = c$, i.e. for massless particles we cannot calculate a γ -factor as they always move at the speed of light.

Problem 76: Almost massless particles

Neutrinos are generally considered to be massless or almost massless. If it is given that $\sqrt{1+x^2} \approx 1 + \frac{x^2}{2}$ for small x ; give an approximate expression for the energy E of an almost massless particle in terms of its momentum p and its mass m .

For tachyons the situations becomes rather disturbing. If we would attempt to associate a momentum and energy with them they would require imaginary masses!

Problem 77: Tachyons revisited

Consider a massive particle with a spacelike momentum.

a: Draw such a situation in a frame.

b: Give a short calculation showing that for such "tachyons" $m^2 < 0$.

4.3. Relativistic Interactions

The energy and momentum of a particle of mass m are both components of a single spacetime vector

$$\mathbf{p} = m\mathbf{u} . \quad (4.3.1)$$

Now, if $\mathbf{p}_j$ represents the energy-momentum of the j -th particle entering into a collision, and $\mathbf{q}_k$ is the energy-momentum with which the k -th particle exits the collision then we must have

$$\sum_j^N \mathbf{p}_j = \sum_k^{N'} \mathbf{q}_k , \quad (4.3.2)$$

because energy and momentum are both conserved. Yet in contrast to non-relativistic physics the sums do not need to go over the same number of particles as, due to $E = mc^2$ energy can be used to create new particles. To see this we

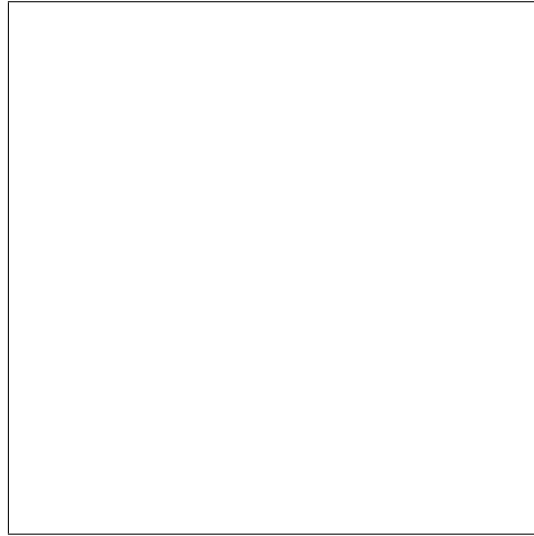


fig. 18: Momentum in- and out-states

introduce the *invariant mass* M_{in} of the incoming particles as

$$M_{in}^2 = \left(\sum_j^N \mathbf{p}_j \right)^2 . \quad (4.3.3)$$

Problem 78: Working with invariant Mass

- a: Why is M_{in} a Lorentz invariant?
- b: Suppose we have two particles of restmass m coming in with momenta $\mathbf{p}_1$ and $\mathbf{p}_2$. Express M_{in} in terms of m , their speeds v_1 and v_2 and the angle between their velocities.
- c: Compare M_{in} in the following two situations: 1) the two particles hit each other head-on with speeds $v_1 = v_2 = 0.375c$, and 2) one of the particles is the "target" at rest and the other particle is the "projectile" that hits the first with a speed $v_2 = 0.75c$.
- d: If you should design an experiment in which as much as energy in collisions should be available for particle creation, which of the two types of collisions from c: would you choose?

Energy-Momentum conservation yields a set of four equations. If we consider all input as *known* we have $4N'$ unknowns as output. So energy-momentum conservation alone is not sufficient to completely describe the collision. Even if we know the rest-masses of all exiting particles, so we know their $\mathbf{q}_k^2 = m_k^2 c^2$, leaving us with only $3N'$ this is still not enough if $N' > 1$. The additional equations we need come from the way the interactions between the particles work. So, as long as we do not know how to describe these we will be forced to look at special cases where we simply prescribe more information about the outgoing particles.

4.3.1. Compton scattering

Consider a collision between a photon and an electron. Let the wave properties of the photon, before and after the collision, be described by the vectors $\mathbf{k}_{in}$ and $\mathbf{k}_{out}$. Energy-momentum conservation then gives us

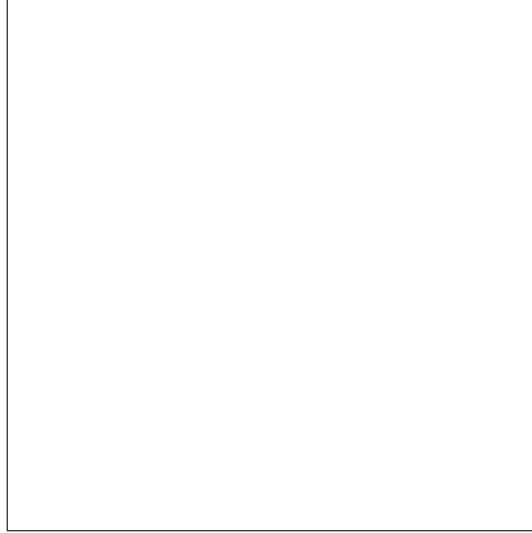


fig. 19: Compton scattering

$$\hbar \mathbf{k}_{in} + \mathbf{p}_e = \hbar \mathbf{k}_{out} + \mathbf{q}_e . \quad (4.3.4)$$

Now suppose the electron is initially at rest, i.e. we describe the collision from the rest-frame of the incoming electron, $\mathbf{p}_e = m_e c \mathbf{e}_0$. Now let the photon come in along the $\mathbf{e}_3$ axis and scatter into the $\mathbf{e}_2 \mathbf{e}_3$ plane. Then we can write for the photon wave vectors

$$\hbar \mathbf{k}_{in} = \hbar k_{in} \{ \mathbf{e}_0 + \mathbf{e}_3 \} , \quad (4.3.5)$$

$$\hbar \mathbf{k}_{out} = \hbar k_{out} \{ \mathbf{e}_0 + \sin(\theta) \mathbf{e}_2 + \cos(\theta) \mathbf{e}_3 \} . \quad (4.3.6)$$

The angle θ is called the *scattering angle*. From energy momentum conservation we now know that

$$\begin{aligned} \mathbf{q}_e &= \mathbf{p}_e + \hbar \mathbf{k}_{in} - \hbar \mathbf{k}_{out} , \\ &= \{ m_e c + \hbar k_{in} - \hbar k_{out} \} \mathbf{e}_0 - \hbar k_{out} \sin(\theta) \mathbf{e}_2 \\ &\quad + \{ \hbar k_{in} - \hbar k_{out} \cos(\theta) \} \mathbf{e}_3 . \end{aligned}$$

As we now that the outcoming particle is an electron, we should enforce this by requiring that

$$\mathbf{q}_e^2 = m_e^2 c^2 . \quad (4.3.7)$$

This leads to an equation for k_{out} in terms of k_{in} , m_e and the angle θ . It reads

$$k_{out} = \frac{k_{in}}{1 + \frac{\hbar}{m_e c} (1 - \cos(\theta)) k_{in}} . \quad (4.3.8)$$

This equation now relates the incoming wavelength of the photon, as $\lambda_{in} = \frac{2\pi}{k_{in}}$ to the wavelength of the outgoing photon that was scattered by an angle θ .

This equation has been tested on many occasions and its success provides an argument in favour of treating photons as *particles* with a momentum $\hbar \mathbf{k}$. It was instrumental in clarifying the wave-particle duality of photons. The length

$$\lambda_c \equiv \frac{\hbar}{m_e c} , \quad (4.3.9)$$

is known as the *Compton wavelength* and serves as a measure for the scales at which quantum mechanics becomes important when applied to electrons. We have derived this equation in the frame in which the initial electron is at rest. If the initial electron were moving than all we would need to do is to transform the expression for the outgoing photon to the required frame.

Usually this process is considered as transferring energy from the incoming photon to the outgoing electron. However, sometimes the reverse happens and a lower energy photon is hit by a high-energy charged particle. This is then called *inverse Compton scattering* and this happens for example when cosmic-ray protons, which have very high energies, collide against the photons from the 3-Kelvin background radiation. The result is that the outgoing photons are usually in the gamma radiation range.

Problem 79: 3K photons and Cosmic Rays

a: Find (on the web) the typical energy of a cosmic radiation proton.

According to Wien's displacement law the most likely wavelength for a photon from a system at temperature T is given by $\lambda = \frac{hc}{4.965kT}$.

b: What is the minimal k_{out} for this photon if it is struck by the proton?

4.3.2. Pair Annihilation

In the process of pair-annihilation an electron and a positron annihilate each other producing photons. One can wonder what the minimum number of photons is that comes out of such a process. In a problem you will prove yourself that the annihilation must produce more than one photon. Here we will consider the case that two photons energy from the reaction.

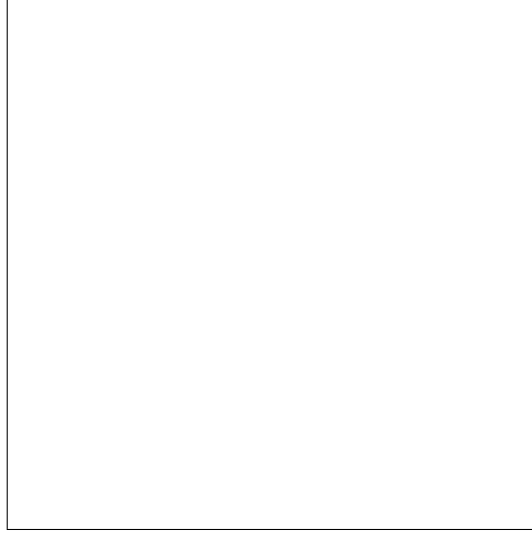


fig. 20: Pair annihilation

Problem 80: Photons from Annihilation

Consider a particle of mass m meeting its anti-particle partner with the same mass. Consider this annihilation from a frame in which the two incoming particles have equal but opposite velocities v along the $\mathbf{e}_1$ -direction.

- a: What is the total energy available for the creation of photons.
- b: What is the total incoming momentum vector.
- c: Why can't there be only a single photon in the out-state? (Hint: check $\mathbf{k}^2 = 0$).

So we have

$$\mathbf{p}_{e-} + \mathbf{p}_{e+} = \hbar \mathbf{k}_1 + \hbar \mathbf{k}_2 , \quad (4.3.10)$$

as the energy-momentum conservation law. We will, again, analyse the process from the rest-frame of the incoming electron. We will assume that the positron comes in along the $\mathbf{e}_3$ -direction, so we have

$$\begin{aligned} \mathbf{p}_{e-} &= mc\mathbf{e}_0 , \\ \mathbf{p}_{e+} &= \frac{E}{c}\mathbf{e}_0 + p\mathbf{e}_3 . \end{aligned}$$

Now of course we do not know the momenta or wavelengths of the out-going photons. But we will restrict photon 1 to move in the $\mathbf{e}_2\mathbf{e}_3$ -plane, so

$$\mathbf{k}_1 = k_1 \{ \mathbf{e}_0 + \sin(\theta)\mathbf{e}_2 + \cos(\theta)\mathbf{e}_3 \} .$$

Now the momentum vector of photon 2 is fixed by the requirement of energy-momentum conservation. We have

$$\begin{aligned} \hbar \mathbf{k}_2 &= \mathbf{p}_{e-} + \mathbf{p}_{e+} - \hbar \mathbf{k}_1 , \\ &= \left(mc + \frac{E}{c} - \hbar k_1 \right) \mathbf{e}_0 - \hbar k_1 \sin(\theta) \mathbf{e}_2 \\ &\quad + (p - \hbar k_1 \cos(\theta)) \mathbf{e}_3 . \end{aligned}$$

From the time-component we can immediately read off what k_2 will be

$$k_2 = mc + \frac{E}{c} - \hbar k_1 , \quad (4.3.11)$$

and so all we need to do is fix k_1 . That is possible due to the fact that we know the photons to be massless. For $\mathbf{k}_1$ this was fixed by hand by defining it as a light like vector. But for $\mathbf{k}_2$ we still need to enforce this. So if we set $\mathbf{k}_2^2 = 0$ we obtain an equation that allows us to express k_1 as a function of the electron-positron mass m and the positron momentum p . It may seem to you that E is also a variable. But because for the positron we also have that $\mathbf{p}_{e+}^2 = m^2 c^2$ the positron energy is in fact

$$E = \sqrt{m^2 c^4 + p^2 c^2} . \quad (4.3.12)$$

We find for the photon k_1

$$k_1 = \frac{m(mc^2 + E)}{\hbar(mc + \frac{E}{c} - p \cos(\theta))} . \quad (4.3.13)$$

It is interesting to check for yourself whether the limits, like for example $p \rightarrow 0$ of this result make sense.

Problem 81: Annihilation at right-angles

Consider the pair annihilation of a proton-antiproton pair.

- a: Draw the process for which $\theta = \frac{\pi}{2}$.
- b: Compute k_1 if it is known that the incoming anti-proton has an energy which is $(n-1)$ -times its rest-energy.
- c: In an experiment the momenta of incoming anti-particles with an unknown mass are not known. Can we still effectively determine the mass of the particles involved from a measurement of photon wave numbers?.

Sometimes these sorts of scattering calculations become so complicated that one resorts to computer power to do the algebra. In that case, checking such limits is a vital tool for assessing whether the computer computed a correct result.

Problem 82: Ultra-relativistic Annihilation

Suppose that the energy of the incoming anti-particle by far exceeds the rest-energy, i.e. $E \gg mc^2$.

- a: Why can we consider the particle to be approximately massless?
- b: Give approximate expressions for k_1 and k_2 .
- c: Sketch the graph of k_1 as a function of θ in this case.

4.4. Currents and Densities

Consider a continuous charge distribution in spacetime. Suppose that according to Bob the charge-density, $\rho(\mathbf{x})$, is stationary. This implies that in Bob's frame the *flow* of charge through spacetime is given by a vector-field

$$\mathbf{j}(\mathbf{x}) = \rho(\mathbf{x})\mathbf{e}_0 . \quad (4.4.1)$$

The charge-density is the time-component of a spacetime vector. For Alice we can find the corresponding components by taking dot-products with her basis vectors. She, for example, measures a charge-density

$$\rho(\mathbf{x})' = \gamma_u \rho(\mathbf{x}) ,$$

whereas she measures the charge to flow producing a current in the $\mathbf{e}'_1$ -direction given by

$$j^1(\mathbf{x})' = -\gamma_u \frac{u}{c} \rho(\mathbf{x}) .$$

It shouldn't be unexpected that she sees the charges moving in the opposite direction, but the fact that she see the density increased by a factor γ_u may be a surprise at first. However it is consistent with the length-contraction which reduces all volumes by $\frac{1}{\gamma_u}$. As a result the total amount of charge, i.e. density times volume, remains invariant.

4.4.1. Relative current-densities

In general we have the following situation. A current of charge or mass, or in fact anything else, is described by a spacetime vectorfield $\mathbf{j}(\mathbf{x})$. With respect to an inertial observer moving with a proper velocity $\mathbf{u}$ this vectorfield can be split up into a density

$$\rho(\mathbf{x}) = \mathbf{J}(\mathbf{x}) \cdot \mathbf{u} , \quad (4.4.2)$$

and a relative current-density

$$\vec{j}(\mathbf{x}) = \frac{\mathbf{J}(\mathbf{x}) \wedge \mathbf{u}}{c} . \quad (4.4.3)$$

We only divide here by c , if we would divide by $\mathbf{J}(\mathbf{x}) \cdot \mathbf{u}$, i.e. by $\rho(\mathbf{x})$ then we also get a meaning full quantity, but only if the spacetime current-density $\mathbf{J}(\mathbf{x})$ is timelike, i.e. $\mathbf{J}(\mathbf{x})^2 > 0$. For in that case the quantity

$$\vec{v}_{flow}(\mathbf{x}) = \frac{\mathbf{J}(\mathbf{x}) \wedge \mathbf{u}}{\mathbf{J}(\mathbf{x}) \cdot \mathbf{u}} ,$$

can be interpreted as the proper velocity of the charge-density $\rho(\mathbf{x})$ at $\mathbf{x}$. But we can have systems in which there is a current-density but no charge-density (like in most common examples of currents through conductors). In such a case the spacetime current-density is spacelike.

Problem 83: Classification of Currents

- a: Show that for a time-like spacetime current-density $\mathbf{J}(\mathbf{x})$ we can always find a frame at $\mathbf{x}$ in which it takes the form $\rho(\mathbf{x})\mathbf{e}_0$.
- b: Show that for a spacelike current-density we can always find a frame at $\mathbf{x}$ for which $\rho(\mathbf{x}) = 0$.
- c: Argue that for a lightlike current-density we can never reach any of these two situations.

The geometric classification of spacetime vectors has physical meaning as well for spacetime current-densities. Yet we should be aware that this meaning refers to *local* frames, i.e. observer frames that were defined *at* $\mathbf{x}$. Such frames may change from one event in spacetime to another. In all the tools that we have created so far, we have always dealt with global frames. This is a shortcoming that we will correct in the very first section of the next chapter. The discussion of local frames is, in part, a discussion of *coordinate systems* on spacetime.

4.5. Problems

In this section you will find a number of problems to be used for revising the material.

Problem 84: Time-dilation and Doppler effect

Explain in words what the connection is between time-dilation and the transversal Doppler effect.

Problem 85: Annihilation

Consider an electron and an anti-electron annihilating each other to form one or more photons. Both particles have mass m . Show that it is not possible that only one single photon is formed.

Problem 86: The zero-momentum frame

Consider two particles with restmasses m_1 and m_2 and proper velocities $\mathbf{u}_1$ and $\mathbf{u}_2$ along the $\mathbf{e}_1$ -direction of Bob.

- Give their *total* energy-momentum $\mathbf{P}$ and express it in terms of Bob's frame.
- Give an expression for $\mathbf{P}^2$ and show that $M^2 \equiv \mathbf{P}^2 > 0$.
- Find a Lorentz transformation such that for the transformed component $P^{1'}$ we have $P^{1'} = 0$.
- Let $\mathbf{e}'_0$ be the transformed time-like unit-vector and show that we have $\mathbf{P}' = M\mathbf{e}'_0$. Compute $\mathbf{e}'_1$.

This frame you've just constructed is called the "zero-momentum" or "centre-of-momentum" frame. It is very useful in studying the properties of relativistic collisions between particles. Often you do not even need to construct it, knowing that you can view things from this frame is basically all you need.

Problem 87: Using active transformations

Let the Lorentz transformation between Alice and Bob be given by $R = \exp(\alpha K_1)$.

- a: Use that $\mathbf{e}'_0 = R\mathbf{e}_0\tilde{R}$ to rederive the relationship between α and u .
- b: Now Bob observes a vector $\mathbf{b}$. Why can you read off the components that Alice measures by expanding $\tilde{R}\mathbf{b}R$ in Bob's basis?
- c: Consider the bivector F given by

$$F = E_1K_1 + E_2K_2 + E_3K_3 + B_1J_1 + B_2J_2 + B_3J_3 .$$

where the numbers E_j and B_k are the results of Bob's measurements. Compute $\tilde{R}FR$ and determine the components that Alice will measure.

Problem 88: Charging up through motion

- a: Use internet resources to find the typical electron-density in a typical piece of conducting material.
- b: How large with the density of positive charge be in the wire?
- c: Find the typical speed at which conduction electrons travel through the wire when a current has set in?
- d: The moving electron-density has now changed. Find the total charge density in the wire as the current flows.

Chapter 5

Forces in Spacetime

In this chapter we will study how to describe forces in Spacetime. In the process we will need to develop some skills and methods in spacetime calculus. We start with that, before we turn to the description of forces in spacetime, which we then specialise to the generic example: electromagnetic forces.

5.1. Charts and local frames

Observers like to draw charts or maps of spacetime, just like geometers like drawing charts of the Earth's surface. A key feature of such charts is that they assign co-ordinates to every point on the Earth.

5.1.1. Charts and parametrisations

Consider an event and its direct neighborhood in spacetime $\mathcal{N}$. Let us denote an arbitrary event in $\mathcal{N}$ by $\mathbf{x}$. A *chart* on this neighbourhood is a map that assigns four real numbers $z^\mu(\mathbf{x})$ to each event in $\mathcal{N}$ such that each event can be uniquely identified by these four number $z^\mu(\mathbf{x})$. These numbers are usually called *coordinates*.

Parametrisations

A related concept, in a sense the *inverse* concept of a chart, is that of a *parametrization*. A parametrisation is a function $\mathbf{x}(z)$ that assigns a unique event to every different array of four real numbers z^μ . Again we will call these four numbers z^μ the coordinates of the event $\mathbf{x}(z)$, though this may be an slight abuse of terminology. When I casually use the term *coordinates* I typically can mean either of the above. Throughout this course we will assume that it is, atleast in principle, possible to cover all of spacetime with a single set of coordinates. Such spacetimes are called *flat*. In the chapters on dislocation theory and gravity we will see spaces and spacetimes where this is not possible.

Local basis vectors

At every spacetime event the variation of a coordinate induces a notion of a space- and/or time-direction, i.e. if we calculate the rate-of-change of $\mathbf{x}(z)$ under variations one of the z^μ we obtain the vector

$$\mathbf{g}_\mu(\mathbf{x}(z)) = \frac{\partial}{\partial z^\mu} \mathbf{x}(z) . \quad (5.1.1)$$

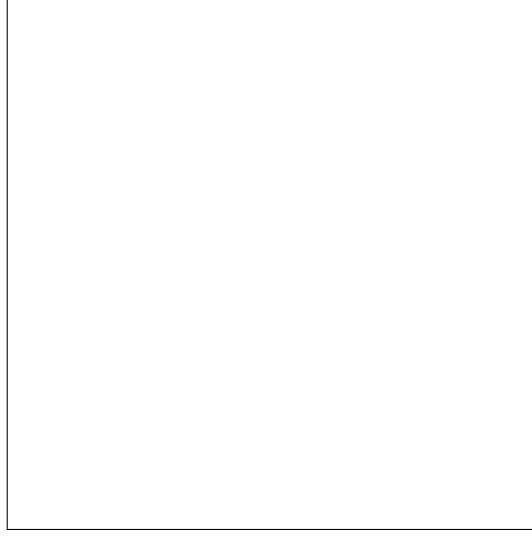


fig. 21: A map from Spacetime to R^4

We will call this set of *local* basis-vectors $\mathbf{g}_\lambda(\mathbf{x})$, at every event $\mathbf{x}(z)$ the *induced* basis by the coordinatesystem.

The induced basis vectors are extremely useful. Suppose we have a particle following a worldline $\mathbf{x}(z(\tau))$ where $z(\tau) = \{z^0(\tau), z^1(\tau), z^2(\tau), z^3(\tau)\}$. The tangent-vector to the worldline, i.e. the proper-velocity $\mathbf{u}(\tau)$, can be expressed as

$$\begin{aligned} \mathbf{u}(\tau) &\equiv \frac{d}{d\tau} \mathbf{x}(\tau) , \\ &= \sum_{\lambda=0}^3 \left\{ \frac{d}{d\tau} z^\lambda(\tau) \right\} \frac{\partial}{\partial z^\lambda} \mathbf{x}(z) , \\ &= \sum_{\lambda=0}^3 \left\{ \frac{d}{d\tau} z^\lambda(\tau) \right\} \mathbf{g}_\lambda(z) . \end{aligned}$$

The nice thing is that all we used was the chainrule, so this form of the tangent vector holds for any coordinate system.

Let us take a look at some of the most useful coordinate systems.

5.1.2. Examples

We start with the simplest coordinate system. Let $\mathbf{e}_\mu$ be an orthonormal basis of some observer, and $\mathbf{x}$ an arbitrary event in spacetime. The component

$$x^\mu = \mathbf{e}^\mu \cdot \mathbf{x} , \tag{5.1.2}$$

defines a set of coordinates on spacetime. Spacetime is parametrised by the array of components $x = \{x^0, x^1, x^2, x^3\}$, as $\mathbf{x}(x)$,

$$\mathbf{x}(x) = \sum_{\lambda=0}^3 x^\lambda \mathbf{e}_\lambda . \tag{5.1.3}$$

Obviously this coordinate system satisfies

$$\mathbf{g}_\lambda \equiv \frac{\partial}{\partial x^\lambda} \mathbf{x}(x) = \mathbf{e}_\lambda . \quad (5.1.4)$$

These coordinates are called Cartesian spacetime coordinates.

Cylindrical coordinates

Let the $\mathbf{e}_\mu$ be the standard cartesian basis-vectors. Now suppose that we parametrize spacetime as follows; any event $\mathbf{x}$ is written as

$$\mathbf{x}(r, \theta, z, ct) = r \{ \cos(\theta) \mathbf{e}_1 + \sin(\theta) \mathbf{e}_2 \} + z \mathbf{e}_3 + ct \mathbf{e}_0 . \quad (5.1.5)$$

The parameter r is the *radial distance* of the event $\mathbf{x}$ to the z -axis. The parameter θ is the angle that the shortest line connecting the z -axis with the event $\mathbf{x}$ makes with the $\mathbf{e}_1$ -direction. Ofcourse ct is just a standard time-parameter and z is the parameter on the z -axis. This parametrization is called *cylindrical* and the coordinates $\{ct, z, \theta, r\}$ that we can assign to every event are called *cylindrical coordinates*. These coordinates are very usefull when discribing systems that are symmetric under rotations around some given axis. We then identify the z -axis with that axis of symmetry.

Given this parametrisation we see that it induces a set of basis-vectors at *every event in spacetime*,

$$\mathbf{g}_t \equiv \frac{\partial}{\partial ct} \mathbf{x}(r, \theta, z, ct) = \mathbf{e}_0 , \quad (5.1.6)$$

$$\mathbf{g}_r \equiv \frac{\partial}{\partial r} \mathbf{x}(r, \theta, z, ct) = \cos(\theta) \mathbf{e}_1 + \sin(\theta) \mathbf{e}_2 , \quad (5.1.7)$$

$$\mathbf{g}_\theta \equiv \frac{\partial}{\partial \theta} \mathbf{x}(r, \theta, z, ct) = r \{ -\sin(\theta) \mathbf{e}_1 + \cos(\theta) \mathbf{e}_2 \} , \quad (5.1.8)$$

$$\mathbf{g}_z \equiv \frac{\partial}{\partial z} \mathbf{x}(r, \theta, z, ct) = \mathbf{e}_3 . \quad (5.1.9)$$

Problem 89: Orthogonal local basis

- a: Show that the induced basis-vectors in the cylindrical coordinate system are orthogonal.
- b: Determine which basis-vectors are normalised.
- c: Determine the induced reciprocal basis in terms of the induced basis.

The induced basis vectors are orthogonal, but not all of them are normalised. We can use these basis-vectors to rewrite the parametrisation as

$$\mathbf{x}(r, \theta, z, ct) = r \mathbf{g}_r + z \mathbf{g}_z + ct \mathbf{g}_t .$$

Notice that there is no $\theta \mathbf{e}_\theta$ term! We can also express the old cartesian coordinates in terms of cylindrical ones by simply reading them off the parametrisation using cartesian basis-vectors, we find for example

$$\begin{aligned} x &= r \cos(\theta) , \\ y &= r \sin(\theta) . \end{aligned}$$

Obviously we also have

$$\begin{aligned} r &= \sqrt{x^2 + y^2} , \\ \theta &= \arctan \frac{y}{x} . \end{aligned}$$

The first is easy to see, the second requires knowing that $\arctan$ is the *inverse* of the $\tan$ -function.

Problem 90: Surface of a Cylinder

Consider a cylindrical coordinate system with coordinates $\{ct, r, \theta, z\}$.

a: Explain why the surfaces of constant r are cylinders.

b: Show that an infinitesimal piece of cylindrical surface at the event specified by $\{ct, r, \theta, z\}$ can be described by the bivector dS given by

$$dS_{\theta z} = d\theta dz \mathbf{g}_\theta(\mathbf{x}) \wedge \mathbf{g}_z(\mathbf{x}) .$$

c: Try to invent an interpretation for the following infinitesimal object

$$d\mathbf{T}_{t\theta z} = c dt \mathbf{g}_t(\mathbf{x}) \wedge dS_{\theta z} .$$

Spherical Coordinates

Starting out from the cartesian basis-vectors, we introduce the following parameters $r = \{ct, r, \theta, \phi\}$ and define

$$\mathbf{x}(r) = ct \mathbf{e}_0 + r \{ \sin(\theta) \cos(\phi) \mathbf{e}_1 + \sin(\theta) \sin(\phi) \mathbf{e}_2 + \cos(\theta) \mathbf{e}_3 \} . \quad (5.1.10)$$

This coordinate system is called *spherical* coordinates. Note that at, for example, $ct = 0$ the parameters ϕ running from 0 to 2π and θ running from 0 to π describe a sphere of radius r . If we combine this for different values of r we find a whole family of spheres filling up space.

Problem 91: Probing spherical coordinates

a: Compute the locally induced basis.

b: Show that this basis is orthogonal.

c: Suppose that a spacetime vector is expanded as $\mathbf{b} = b^0 \mathbf{g}_0(r) + b^r \mathbf{g}_r(r) + b^\theta \mathbf{g}_\theta(r) + b^\phi \mathbf{g}_\phi(r)$. Give an expression for $\mathbf{b} \cdot \mathbf{b}$.

Parabolic coordinates

Starting out from the cartesian basis-vectors, we introduce the following parameters $\xi = \{ct, \xi, \eta, \phi\}$ and define

$$\mathbf{x}(\xi) = ct \mathbf{e}_0 + \frac{1}{\alpha} \sqrt{\xi \eta} \cos(\phi) \mathbf{e}_1 + \frac{1}{\alpha} \sqrt{\xi \eta} \sin(\phi) \mathbf{e}_2 + \frac{1}{2\alpha} (\xi - \eta) \mathbf{e}_3 . \quad (5.1.11)$$

This coordinate system is called *parabolic* coordinates. Note that at, for example, $ct = 0$ the parameter ϕ running from 0 to 2π describes a circle in the $\mathbf{e}_1 \mathbf{e}_2$ -plane with a radius $\frac{1}{\alpha} \sqrt{\xi \eta}$. If we combine this for different values of η for a fixed value

of ξ we find parabolae around the $\mathbf{e}_3$ -axis with their tops and bottoms at $\eta = 0$.

Problem 92: Probing parabolic coordinates

- a: Explain why the parameters ξ and η must be larger or equal to 0.
- b: Compute the locally induced basis.
- c: Show that this basis is orthogonal.

There are many more coordinate systems for space and all can be more or less useful depending on the context of the physical problem you are studying. Some of the lesser wellknown ones are the *oblate spheroidal*, the *prolate spheroidal* and the *parabolic cylindrical* coordinates.

Problem 93: Help!

- a: Use the help-functions of a software programme like Mathematica, or web resources, to find out what the last three coordinate systems are.
- b: Write a parametrization of $\mathbf{x}$ in each of these coordinate systems.

Hyperbolic coordinates in Spacetime

The previous three examples left the time-coordinate cartesian. There is no particular reason to do so, although we must take care when interpreting the physical "side effects" of a particular choice of coordinates. As said before, the choice of coordinate system does not represent any physical approximation or assumption. However, in many ways, the choice of coordinates sometimes appears to be a choice of *observer*. Sometimes different observers find seemingly different results, although the underlying laws of nature are observer-independent. We will see an example here.

Consider the following parameters, $\rho = \{\rho, \alpha, \theta, \phi\}$ in the parametrization

$$\begin{aligned} \mathbf{x}(\rho) = & \rho \cosh(\alpha) \mathbf{e}_0 + \rho \sinh(\alpha) [\sin(\theta) \cos(\phi) \mathbf{e}_1 \\ & + \sin(\theta) \sin(\phi) \mathbf{e}_2 + \cos(\theta) \mathbf{e}_3] . \end{aligned} \quad (5.1.12)$$

At the basis of this parametrisation is a spherical coordinate system where the radial coordinate has been replaced by $\rho \sinh(\alpha)$, from which we can deduce that α can only take positive values. For every combination of α and ρ the coordinates θ and ϕ describe a time-like spherical shell in the future lightcone of the origin. This coordinate system can only be used on a quarter of Spacetime, but there it is perfectly alright.

Problem 94: Hyperbolic past coordinates

- a: Adapt the hyperbolic parametrization from the text to describing the interior of the backward lightcone.
- b: Compute the induced basis and establish its orthogonality.
- c: If vectors are expanded as $\mathbf{b} = b^\rho \mathbf{g}_\rho + b^\alpha \mathbf{g}_\alpha + b^\theta \mathbf{g}_\theta + b^\phi \mathbf{g}_\phi$ then show that we can write the spacetime dot-product in terms of components as

$$\mathbf{a} \cdot \mathbf{b} = \sum_{\mu, \nu} g_{\mu\nu} a^\mu b^\nu .$$

- d: Compute the matrix elements $g_{\mu\nu}$.

The Rindler wedge

Another variant of this coordinate system is wellknown in the relativity literature. For the so-called Rindler-wedge, i.e. the spacelike quarter right of the origin we can use

$$\begin{aligned} \mathbf{x}(\rho) &= \rho \sinh(\alpha) \mathbf{e}_0 \\ &+ \rho \cosh(\alpha) [\sin(\theta) \cos(\phi) \mathbf{e}_1 + \sin(\theta) \sin(\phi) \mathbf{e}_2 + \cos(\theta) \mathbf{e}_3] . \end{aligned} \quad (5.1.13)$$

Now α can take any value and for fixed α and ρ the coordinates θ and ϕ describe spacelike spherical shells. Lines of constant ρ, θ and ϕ describe objects moving under a constant radial acceleration.

Problem 95: Constant velocity in the Rindler wedge?

Consider a particle, whose worldline in Rindler-wedge coordinates is described by $z^\mu(\tau) = \{\rho(0), \alpha_0 \tau, \theta(0), \phi(0)\}$.

- a: Sketch this worldline in a Spacetime diagram.
- b: Calculate the proper-velocity of this particle.
- c: Why can we say that the velocity of the particle *seems* constant in the local coordinates?

The fact that in non-cartesian coordinate systems involving time the time-dependence of positions of physical objects is partly due to the coordinate system may seem terribly confusing. This is probably also one reason why they are hardly discussed anywhere in introductory texts. In fact, generically we seem to have trouble with picturing what time-dependent coordinates really mean. Nevertheless they can be useful when, for example, we would need to consider geometries that are time-dependent. Suppose for example that you want to study particle physics in the background of some expanding bubble of new phase. As we saw earlier, such a bubble can be modelled as an expanding spherical shell, i.e. a spacelike hyperbola. In the Rindlerwedge these are natural objects!

Problem 96: Surface of a moving sphere

Consider the Rindler wedge coordinate system with coordinates $\{\rho, \alpha, \theta, \phi\}$.

- a: Explain why the surfaces of constant ρ and constant α are spheres with a radius $\rho \cosh(\alpha)$.
- b: Show that, when compared to a standard cartesian coordinate system around the same origin, these spheres seem to expand in the sense that the sphere of fixed ρ around the origin at time $ct = \rho \sinh(\alpha)$ indeed has a radius given by $r = \sqrt{\rho^2 + c^2 t^2}$. (Hint: use the identity satisfied by the hyperbolic functions to show this).
- c: Show that an infinitesimal piece of spherical surface at the event specified by $\{\rho, \alpha, \theta, \phi\}$ can be described by the bivector dS given by

$$dS_{\theta\phi} = d\theta d\phi \mathbf{g}_\theta(\mathbf{x}) \wedge \mathbf{g}_\phi(\mathbf{x}) .$$

- d: Give arguments for the interpretation of the infinitesimal object

$$d\mathbf{T}_{\alpha\theta\phi} = d\alpha \mathbf{g}_\alpha(\mathbf{x}) \wedge dS_{\theta\phi} .$$

as a infinitesimal piece of surface on a *moving spherical shell*.

5.2. Differentiation in Spacetime: ∇ .

In our study of field theories, in particular of the electromagnetic field, we will need a mathematical way of talking about *changes* in these fields when we move from one spacetime event to another. The *vector-derivative* ∇ is just the thing we need.

Consider a cartesian parametrisation of spacetime

$$\mathbf{x}(x) = \sum_{\nu=0}^3 x^\nu \mathbf{e}_\nu .$$

If we have functions depending on spacetime position, for example a scalar function $f(\mathbf{x})$, we can differentiate with respect to any of the components x^μ . The values that the function $f(\mathbf{x})$ takes along the worldline of a moving particle is given by the function $f(\mathbf{x}(\tau))$. If we differentiate this function with respect to τ and apply the chainrule we find

$$\frac{d}{d\tau} f(\mathbf{x}(\tau)) = \sum_\mu \left[\frac{d}{d\tau} x^\mu(\tau) \right] \frac{\partial}{\partial x^\mu} f(\mathbf{x}(\tau)) ,$$

which we would like to interpret as the result of multiplying the proper velocity with a *vector-derivative* ∇ . If we use the reciprocal frame $\mathbf{e}^\nu$ we can write

$$\begin{aligned} \frac{d}{d\tau} f(\mathbf{x}(\tau)) &= \sum_\mu \sum_\nu \left[\frac{d}{d\tau} x^\mu(\tau) \mathbf{e}_\mu \right] \cdot \mathbf{e}^\nu \frac{\partial}{\partial x^\nu} f(\mathbf{x}(\tau)) , \\ &= \mathbf{u}(\tau) \cdot \nabla f(\mathbf{x}(\tau)) . \end{aligned}$$

In this way the partial derivatives arise by taking the component in one particular direction of a vector-derivative ∇ ,

$$\frac{\partial}{\partial x^\mu} f(\mathbf{x}) = \mathbf{e}_\mu \cdot \nabla f(\mathbf{x}) . \quad (5.2.1)$$

The vector derivative is defined as

$$\nabla f(\mathbf{x}) = \sum_\mu \mathbf{e}^\mu \frac{\partial}{\partial x^\mu} f(\mathbf{x}) . \quad (5.2.2)$$

The reciprocal basis vectors appear in the definition of the vector-derivative. Recall that they are defined so that they satisfy

$$\mathbf{e}_\mu \cdot \mathbf{e}^\nu \equiv \delta_\mu^\nu .$$

This definition seems to lean heavily on the use of a cartesian coordinate system and cartesian basis vectors, but that is not the case. Instead of trying out how to compute things for an arbitrary coordinate system we will prove that non-cartesian coordinates work the same way.

Non-Cartesian Coordinates

Suppose we would express the function $f(\mathbf{x}(x))$ in terms of other coordinates $z_\mu(x)$.

The partial derivatives can then be re-expressed as partial derivatives with respect to the z_μ ,

$$\frac{\partial}{\partial x^\lambda} f(z(x)) = \sum_\mu \frac{\partial z^\mu(x)}{\partial x^\lambda} \frac{\partial}{\partial z^\mu} f(z(x)) . \quad (5.2.3)$$

Consequently our vector derivative will become

$$\begin{aligned} \nabla f(\mathbf{x}(x)) &= \sum_{\mu,\lambda} \mathbf{e}^\lambda \frac{\partial z^\mu(x)}{\partial x^\mu} \frac{\partial}{\partial z^\mu} f(z) , \\ &= \sum_{\mu=0}^3 \mathbf{g}^\mu(z) \frac{\partial}{\partial z^\mu} f(z) . \end{aligned}$$

Here we used *reciprocal* basis vectors $\mathbf{g}^\nu$ defined as

$$\begin{aligned} \mathbf{g}^\mu(z) &= \sum_{\lambda=0}^3 \mathbf{e}^\lambda \frac{\partial z^\mu(x)}{\partial x^\lambda} , \\ &= \nabla z^\mu(x) . \end{aligned} \quad (5.2.4)$$

But are they really reciprocal to the $\mathbf{g}_\nu(z)$ we introduced earlier?

Note that from the parametrization $\mathbf{x}(z)$ we had the induced basis

$$\mathbf{g}_\nu(z) = \frac{\partial}{\partial z^\nu} \mathbf{x}(z) .$$

Above we saw that every set of spacetime coordinates also induces a reciprocal basis at every spacetime point

$$\mathbf{g}^\mu(z) = \nabla z^\mu(x) .$$

If these two sets are truly reciprocal at the spacetime event $\mathbf{x}$ we should have

$$\mathbf{g}_\mu(\mathbf{x}) \cdot \mathbf{g}^\nu(\mathbf{x}) = \delta_\mu^\nu . \quad (5.2.5)$$

In a problem you will prove just that.

Problem 97: Yes it is reciprocal

a: Give an expression for $\mathbf{g}_\mu(\mathbf{x}) \cdot \mathbf{g}^\nu(\mathbf{x})$ in terms of partial derivatives.

b: Why do you think can we say that

$$\sum_\nu \frac{\partial x^\mu}{\partial z^\nu} \frac{\partial z^\nu}{\partial x^\sigma} = \frac{\partial x^\mu}{\partial x^\sigma} .$$

c: Why is $\frac{\partial x^\mu}{\partial x^\sigma} = 1$ when $\mu = \sigma$ and 0 otherwise?

d: Complete the proof.

Now all that we need to do is see what happens when we apply the vector-derivative to various functions such as scalar-fields, vector-fields, etc.

5.2.1. Scalarfields and Gradients

expansion of the vector-derivative in terms of the partial derivatives, in a cartesian coordinate system, is

$$\begin{aligned}\nabla f(\mathbf{x}(x)) &= \sum_{\mu} \mathbf{e}^{\mu} \frac{\partial}{\partial x^{\mu}} f(\mathbf{x}) , \\ &= \mathbf{e}_0 \frac{\partial}{\partial x^0} f(\mathbf{x}) - \mathbf{e}_1 \frac{\partial}{\partial x^1} f(\mathbf{x}) - \mathbf{e}_2 \frac{\partial}{\partial x^2} f(\mathbf{x}) - \mathbf{e}_3 \frac{\partial}{\partial x^3} f(\mathbf{x}) .\end{aligned}$$

Note that this derivative has minus signs for the spatial components due to the fact that for spacelike vectors $\mathbf{e}^j = -\mathbf{e}_j$.

Problem 98: Restoring a vector

Consider the function $h(\mathbf{x}) = \frac{1}{2}\mathbf{x}^2$.

- a: Write out this function in terms of components of $\mathbf{x}$.
- b: Use the vector-derivative to show that $\nabla h(\mathbf{x}) = \mathbf{x}$.

Applying the vector-derivative to a scalar function yields a vector-field. This vectorfield is called *the gradient* of $f(\mathbf{x})$. It has many important applications. For example, the induced reciprocal basis of a coordinate system is a set of gradients. The gradient specifies the direction in which the functions shows the largest increase. It is always perpendicular to the spaces of constant $f(\mathbf{x})$. So for the induced reciprocal basis-vectors $\mathbf{g}^{\nu}(z)$ points in the direction in which the coordinate has the fastest increase and it is perpendicular to spaces of constant coordinate value.

Problem 99: ∇ and spherical coordinates

Consider a cartesian coordinate system on spacetime $x = \{ct, x, y, z\}$. Let the radial distance, $r(\mathbf{x})$, of an event be defined by

$$r(\mathbf{x}) = \sqrt{x^2 + y^2 + z^2} .$$

- a: Compute $\nabla r(\mathbf{x})$ and show that it is perpendicular to spheres of constant r .
- b: Introduce spherical coordinates and check that $\nabla r(\mathbf{x})$ is perpendicular to all $\mathbf{g}_{\mu}(\mathbf{x})$ except $\mathbf{g}_r(\mathbf{x})$.
- c: Compute $\nabla \frac{1}{r(\mathbf{x})}$ in spherical coordinates.

The gradient of a function is a *vector*-field. It is the first example of a class of functions we will study next.

Problem 100: Gradient in parabolic coordinates

Consider the parabolic coordinate system given in the previous section.

- a: Compute the induced reciprocal basis-vectors. (Hint: use the fact that the induced basis vectors are orthogonal!)
- b: Suppose we have a scalar function $f(u, v, \phi)$ give the expression for ∇f .

5.2.2. Vectorfields and Sources

In the case of vectorfields the derivative provides information that can be interpreted in terms of the *sources* of the vectorfield. The notion of a source of a field is very important when we are going to discuss how and why electromagnetic fields arise in nature.

So let $\mathbf{b}(\mathbf{x})$ be a general spacetime vectorfield, and let us take cartesian coordinates for simplicity. Then we can compute

$$\begin{aligned}\nabla \mathbf{b}(\mathbf{x}) &= \sum_{\mu, \nu} \mathbf{e}^\mu \frac{\partial}{\partial x^\mu} \mathbf{e}_\nu b^\nu(\mathbf{x}) , \\ &= \sum_{\mu, \nu} [\mathbf{e}^\mu \mathbf{e}_\nu] \left[\frac{\partial}{\partial x^\mu} b^\nu(\mathbf{x}) \right] , \\ &= \left[\sum_\mu \frac{\partial}{\partial x^\mu} b^\mu(\mathbf{x}) \right] + \frac{1}{2} \left[\sum_{\mu, \nu} \left(\frac{\partial}{\partial x^\mu} b^\nu(\mathbf{x}) - \frac{\partial}{\partial x^\nu} b^\mu(\mathbf{x}) \right) \mathbf{e}^\mu \wedge \mathbf{e}_\nu \right] .\end{aligned}$$

The separate parts of this expression are often very usefull. Because of that we define the bivector

$$\begin{aligned}Curl[\mathbf{b}(\mathbf{x})] &\equiv \frac{1}{2} \sum_{\mu, \nu} \left[\frac{\partial}{\partial x^\mu} b^\nu(\mathbf{x}) - \frac{\partial}{\partial x^\nu} b^\mu(\mathbf{x}) \right] \mathbf{e}^\mu \wedge \mathbf{e}_\nu , \\ &= \nabla \wedge (\mathbf{b}(\mathbf{x}))\end{aligned}\tag{5.2.6}$$

and the scalar

$$Div[\mathbf{b}(\mathbf{x})] = \nabla \cdot \mathbf{b}(\mathbf{x}) = \left[\sum_\mu \frac{\partial}{\partial x^\mu} b^\mu(\mathbf{x}) \right] .\tag{5.2.7}$$

The curl of a vector field is a measure for how *rotational* the field is. In



fig. 22: A field with non-zero curl

ordinary space this is fairly easy to picture: the atmospheric flow of air in tornado's or tropical storm systems are good examples of rotational vectorfields. The divergence of a vectorfield is a measure for the strength of the *sources* of the field.

Again this is fairly easy to picture in ordinary space: if you consider the flow of water in the highest layers of the oceans you may know that in certain areas these flows have sources or sinks as warmer water rises from the deep regions of the ocean towards the surface, or colder water sinks from the surface into the deeper regions. These phenomena act as sources and sinks for the surface waters and give rise to a non-zero divergence in the flow of surface waters. In Electrodynamics

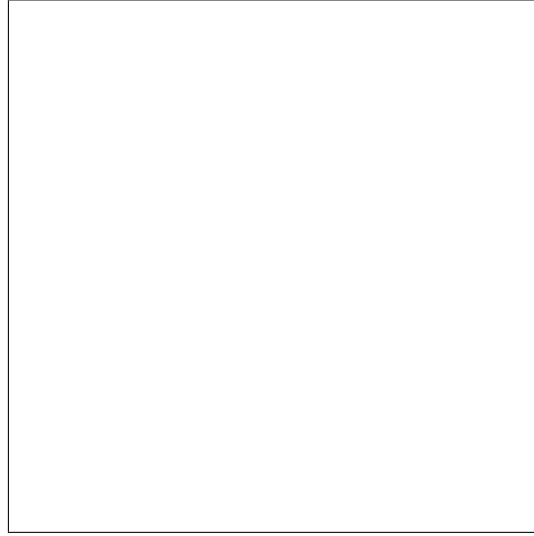


fig. 23: A charge is a source (+) or sink (-) of the electric field

the situation is similar: electric charge acts as a source or sink for the electric fields.

$\nabla \wedge \nabla$ and $\nabla \cdot \nabla$

If we start out from a vectorfield that is a gradient, $\mathbf{b}(\mathbf{x}) = \nabla\beta(\mathbf{x})$, then we get

$$\begin{aligned}\nabla\mathbf{b}(\mathbf{x}) &= \nabla\nabla\beta(\mathbf{x}) = \\ &= \nabla \cdot \nabla\beta(\mathbf{x}) \\ &= \left[\sum_{\mu=0}^3 (\mathbf{e}^\mu)^2 \frac{\partial^2}{\partial x^{\mu 2}} \beta(\mathbf{x}) \right] .\end{aligned}\tag{5.2.8}$$

If you check out the signs in the last line you see that the so-called *wave equation*

$$\left(\frac{\partial^2}{\partial x^{02}} - \frac{\partial^2}{\partial x^{12}} - \frac{\partial^2}{\partial x^{22}} - \frac{\partial^2}{\partial x^{32}} \right) \beta(\mathbf{x}) = 0 ,\tag{5.2.9}$$

can be rewritten in spacetime algebra as

$$\nabla^2\beta(\mathbf{x}) = 0 .\tag{5.2.10}$$

This result shows upon closer inspection a very general and useful side-result,

$$\nabla \wedge \nabla\beta(\mathbf{x}) = 0 .\tag{5.2.11}$$

Now it is time to see some applications of all this mathematical stuff on differentiation in Spacetime.

5.3. Applications in Classical Dynamics

From non-relativistic classical mechanics you will know the concept of *potential-energy*. As we have seen in the chapter on kinematics, *kinetic* energy turned out to be part of an energy-momentum vector $\mathbf{p}(\tau)$. From the principles of special relativity it then follows that potential energy should likewise be part of a spacetime vector that I shall call a potential energy-momentum vector.

The total energy-momentum vector of a particle of restmass m can thus include the contributions from the kinetic energy-momentum $m\mathbf{u}(\tau)$ and a potential energy-momentum from a vectorfield $\mathbf{V}(\mathbf{x})$. In equations,

$$\mathbf{p} = m\mathbf{u} + \mathbf{V}(\mathbf{x}) . \quad (5.3.1)$$

Now what can we say about the forces acting on such a particle as a result of this?

In the restframe of the moving particle the force is caused, like in non-relativistic mechanics, by the *gradient* of the potential-energy. The energy in the restframe caused by the potential energy-momentum vector $\mathbf{V}(\mathbf{x})$ is given by

$$U_{rest}(\mathbf{x}(\tau)) = \mathbf{u}(\tau) \cdot \mathbf{V}(\mathbf{x}(\tau)) .$$

Problem 101: Frame-free Energy revisited

Go back to the previous chapter on kinematics and check that the above equation for the potential energy in the rest-frame of the particle is consistent with the frame-free kinetic energy defined in the previous chapter.

The force-term in the rest-frame is then minus the gradient of this energy, so

$$\sum_{j=1}^3 -\mathbf{e}_j \frac{\partial}{\partial x^j} U_{rest}(\mathbf{x}(\tau)) \rightarrow \nabla \mathbf{u}(\tau) \cdot \mathbf{V}(\mathbf{x}(\tau)) .$$

In this last step a time-derivative is also added to accomodate the possibility that the restframe potential-energy is actually time-dependent. This *force* is the cause of the change in the *total* momentum, i.e. kinetic momentum and potential energy-momentum. The equation that we find this way is

$$\frac{d}{d\tau} \mathbf{p}_T(\tau) = \nabla \mathbf{u} \cdot \mathbf{V}(\mathbf{x}) . \quad (5.3.2)$$

If we assume the rest mass is a constant, using the chainrule to get

$$\frac{d}{d\tau} \mathbf{V}(\mathbf{x}(\tau)) = (\mathbf{u}(\tau) \cdot \nabla) \mathbf{V}(\mathbf{x}) ,$$

we can rewrite this as

$$m \frac{d}{d\tau} \mathbf{u} = \nabla \mathbf{u} \cdot \mathbf{V}(\mathbf{x}) - \mathbf{u}(\tau) \cdot \nabla \mathbf{V}(\mathbf{x}) . \quad (5.3.3)$$

So we have found a relativistic equation of motion. Below we will see that the expression on the righthandside actually contains a bivector forcefield $\nabla \wedge \mathbf{V}(\mathbf{x})$.

5.3.1. Forces are Bivectors

The righthandside of the relativistic equation of motion hides a bivector force which can be retrieved from

$$\nabla(\mathbf{u} \cdot \mathbf{V}(\mathbf{x})) - (\mathbf{u}(\tau) \cdot \nabla)\mathbf{V}(\mathbf{x}) = [\nabla \wedge \mathbf{V}(\mathbf{x}), \mathbf{u}] .$$

Problem 102: Show it

Write out both sides of the above equation and show that it is indeed true.

For all fundamental physical interactions the potential energy-momentum vector $\mathbf{V}$ is caused by the fact that the particle carries a charge q and interacts with a vector-potential $\mathbf{A}$ such that

$$\begin{aligned} \mathbf{V}(\mathbf{x}) &= q\mathbf{A}(\mathbf{x}) , \\ F &= \nabla \wedge \mathbf{A}(\mathbf{x}) , \\ m \frac{d}{d\tau} \mathbf{u} &= q[F(\mathbf{x}(\tau)), \mathbf{u}(\tau)] . \end{aligned} \tag{5.3.4}$$

We have found a bivector field to represent a *spacetime force*. The bivector force revealed itself as the *curl* of a vectorfield in spacetime, the curl of a potential energy-momentum!

5.3.2. Bivector Forces and Energy-Momentum

The kinetic energy of a moving massive particle is the time-component of its energy-momentum vector. Hence, according to an observer using the orthonormal basis $\mathbf{e}_\mu$ we have

$$\begin{aligned} \frac{d}{d\tau} \frac{E(\tau)}{c} &= m \frac{d}{d\tau} \mathbf{e}^0 \cdot \mathbf{u} = \mathbf{e}^0 \cdot \mathbf{m} \frac{d}{d\tau} \mathbf{u} , \\ &= q \mathbf{e}^0 \cdot [\mathbf{F}(\mathbf{x}(\tau)), \mathbf{u}(\tau)] . \end{aligned}$$

The inertial observer performing measurement on the electromagnetic field F , using the cartesian frame $\mathbf{e}_\mu$ expands the bivector field as

$$F(\mathbf{x}) = \sum_{j=1}^3 E_j(\mathbf{x}) K_j + B_j(\mathbf{x}) J_j , \tag{5.3.5}$$

where the coefficients $E_j(\mathbf{x})$ and $B_j(\mathbf{x})$ are scalar fields and the K_j and J_j are the usual timelike and spacelike bivectors. This is equivalent to the following way of defining the electric and magnetic fields

$$F_E \equiv \frac{1}{2} [F - \mathbf{u} F \mathbf{u}] , \tag{5.3.6}$$

$$F_M \equiv \frac{1}{2} [F + \mathbf{u} F \mathbf{u}] , \tag{5.3.7}$$

for an observer with a proper velocity $\mathbf{u}$.

Problem 103: Electric and Magnetic fields and commutation

Consider some observer moving with a proper velocity $\mathbf{u}$ in some electromagnetic field F .

a: Give the formal expressions for F_E and F_M as measured by this observer in terms of F and $\mathbf{u}$.

b: Show that the electric part F_E satisfies $F_E \mathbf{u} = -\mathbf{u} F_E$.

c: Show that the magnetic part F_M satisfies $F_M \mathbf{u} = \mathbf{u} F_M$.

As a result we find that for measurements by an observer whose proper-velocity is $\mathbf{e}_0$ we get

$$\begin{aligned} \frac{d}{d\tau} \frac{E(\tau)}{c} &= q \vec{v}(\tau) \cdot \vec{E}(\mathbf{x}) , \\ \vec{E}(\mathbf{x}) &= \sum_{j=1}^3 E_j(\mathbf{x}) K_j , \\ \vec{v}(\tau) &= \mathbf{u}(\tau) \wedge \mathbf{e}_0 . \end{aligned}$$

This shows that it are the *electric fields* that do work, i.e. change the kinetic energy of the particle, and not the magnetic fields. The magnetic interactions only affect the direction of the particle's relative velocity but not its magnitude. We saw that forces in Spacetime reveal themselves as bivector-fields. But how does this connect to the non-relativistic notion of having electric and magnetic vector-fields? I want to clarify this important conceptual point first. Then we will go on to ask ourselves how these fields are connected with their sources.

5.4. Electromagnetic Fields and Potentials

In "three dimensional physics plus time" the electric and magnetic fields are considered to be *vectors* how can that be reconciled with them being bivectors in Spacetime? Using the pseudo-scalar i we can relate the J_j and the K_j through

$$iJ_j = K_j ,$$

and so, with $i^2 = -1$, we can also write F as

$$F = \sum_{j=1}^3 (E_j(\mathbf{x}) - iB_j(\mathbf{x})) K_j = \vec{E}(\mathbf{x}) - i\vec{B}(\mathbf{x}) .$$

So both the electric and magnetic fields can be associated algebraically with the time-like bivectors, i.e. with *relative vectors*. Non-relativistic physics *simply does not distinguish* between vectors in spacelike directions such as $\mathbf{e}_1$ and *relative vectors* in a spacelike direction such as $\mathbf{e}_1 \mathbf{e}_0$ which really is a bivector. Non-relativistic physics is blind to $\mathbf{e}_0$ and thus puts different things in the same box.

Now we have seen that the electromagnetic fields can be derived from a vector-potential $\mathbf{A}(\mathbf{x})$. Can we also translate that directly into electric and magnetic fields? This is an observer-dependent question, and thus it has an observer-dependent *yes* as an answer. It works as follows; first we split the vector-derivative via

$$\mathbf{e}_0 \nabla = \mathbf{e}_0 \cdot \nabla + \mathbf{e}_0 \wedge \nabla = \frac{\partial}{\partial t} - \vec{\nabla} . \quad (5.4.1)$$

We also have

$$\mathbf{e}_0 \mathbf{A}(\mathbf{x}) = \mathbf{e}_0 \cdot \mathbf{A}(\mathbf{x}) + \mathbf{e}_0 \wedge \mathbf{A}(\mathbf{x}) = \Phi(\mathbf{x}) - \vec{A}(\mathbf{x}) .$$

Putting these together yields

$$\begin{aligned} \nabla \mathbf{A}(\mathbf{x}) &= \nabla \mathbf{e}_0^2 \mathbf{A}(\mathbf{x}) , \\ &= \left\{ \frac{\partial}{\partial t} + \vec{\nabla} \right\} \{ \Phi(\mathbf{x}) + \vec{A}(\mathbf{x}) \} , \\ &= \left\{ -\frac{\partial}{\partial t} \vec{A}(\mathbf{x}) + \vec{\nabla} \Phi(\mathbf{x}) \right\} + [\vec{\nabla}, \vec{A}(\mathbf{x})] . \end{aligned}$$

From this we can identify the electric and magnetic fields as the timelike and spacelike bivector-parts

$$\vec{E}(\mathbf{x}) = -\frac{\partial}{\partial t} \vec{A}(\mathbf{x}) + \vec{\nabla} \Phi(\mathbf{x}) , \quad (5.4.2)$$

$$\vec{B}(\mathbf{x}) = i[\vec{\nabla}, \vec{A}(\mathbf{x})] . \quad (5.4.3)$$

Problem 104: The Coulomb field

Consider an observer using the standard orthonormal basis and measuring a vector-potential of the form $\mathbf{A}(\mathbf{x}) = \frac{Q}{\sqrt{(\mathbf{e}_0 \wedge \mathbf{x})^2}} \mathbf{e}_0$. In this form the field is called a Coulomb field.

- a: Choose spherical or cartesian coordinates and express $\mathbf{A}(\mathbf{x})$ in terms of those coordinates.
- b: Compute $F = \nabla \wedge \mathbf{A}(\mathbf{x})$.
- c: Determine the electric and magnetic fields according to the observer whose time-like unitvector is the $\mathbf{e}_0$.

Power in Potentials

At the end of the previous chapter we saw that the rate-of-change of the kinetic energy of a charged particle in an external electromagnetic field F was given by

$$\frac{d}{d\tau} \frac{E(\tau)}{c} = q \vec{v}(\tau) \cdot \vec{E}(\mathbf{x}) . \quad (5.4.4)$$

If we now combine the expression of $\vec{E}(\mathbf{x})$ in terms of the potentials we get

$$\frac{d}{d\tau} \frac{E(\tau)}{c} = q \vec{v}(\tau) \cdot \left\{ -\frac{\partial}{\partial t} \vec{A}(\mathbf{x}) + \vec{\nabla} \Phi(\mathbf{x}) \right\} , = -\frac{d}{d\tau} \{ \Phi(\mathbf{x}) - \vec{v}(\tau) \cdot \vec{A}(\mathbf{x}) \} = -\frac{d}{d\tau} \mathbf{u}(\tau) \cdot \mathbf{A}(\mathbf{x}) .$$

In other words, the rate-of-change of the kinetic energy of the particle is equal to minus the rate-of-change of potential energy in the frame of the moving charge. This is the result that also was the starting point of our discussion of spacetime forces.

5.5. Sources of the Electromagnetic Fields

The vector-derivative can operate on any type of function. Physically most interesting are scalar fields, vector fields and bivector fields. As we have seen in the example of vector-fields, the vector-derivative connects the vectorfields to its sources and curls. The same thing happens for bivector fields such as the electromagnetic field. Let me first give the space+time version of the the laws that govern the electromagnetic field in *words*:

- Electric charges are sources (positive charge) or sinks (negative charge) of the electric field;
- Magnetic fields have no sources or sinks;
- The rate of change of magnetic fields causes electric fields to become curled;
- Electric current-density and the rate of change of electric fields causes the magnetic fields to become curled.

Now you might expect that when formulated in Spacetime, so without the observer-dependent split between space and time, this can be a bit more efficient; and so it is. In spacetime these four laws become two

- The electric spacetime current density $\mathbf{J}$ is the source of the electromagnetic field F ;
- There is no magnetic source.

Due to the fact that divergence *and* curl are both algebraically contained in the single ∇ symbol, the mathematical expression of these two laws is reduced even further to a single statement

$$\nabla F(\mathbf{x}) = 4\pi\mathbf{J}(\mathbf{x}) .$$

Try to grasp how immensely powerfull our mathematical language has become by now: this simple law with only three symbols encodes practically all the physics of electromagnetism.

Problem 105: Div and Curl for bivectors

Consider the simple bivectorfield $F(\mathbf{x}) = E_1(\mathbf{x})K_1$.

a: Calculate $\nabla F(\mathbf{x})$.

b: Identify the vector- and the trivector-part of $\nabla F(\mathbf{x})$.

c: Why can we call the vector-part the "divergence" of the bivector-field, and the tri-vector part the "curl of the bivectorfield"?

Now we are going to dissect this single equation into its various meaningful parts. A straightforward calculation gives

$$\nabla F = \sum_{\mu} \mathbf{e}^{\mu} \frac{\partial}{\partial x^{\mu}} \sum_{j=1}^3 (E_j(\mathbf{x}) - iB_j(\mathbf{x}))K_j ,$$

$$\begin{aligned}
&= \sum_{\mu} \sum_{j=1}^3 \mathbf{e}^{\mu} \left(\frac{\partial}{\partial x^{\mu}} E_j(\mathbf{x}) - i \frac{\partial}{\partial x^{\mu}} B_j(\mathbf{x}) \right) K_j , \\
&= \sum_{\mu, j} \left(\frac{\partial}{\partial x^{\mu}} E_j(\mathbf{x}) + i \frac{\partial}{\partial x^{\mu}} B_j(\mathbf{x}) \right) \mathbf{e}^{\mu} K_j .
\end{aligned}$$

Now the *algebraic* nature of the various contributions to ∇F is determined by the possible outcomes of the products $\mathbf{e}^{\mu} K_j$. These can be computed rather easily, we find:

$$\begin{aligned}
\mathbf{e}^j K_j &= \mathbf{e}_0 = \mathbf{e}^0 , \\
\mathbf{e}^0 K_j &= -\mathbf{e}_j = \mathbf{e}^j , \\
\mathbf{e}^i K_j &= \mathbf{e}^0 \mathbf{e}^i \mathbf{e}^j , \quad i \neq j .
\end{aligned}$$

Problem 106: Verify this

Verify the above relations.

So we find vector-contributions (the first two results above) and trivector contributions (the last result). So the derivative ∇F contains a vector-part that we will denote at $\nabla \cdot F$. It also contains a trivector part that we then denote $\nabla \wedge F$. Maxwell's law separated in vector- and trivector parts reads

$$\begin{aligned}
\nabla \cdot F(\mathbf{x}) &= 4\pi \mathbf{J}(\mathbf{x}) , \\
\nabla \wedge F(\mathbf{x}) &= 0 .
\end{aligned}$$

These are the *divergence* and *curl* of the bivectorfield F . The divergence again represents a source and in the equations for the electromagnetic field this will be the electric current-density which is a vector-field. For the curl the situation seems more difficult; however there is a nice bit of mathematics at work here. The fact that $\nabla \wedge F = 0$ actually guarantees that we are allowed to write the electromagnetic field F as the curl of a vector-potential $F = \nabla \wedge A$.

Problem 107: $\nabla \wedge \nabla$

Consider a vector field $\mathbf{b}(\mathbf{x})$.

a: Calculate the bivector-field $F = \nabla \wedge \mathbf{b}(\mathbf{x})$.

b: Use the fact that $\nabla \wedge F$ is the trivector-part of ∇F to calculate $\nabla \wedge F$ and show that it vanishes.

In more traditional texts on *relativistic* or *covariant* classical electrodynamics you will find *two* equations for F and not *one single* Maxwell's law. The reason is that traditional texts do not use geometric product which incorporates the wedge-product. Many treatises on electrodynamics however even treat the spacetime picture as an add-on reserved for a final chapter. An approach that also makes an observer-dependent spacetime split yields *four* Maxwell's equations. To see how this comes about we first separate the electric and magnetic fields by writing

$$F(\mathbf{x}) = E(\mathbf{x}) - iB(\mathbf{x}) ,$$

and consider both bivectors $E(\mathbf{x})$ and $B(\mathbf{x})$ to be strictly timelike (so this involves a choice of frame). If we now apply the ∇ , take into account that i anti-commutes

with vectors, split vector- and trivector parts, and finally use that i turns vectors into trivectors and vice versa, we obtain

$$\begin{aligned}\nabla F(\mathbf{x}) &= \nabla E(\mathbf{x}) + i\nabla B(\mathbf{x}) \\ &= \nabla \cdot E(\mathbf{x}) + \nabla \wedge E(\mathbf{x}) + i\nabla \cdot B(\mathbf{x}) + i\nabla \wedge B(\mathbf{x}) , \\ &= [\nabla \cdot E(\mathbf{x}) + i\nabla \wedge B(\mathbf{x})] + [\nabla \wedge E(\mathbf{x}) + i\nabla \cdot B(\mathbf{x})] .\end{aligned}$$

If we again write Maxwell's law in terms of vector- and trivector parts, but now with the split in electric and magnetic fields as well we find

$$\begin{aligned}\nabla \cdot E(\mathbf{x}) + i\nabla \wedge B(\mathbf{x}) &= 4\pi\mathbf{J}(\mathbf{x}) , \\ \nabla \wedge E(\mathbf{x}) + i\nabla \cdot B(\mathbf{x}) &= 0 .\end{aligned}$$

The last equation is a trivector equation, but by another multiplication by i it too becomes a vector equation,

$$i\nabla \wedge E(\mathbf{x}) - \nabla \cdot B(\mathbf{x}) = 0 .$$

As a last step in our space-time splitting of these equations we take the geometric product of these equations with the time-like basis-vector $\mathbf{e}_0$. To do this efficiently it is very useful to notice that the pseudo-scalar also has the following interesting property

$$\begin{aligned}(i\mathbf{a}) \wedge \mathbf{b} &= i(\mathbf{a} \cdot \mathbf{b}) , \\ (i\mathbf{a}) \cdot \mathbf{b} &= i(\mathbf{a} \wedge \mathbf{b}) ,\end{aligned}$$

Problem 108: i anti-commutes with vectors

Show that the above results only rely on the fact that i anti-commutes with vectors.

As Maxwell's equations are now all vector-equations, the product with $\mathbf{e}_0$ will contain a scalar part and a bivector part. Using $\mathbf{e}_0 \cdot \mathbf{J}(\mathbf{x}) = \rho(\mathbf{x})$ we get four equations in total:

$$\begin{aligned}\mathbf{e}_0 \cdot [\nabla \cdot E(\mathbf{x})] &= 4\pi\rho(\mathbf{x}) , \\ \mathbf{e}_0 \wedge [\nabla \cdot E(\mathbf{x})] - i(\mathbf{e}_0 \cdot [\nabla \wedge B(\mathbf{x})]) &= 4\pi\mathbf{e}_0 \wedge \mathbf{J}(\mathbf{x}) , \\ -\mathbf{e}_0 \cdot [\nabla \cdot B(\mathbf{x})] &= 0 , \\ -i(\mathbf{e}_0 \cdot [\nabla \wedge E(\mathbf{x})]) - \mathbf{e}_0 \wedge [\nabla \cdot B(\mathbf{x})] &= 0 .\end{aligned}$$

Three of these four equations have been named after the scientists who have contributed greatly to the final formulation of Maxwell's equations. The first is called Gauss-law, the second is called Ampere's law, the third has no name and the fourth is Faraday's law. These four laws will take up much of the remainder of this course on electrodynamics.

Problem 109: Show it

The above form of Maxwell's four equations have been obtained due to the fact that a few terms vanish. In this exercise you will check that.

a: Show, without computations but with algebraic arguments only, that $\mathbf{e}_0 \wedge [\nabla \wedge B(\mathbf{x})] = 0$ and that this term has dropped from the first of Maxwell's equations.

b: Show that $\mathbf{e}_0 \wedge [\nabla \wedge E(\mathbf{x})] = 0$ and that this term has dropped from the third of Maxwell's equations.

We can take these laws a few steps further still. First note that we can split the vector derivative

$$\begin{aligned}\mathbf{e}_0 \nabla &= \mathbf{e}_0 \cdot \nabla + \mathbf{e}_0 \wedge \nabla , \\ &= \frac{\partial}{\partial t} + \vec{\nabla} ,\end{aligned}$$

where we used $\vec{\nabla}$ to represent the relative vector $\mathbf{e}_0 \wedge \nabla$. If we insert these expressions into the Maxwell equations we have derived we get

$$\begin{aligned}\vec{\nabla} \cdot E(\mathbf{x}) &= 4\pi\rho(\mathbf{x}) , \\ \frac{\partial}{\partial t}E(\mathbf{x}) - i\vec{\nabla} \wedge B(\mathbf{x}) &= 4\pi\vec{J}(\mathbf{x}) , \\ \vec{\nabla} \cdot B(\mathbf{x}) &= 0 , \\ i\vec{\nabla} \wedge E(\mathbf{x}) + \frac{\partial}{\partial t}B(\mathbf{x}) &= 0 .\end{aligned}\tag{5.5.1}$$

These laws are now clearly identifiable as the four Maxwell equations as they appear in the standard texts.

Before we attempt to solve these equations, i.e. find the field F when the sources are given, we will first look at the dynamics of charged, spinning particles in external electromagnetic fields.

Chapter 6

Dynamics in Electromagnetic fields

In this chapter we will discuss the dynamics of charged particles in Electromagnetic fields. One important new development will come from the realization that the eigen-rotor contains more, and even more physical, information. Not only does it determine the proper velocity of a particle, but also the orientation of its *spin*. The spin of a particle is a physical property that leads us onto the brink of quantum mechanics.

6.1. Elementary dynamics of charged particles

Recall that in the chapter on kinematics we already found that due to the constant spacetime length of the proper velocity of a particle $\mathbf{u}(\tau)$, we could introduce the eigen-rotor $\Lambda(\tau)$ which was such that for a particle with proper velocity $\mathbf{u}(\tau)$ we could write

$$\mathbf{u}(\tau) = \Lambda(\tau)\mathbf{u}(0)\tilde{\Lambda}(\tau) .$$

As a result upon the introduction of the bivector

$$\Omega(\tau) = 2\left[\frac{d}{d\tau}\Lambda(\tau)\right]\tilde{\Lambda}(\tau) ,$$

the proper acceleration $\mathbf{a}(\tau)$ is related to the proper velocity $\mathbf{u}(\tau)$ through

$$\mathbf{a}(\tau) = [\Omega(\tau), \mathbf{u}(\tau)] . \quad (6.1.1)$$

As accelerations are the result of the influence of external forces, it seems very reasonable to assume that they can be represented by a bivector-field F , to which a particle with a restmass m and the appropriate charge q couples. Its proper velocity will then satisfy the following *equation of motion*

$$m\frac{d}{d\tau}\mathbf{u}(\tau) = q[F(\mathbf{x}(\tau)), \mathbf{u}(\tau)] . \quad (6.1.2)$$

This is indeed the *proper* generalisation of Newton's second law to spacetime. Now when you look at the general form of this equation it bears an interesting resemblance to the paths of extremal length we discussed in the previous chapter. We

will see later in this chapter that this link can indeed be established.

Problem 110: Energy and Work

Consider a particle with a proper momentum $\mathbf{p}(\tau) = m\mathbf{u}(\tau)$, a charge q , and an outside observer using the orthonormal basis $\mathbf{e}_\mu$.

a: Show that the proper-time rate-of-change of the kinetic energy measured by the observer using is given by $q\mathbf{e}_0 \cdot [F(\mathbf{x}(\tau)), \mathbf{u}(\tau)]$.

b: Show that $\mathbf{a} \cdot [F(\mathbf{x}), \mathbf{b}] = -\mathbf{b} \cdot [F(\mathbf{x}), \mathbf{a}]$.

c: Expand the bivector F in terms of the basis $\mathbf{e}_\mu$.

d: Show that only the timelike parts in F contribute to the proper time rate-of-change of the kinetic energy of the particle.

Here we first study one of the merits of this eigen-rotor approach. The eigen-rotor of a particle moving under the influence of the force $F(\mathbf{x})$ satisfies

$$\Omega(\tau) = \frac{q}{2m} F(\mathbf{x}) ,$$

which we can rewrite as

$$\frac{d}{d\tau} \Lambda(\tau) = \frac{q}{2m} F \Lambda(\tau) . \quad (6.1.3)$$

This is the so-called rotor-equation. In some cases the rotor equation is easier to solve than the equation of motion. In the next chapter you will see that this equation has a *formal* general solution. The solution is general in that it can be satisfied for any bivector field $F(\mathbf{x})$, but it is *formal* in the sense that it is often impossible to calculate explicitly.

Now note that if we write

$$\mathbf{u}(\tau) = \Lambda(\tau) \mathbf{u}(0) \tilde{\Lambda}(\tau) ,$$

we obviously have

$$\mathbf{u}(0) = \Lambda(0) \mathbf{u}(0) \tilde{\Lambda}(0) ,$$

and so $\Lambda(0) = 1$. So the rotor $\Lambda(\tau)$ satisfies the rotor-equation and has a standard initial condition

$$\begin{aligned} \frac{d}{d\tau} U(\tau) &= \frac{q}{2m} F(\mathbf{x}(\tau)) U(\tau) , \\ U(0) &= 1 . \end{aligned}$$

Constant Electromagnetic Fields

If the electromagnetic field is constant in Spacetime, solving the rotor-equation becomes almost trivial. For constant fields we can immediately write down the result

$$\Lambda(\tau) = \exp\left(\frac{q\tau}{2m} F\right) , \quad (6.1.4)$$

and a straightforward differentiation shows that this is indeed the correct solution. The proper velocity of such a particle is now given by

$$\mathbf{u}(\tau) = \exp\left(\frac{q\tau}{2m} F\right) \mathbf{u}(0) \exp\left(-\frac{q\tau}{2m} F\right) .$$

To determine the worldline we need to integrate this expression. There is a general method for this that we will treat in the next section.

6.1.1. Examples

Here we will look at some examples to get a better understanding of how these equations work.

Constant Electric Fields

Let us first consider a simple situation, a particle in a constant, purely electric field. Subsequently we will deal with a purely magnetic field and notice that the method of solution is practically identical.

Consider some observer who chooses his $\mathbf{e}_1$ -direction such that the constant electric field becomes

$$F(\mathbf{x}) = E\mathbf{e}_1\mathbf{e}_0 . \quad (6.1.5)$$

All we need to do is to compute

$$\begin{aligned} \Lambda(\tau) &= \exp\left(\frac{q\tau}{2m}F\right) , \\ &= \cosh\left(\frac{qE\tau}{2m}\right) + \sinh\left(\frac{qE\tau}{2m}\right)\mathbf{e}_1\mathbf{e}_0 . \end{aligned}$$

Substituting this into the equation for the proper velocity, where we choose the particle to be initially at rest, $\mathbf{u}(0) = c\mathbf{e}_0$, gives

$$\mathbf{u}(\tau) = c \cosh\left(\frac{qE\tau}{m}\right)\mathbf{e}_0 + c \sinh\left(\frac{qE\tau}{m}\right)\mathbf{e}_1 .$$

This spacetime velocity can now be integrated to give the worldline

$$\begin{aligned} \mathbf{x}(\tau) &= \int_0^\tau d\tau' \mathbf{u}(\tau') , \\ &= \mathbf{x}(0) + \frac{mc}{qE} \left\{ \sinh\left(\frac{qE\tau}{m}\right)\mathbf{e}_0 + [\cosh\left(\frac{qE\tau}{m}\right) - 1]\mathbf{e}_1 \right\} . \end{aligned} \quad (6.1.6)$$

If we draw a spacetime diagram of this motion we find a spacelike hyperbola.

Problem 111: Verify it

- a: Use Mathematica to verify the above results for the proper-velocity and the worldline as functions of the proper time τ .
- b: Make a number of parametric plots of these worldlines for different values of, for example, E or the initial position and/or velocity.

A constant Magnetic field

This ofcourse works in exactly the same way as the calculation above. Now we may for simplicity assume that the observer has chosen his basis such that

$$F(\mathbf{x}) = BJ_1 . \quad (6.1.7)$$

Again, all we need to do is to compute

$$\begin{aligned} \Lambda(\tau) &= \exp\left(\frac{q\tau}{2m}F\right) , \\ &= \cos\left(\frac{qB\tau}{2m}\right) + \sin\left(\frac{qB\tau}{2m}\right)\mathbf{e}_2\mathbf{e}_3 . \end{aligned}$$

If we now choose a initial velocity we should be aware that a particle which is initially at rest will not experience a force from the magnetic field. If we would choose

$$\mathbf{u}(0) = \gamma\{c\mathbf{e}_0 + v\mathbf{e}_1\} ,$$

we find

$$\mathbf{u}(\tau) = \mathbf{u}(0) .$$

This is not terribly interesting, But if we choose

$$\mathbf{u}(0) = \gamma\{c\mathbf{e}_0 + v\mathbf{e}_2\} ,$$

we get

$$\mathbf{u}(\tau) = \gamma\{c\mathbf{e}_0 + v[\cos(\frac{qB\tau}{m})\mathbf{e}_2 + \sin(\frac{qB\tau}{m})\mathbf{e}_3]\} .$$

The particle makes circular orbits around the magnetic field lines.

Problem 112: The worldline

- a: Use Mathematica to compute the worldline corresponding to the proper velocity above.
- b: Make a parametric plot of the proper velocity for different initial velocities and a fixed $\omega = \frac{qB}{m}$, do the same for the worldlines.
- c: Explain why the angular speed ω is defined in the restframe of the particle and that the outside observer using the orthonormal frame $\mathbf{e}_\mu$ observes an angular speed $\gamma\omega$.

Parallel Fields

As we have seen, when electric and magnetic fields are either pure, or perpendicular the problem of finding a particle trajectory is solved in a relatively straightforward fashion. This changes a little bit when the fields have a more general relative orientation. In that case the motion of the charged particle will display two components called *drift* and *gyration*. In the simplest case we have magnetic and electric parts which commute, for example

$$F(\mathbf{x}) = E\mathbf{e}_1\mathbf{e}_0 + B\mathbf{e}_2\mathbf{e}_3 . \quad (6.1.8)$$

Because of the fact that they commute we can factorise the exponential,

$$\begin{aligned} \Lambda(\tau) &= \exp\left(\frac{q\tau}{2m}F\right) , \\ &= \exp\left(\frac{q\tau}{2m}E\mathbf{e}_1\mathbf{e}_0\right) \exp\left(\frac{q\tau}{2m}B\mathbf{e}_2\mathbf{e}_3\right) . \end{aligned}$$

To understand the result again consider the initial velocity

$$\mathbf{u}(0) = \gamma\{c\mathbf{e}_0 + v\mathbf{e}_2\} .$$

The motion will split up into a rotational part in the $\mathbf{e}_2\mathbf{e}_3$ -plane and an accelerated motion in the $\mathbf{e}_1$ -direction as follows

$$\mathbf{u}(\tau) = \gamma\{c \cosh(\frac{qE\tau}{m})\mathbf{e}_0 + c \sinh(\frac{qE\tau}{m})\mathbf{e}_1 + v[\cos(\frac{qB\tau}{m})\mathbf{e}_2 + \sin(\frac{qB\tau}{m})\mathbf{e}_3]\} .$$

The particle makes helical orbits around the magnetic field lines.

Problem 113: Plotting worldlines

- a: Choose various values for E and B and study the worldlines of particles with $v = 0.5c$ by making parametric plots with Mathematica.
b: What happens if the initial velocity is along the $\mathbf{e}_1$ -direction?

In this example the electric and magnetic fields are parallel. In general there is no reason for such a restriction and thus the motion of a charge will be a more involved combination of rotational and accelerated motion.

Null Fields

Before we return to the formal development of our ideas about dynamics, let us study one more example. The electromagnetic field F is called a *null-field* if it satisfies $F^2 = 0$. In such a case the solution of the rotor equation becomes exceptionally simple,

$$\begin{aligned}\Lambda(\tau) &= \exp\left(\frac{q\tau}{2m}F\right), \\ &= 1 + \frac{q\tau}{2m}F.\end{aligned}$$

Hence, the proper velocity is also easily expressed in terms of the initial velocity as

$$\begin{aligned}\mathbf{u}(\tau) &= \left[1 + \frac{q\tau}{2m}F\right]\mathbf{u}(0)\left[1 - \frac{q\tau}{2m}F\right], \\ &= \mathbf{u}(0) + \frac{q\tau}{2m}[F, \mathbf{u}(0)] - \frac{q^2\tau^2}{4m^2}F\mathbf{u}(0)F.\end{aligned}$$

Integrating this equation with respect to the proper time τ is also straightforward.

Problem 114: Motion in null-fields

- a: Show that the field $F = f\mathbf{e}_1(\mathbf{e}_0 + \mathbf{e}_3)$ is a null-field.
b: Use the expressions above and apply it to the case where a charged particle starts out at rest at $\tau = 0$ and moves through the field F . Find the expression for $\mathbf{u}(\tau)$ and an expression for $\mathbf{x}(\tau)$.
c: Draw a parametric plot of a number of such worldlines.
d: Is it possible to choose the initial velocity in such a way that no acceleration occurs?

Such null-fields are not uncommon in situations where electrical charges are exposed to electromagnetic radiation. For example electrons caught in high-intensity laserbeams are in such a situation. Although the electromagnetic wavefields are not constant we will see in a later section that the fact that they are also null-fields greatly simplifies the solution of the dynamical problem of finding the trajectories of charged particles in such fields.

6.2. Dynamics in arbitrary Electromagnetic Fields

In this section we take a closer look at a number of examples where an exact solution for the rotor-equation can be found. Next to that we will also derive a

general, formal solution which can be used as a starting point for approximations. The material in this chapter will not be covered in class, but rather serves as source material for your own projects.

6.2.1. Arbitrary constant Electromagnetic Fields

Integrating the rotor solution for the proper velocity to find the worldline is a bit cumbersome as long as the vector is sandwiched between two rotors. We can do something about this by separating the vector into a part that commutes, and a part that anti-commutes with F . Geometrically this would imply splitting $\mathbf{u}(0)$ into terms parallel and perpendicular to the plane represented by the bivector F . But wait!! Does the bivector F really represent a plane, i.e. is it a 2-blade? Ofcourse, in general it will not. But we will be able to split F into two planes that are perpendicular to eachother, and thus represented by bivectors that commute with one another. The decomposition is very usefull in simplifying the solution of problems. But what is even better, it is coordinate-free. The intermediate steps of this derivation are also part of the exercises at the end.

We start of by noting that as F is obviously even, so must F^2 be. Because F is a bivector, F^2 can only consist of scalars and pseudo-scalars. So we define

$$F^2 = a \exp [ib] , \quad (6.2.1)$$

where $a > 0$, and a unit-bivector $\hat{F}$, defined as

$$\hat{F} = \frac{1}{\sqrt{a}} \exp \left(-\frac{i}{2} b \right) F , \quad (6.2.2)$$

thus satisfies $\hat{F}^2 = 1$. Using this electromagnetic field one can write

$$F = \sqrt{a} \cos \left(\frac{1}{2} b \right) \hat{F} + \sqrt{a} \sin \left(\frac{1}{2} b \right) i \hat{F} . \quad (6.2.3)$$

It is easy to check that F and iF commute, so we may write as our final result,

$$\exp \left(\frac{q\tau}{2m} F \right) = \exp \left(\frac{q\tau}{2m} \sqrt{a} \cos \left(\frac{1}{2} b \right) \hat{F} \right) \exp \left(\frac{q\tau}{2m} \sqrt{a} \sin \left(\frac{1}{2} b \right) i \hat{F} \right) . \quad (6.2.4)$$

This decomposition obviously fails if $F^2 = 0$. Note that as scalars a and b are Lorentz invariants. So they can be computed in any frame. If we take the observers restframe and expand in terms of electric and magnetic fields we obtain a highly interesting result

$$a \cos (b) = \sum_{j=1}^3 E_j^2 - B_j^2 , \quad (6.2.5)$$

$$a \sin (b) = 2 \sum_{j=1}^3 E_j B_j . \quad (6.2.6)$$

We find that when the two fields are of *equal magnitude and perpendicular* then we have a serious problem and the procedure mentioned above will not work. If F

is such that $F^2 = 0$ we will call it a *null-field*. But we have seen how to calculate the rotor if F is a null-field, so it represents no real difficulty.

If $b = 0, \pi$ we see that the electric and magnetic fields are perpendicular. The bivector F squares into a real number without any pseudo-scalar part. Such a bivector is called *simple*. As a result $\hat{F}$ is a timelike bivector and thus properly represents a timelike plane, the rotor is generated by

$$\begin{aligned} \sqrt{a}\hat{F} , \quad b = 0 , \text{ or} \\ \sqrt{ai}\hat{F} , \quad b = \pi . \end{aligned}$$

In the first case $F^2 = a^2 > 0$ and thus the rotor is essentially a boost. The electric fields dominate the magnetic fields. In the second case $F^2 = -a^2 < 0$ so the rotor is basically an ordinary rotation. The magnetic interaction dominates the electric fields.

Now that we have found an algebraic way of splitting an arbitrary, constant, electromagnetic bivector F into two commuting parts we are finally able to compute the worldline of a particle in an arbitrary but constant electromagnetic field.

The introduction of the unit-bivector $\hat{F}$ allows us to define a split of the initial velocity $\mathbf{u}(0)$ with respect to this unit-bivector,

$$\mathbf{u}_{//} = \frac{1}{2}[\mathbf{u} - \hat{F}\mathbf{u}\hat{F}] , \quad (6.2.7)$$

$$\mathbf{u}_{\perp} = \frac{1}{2}[\mathbf{u} + \hat{F}\mathbf{u}\hat{F}] . \quad (6.2.8)$$

Now a vector that commutes with a bivector is *not affected* by the transformation generated by that bivector. So we get

$$\begin{aligned} \mathbf{v}(\tau) = & \exp\left(\frac{q\tau}{m}\sqrt{a}\cos\left(\frac{1}{2}b\right)\hat{F}\right)\mathbf{u}_{//}(0) \\ & + \exp\left(\frac{q\tau}{m}\sqrt{a}\sin\left(\frac{1}{2}b\right)i\hat{F}\right)\mathbf{u}_{\perp}(0) . \end{aligned} \quad (6.2.9)$$

The final step is a straightforward integration. Using that $\hat{F}^{-1} = \hat{F}$, it yields the trajectory of the particle

$$\begin{aligned} \mathbf{x}(\tau) = & \mathbf{x}(0) + \left\{ \frac{\exp\left(\frac{q\tau}{m}\sqrt{a}\cos\left(\frac{1}{2}b\right)\hat{F}\right) - 1}{\frac{q}{m}\sqrt{a}\cos\left(\frac{1}{2}b\right)} \right\} \hat{F}\mathbf{u}_{//}(0) \\ & + \left\{ \frac{\exp\left(\frac{q\tau}{m}\sqrt{a}\sin\left(\frac{1}{2}b\right)i\hat{F}\right) - 1}{\frac{qi}{m}\sqrt{a}\sin\left(\frac{1}{2}b\right)} \right\} \hat{F}\mathbf{u}_{\perp}(0) . \end{aligned} \quad (6.2.10)$$

This rather formidable und perhaps un insightful equation represents the general solution to the problem of finding the trajectory of a classical, charged, particle in an arbitrary, constant, electromagnetic fields. Yet, this is one of the few equations in this reader that may qualify as a *formula*; this is an all purpose equation that basically you only need to fill out with a given field. The procedure would work as follows: Suppose you are given an F , then

- Compute F^2 , if $F^2 = 0$ resort to the null-field solution, otherwise determine a and b ;

- Compute $\hat{F}$;
- Compute $\mathbf{u}_\perp(0)$ and $\mathbf{u}_\parallel(0)$;
- Substitute everything in the general solution and carefully calculate the final result, especially when $a = 0$ or $b = 0$.

6.2.2. Formal Solution

If the fields are not constant exact solutions are generally beyond the reach of any straightforward general methods. One can either focus on isolated examples where an exact solution is available, or try to derive a *formal* general solution which is a starting point for approximations. We will start with the latter and derive a general but highly formal solution that is useless for practical calculations but usefull as a point of departure when making approximations. Then we will look at two exactly soluble problems.

Now let us be bold and construct a formal solution for the general case. If we intergrate the rotor-equation from $\tau = 0$ to τ we get

$$\Lambda(\tau) - \Lambda(0) = \frac{q}{2m} \int_0^\tau d\tau' F(\mathbf{x}(\tau')) \Lambda(\tau') . \quad (6.2.11)$$

Now in principle we haven't gained anything yet. But perseverance will now pay of as follows,

$$\begin{aligned} \Lambda(\tau) &= \Lambda(0) + \frac{q}{2m} \int_0^\tau d\tau' F(\mathbf{x}(\tau')) \Lambda(\tau') , \\ &= \Lambda(0) + \frac{q}{2m} \int_0^\tau d\tau' F(\mathbf{x}(\tau')) \left[\Lambda(0) + \frac{q}{2m} \int_0^{\tau'} d\tau'' F(\mathbf{x}(\tau'')) \Lambda(\tau'') \right] , \\ &= \Lambda(0) + \frac{q}{2m} \int_0^\tau d\tau' F(\mathbf{x}(\tau')) \Lambda(0) \\ &\quad + \left(\frac{q}{2m} \right)^2 \int_0^\tau d\tau' \int_0^{\tau'} d\tau'' F(\mathbf{x}(\tau')) F(\mathbf{x}(\tau'')) \Lambda(\tau'') , \end{aligned}$$

Note that the first two terms can be computed. The third term still contains the unknown $\Lambda(\tau)$, but ofcourse we could do the trick again and insert the equation into itself once more. If we do so we find that this third term becomes

$$\begin{aligned} &\left(\frac{q}{2m} \right)^2 \int_0^\tau d\tau' \int_0^{\tau'} d\tau'' F(\mathbf{x}(\tau')) F(\mathbf{x}(\tau'')) \Lambda(0) \\ &+ \left(\frac{q}{2m} \right)^3 \int_0^\tau d\tau' \int_0^{\tau'} d\tau'' \int_0^{\tau''} d\tau''' F(\mathbf{x}(\tau')) F(\mathbf{x}(\tau'')) F(\mathbf{x}(\tau''')) \Lambda(\tau''') . \end{aligned}$$

So we have again found a term that we can compute followed by another term which contains the unknown $\Lambda(\tau''')$. But note that the three computable terms we now have are

$$\begin{aligned} \Lambda(\tau) &\approx \Lambda(0) + \frac{q}{2m} \int_0^\tau d\tau' F(\mathbf{x}(\tau')) \Lambda(0) \\ &\quad + \left(\frac{q}{2m} \right)^2 \int_0^\tau d\tau' \int_0^{\tau'} d\tau'' F(\mathbf{x}(\tau')) F(\mathbf{x}(\tau'')) \Lambda(0) , \\ &= \left[1 + \frac{q}{2m} \int_0^\tau d\tau' F(\mathbf{x}(\tau')) \right. \\ &\quad \left. + \left(\frac{q}{2m} \right)^2 \int_0^\tau d\tau' \int_0^{\tau'} d\tau'' F(\mathbf{x}(\tau')) F(\mathbf{x}(\tau'')) \right] \Lambda(0) . \end{aligned}$$

If we keep on going we will eventually be able to construct the complete solution. We just would need a nice way to write it down. Study the third term in

$$U(\tau) = 1 + \frac{q}{2m} \int_0^\tau d\tau' F(\mathbf{x}(\tau')) + \left(\frac{q}{2m}\right)^2 \int_0^\tau d\tau' \int_0^{\tau'} d\tau'' F(\mathbf{x}(\tau')) F(\mathbf{x}(\tau'')) . \quad (6.2.12)$$

The product of the two bivector fields $F(\mathbf{x}(\tau')) F(\mathbf{x}(\tau''))$ is such that as $\tau'' < \tau'$ the bivectorfield at the *earlier proper time* appears first. We say that the product is *proper-time ordered*. Let us introduce the time-ordering symbol $T[\dots]$ that orders every product of bivectors inside the brackets [...] in such a way that the bivector at the earliest time appears first and the others follow in the order of their times from early to late. Then it is easy to check that we can write

$$\begin{aligned} \int_0^\tau d\tau' \int_0^{\tau'} d\tau'' F(\mathbf{x}(\tau')) F(\mathbf{x}(\tau'')) &= \frac{1}{2} \int_0^\tau d\tau' \int_0^\tau d\tau'' T[F(\mathbf{x}(\tau')) F(\mathbf{x}(\tau''))] . \\ &= \frac{1}{2} T\left[\int_0^\tau d\tau' \int_0^\tau d\tau'' F(\mathbf{x}(\tau')) F(\mathbf{x}(\tau''))\right] . \end{aligned}$$

Using this and continuing to compute higher-order terms in the solution gives us a full solution,

$$U(\tau) = T\left[\exp\left(\frac{q}{2m} \int_0^\tau d\tau' F(\mathbf{x}(\tau'))\right)\right] . \quad (6.2.13)$$

It is the fact that the values of the bivector field $F(\mathbf{x}(\tau'))$ at different positions do not need to commute that makes the computations hard and tedious. The mathematical structure of the solution is however clear. Approximation methods would typically focus on problems involving fields like $F(\mathbf{x} = F_0 + \delta F(\mathbf{x}))$ where F_0^2 is a constant field for which a solution is known and which is very large compared to the *perturbation* $\delta F(\mathbf{x})^2$. One can then also study separately cases where F_0 and $\delta F(\mathbf{x})$ do or do not commute.

6.2.3. A point charge in a Wavefield

Now let us consider our first example of an electromagnetic field that depends on spacetime and still yields a solvable problem. Let $\mathbf{k}$ be a null-vector, i.e. $\mathbf{k}^2 = 0$ and let $\mathbf{e}_\perp$ be perpendicular to $\mathbf{k}$, i.e. $\mathbf{k} \cdot \mathbf{e}_\perp = 0$. We will see somewhat later in the course that the electromagnetic field given by

$$\begin{aligned} F(\mathbf{x}) &= A \mathbf{k} \mathbf{e}_\perp \cos(\mathbf{k} \cdot \mathbf{x}) , \\ &= F_0 \cos(\mathbf{k} \cdot \mathbf{x}) , \end{aligned} \quad (6.2.14)$$

represents an electromagnetic wave, with an amplitude A , travelling in the direction specified by the wavevector $\mathbf{k}$. A quick calculation reveals that for this field $F^2 = 0$ and thus we may set up the null-field solution although we should take into account that the field depends on $\mathbf{x}(\tau)$, so

$$U(\tau) = 1 + \frac{q}{2m} \int_0^\tau d\tau' F_0 \cos(\mathbf{k} \cdot \mathbf{x}(\tau')) . \quad (6.2.15)$$

So we face the problem that we must know the trajectory $\mathbf{x}(\tau)$ in order to proceed. However, there is a conserved quantity that will allow us to bypass this difficulty

and proceed with the solution. As is not hard to check, we have

$$\mathbf{k}F_0 = F_0\mathbf{k} = 0 , \quad (6.2.16)$$

and so the wavevector $\mathbf{k}$ is not affected by the transformation generated by $F(\mathbf{x})$. We can use this for the following clever calculation,

$$\begin{aligned} \mathbf{k} \cdot \mathbf{u}(\tau) &= \mathbf{k} \cdot \{U(\tau)\mathbf{u}(0)\tilde{U}(\tau)\} , \\ &= \{U(\tau)\mathbf{k}\tilde{U}(\tau)\} \cdot \{U(\tau)\mathbf{u}(0)\tilde{U}(\tau)\} , \\ &= U(\tau)\{\mathbf{k} \cdot \mathbf{u}(0)\}\tilde{U}(\tau) , \\ &= \mathbf{k} \cdot \mathbf{u}(0) . \end{aligned}$$

The dotproduct between wavevector and proper velocity of the charged particle is conserved! This dot-product is just the *frequency* of the wave as observed by an observer moving along with the particle! So let us call

$$\omega_0 = \mathbf{k} \cdot \mathbf{u}(0) . \quad (6.2.17)$$

Now notice that if we integrate the both sides of the previous calculations with respect to τ we get

$$\begin{aligned} \int_0^\tau d\tau' \mathbf{k} \cdot \mathbf{u}(\tau') &= \int_0^\tau d\tau' \omega_0 \rightarrow \\ \mathbf{k} \cdot \mathbf{x}(\tau) - \mathbf{k} \cdot \mathbf{x}(0) &= \tau \omega_0 . \end{aligned}$$

So we find the rather nice result that

$$\mathbf{k} \cdot \mathbf{x}(\tau) = \tau \omega_0 + \mathbf{k} \cdot \mathbf{x}(0) . \quad (6.2.18)$$

Consequently we actually can perform the required integration! For we simply have

$$\begin{aligned} U(\tau) &= 1 + \frac{q}{2m} \int_0^\tau d\tau' F_0 \cos(\mathbf{k} \cdot \mathbf{x}(\tau)) , \\ &= 1 + \frac{q}{2m} \int_0^\tau d\tau' F_0 \cos(\tau' \omega_0 + \mathbf{k} \cdot \mathbf{x}(0)) , \\ &= 1 + \frac{q}{2m\omega_0} F_0 \{\sin(\tau \omega_0 + \mathbf{k} \cdot \mathbf{x}(0)) - \sin(\mathbf{k} \cdot \mathbf{x}(0))\} . \end{aligned} \quad (6.2.19)$$

This rotor has now been explicitly calculated and can be used to study the motion of charged particles in any electromagnetic planewave. The simplicity of the solution is remarkable, isn't it.

One application for such a solution arises when we want to study the scattering of electromagnetic waves on free electrons. In that case the particles are assumed to move as specified by the rotor we just calculated. This result is then used to calculate the radiation emitted by the electron due to its motion under the influence of the electromagnetic wave. We will do this computation in a later chapter after we have figured out how moving charges produce radiation. This process is called *Thompson scattering*.

6.2.4. The relativistic Coulomb problem

The *classical* hydrogen atom doesn't exist for real but it is a useful laboratory to test our ideas on the classical dynamics of charged particles. Next to that, the motion of an electron bound to a proton is similar to that of a mass bound to a much heavier body by the forces of Newtonian gravity. Hence, the classical hydrogen atom can also be seen as a model system to estimate relativistic corrections to planetary motion. These usually turn out to be negligible.

The electron, with charge $-e$, moves in a hydrogen atom under the influence of a vector-potential

$$\mathbf{A}(\mathbf{x}) = \mathbf{e}_0 \frac{kQ}{r} , \quad (6.2.20)$$

which is the so-called *Coulomb potential*. Here Q is the charge of the proton, the constant k depends on your choice of units and in SI it is $\frac{1}{4\pi\epsilon_0}$. Finally r is the distance between the electron and the proton and $\mathbf{e}_0$ is the time-like unitvector of the proton's restframe. As a result of this interaction the electron has a potential energy-momentum given by

$$\mathbf{V}(\mathbf{x}) \equiv -e\mathbf{A}(\mathbf{x}) = -\frac{keQ}{r}\mathbf{e}_0 . \quad (6.2.21)$$

The electromagnetic bivectorfield that the electron experiences is

$$\begin{aligned} F(\mathbf{x}) &= \nabla \wedge \mathbf{A}(\mathbf{x}) , \\ &= \nabla \wedge \mathbf{e}_0 \frac{kQ}{r} , \\ &= \mathbf{e}_0 \wedge \hat{\mathbf{r}} \frac{kQ}{r^2} , \\ &= \mathbf{e}_0 \hat{\mathbf{r}} \frac{kQ}{r^2} , \end{aligned} \quad (6.2.22)$$

where $\hat{\mathbf{r}}$ is the radial-unit vector of a spherical coordinate system around the proton. Now that we have the electromagnetic field, all we need to do is solve the following rotor equation,

$$\frac{d}{d\tau}\Lambda(\tau) = -\mathbf{e}_0 \hat{\mathbf{r}}(\tau) \frac{ekQ}{2mr(\tau)^2} \Lambda(\tau) . \quad (6.2.23)$$

Note that there are two major problems with solving this;

- The bivector F depends on the particle trajectory;
- The bivectors $F(\mathbf{x}(\tau))$ at different proper times τ on the worldline probably do not commute as in general $\hat{\mathbf{r}}(\tau_1) \neq \hat{\mathbf{r}}(\tau_2)$ if $\tau_1 \neq \tau_2$.

Is there any hope?

Again there is a conserved quantity that saves the day; Angular momentum L . It is a bivector quantity that was defined non-relativistically as the wedge between momentum and position. For our purposes we have to adapt this a little. First note that the bivector

$$\vec{r}(\tau) \equiv \mathbf{e}_0 \wedge \mathbf{x}(\tau) , \quad (6.2.24)$$

contains the spacelike part of the electron position. Similarly the bivector

$$\vec{p}(\tau) \equiv \mathbf{e}_0 \wedge \mathbf{p}(\tau) , \quad (6.2.25)$$

contains the spacelike part of the energy-momentum vector of the electron. The bivector L can now be constructed using the commutator,

$$L(\tau) \equiv \frac{1}{2} \{ \vec{r}(\tau) \vec{p}(\tau) - \vec{p}(\tau) \vec{r}(\tau) \} = [\vec{r}(\tau), \vec{p}(\tau)] . \quad (6.2.26)$$

To show that it is conserved all you need to do is differentiate this $L(\tau)$ with respect to proper time and use that

$$\begin{aligned} \frac{d}{d\tau} \vec{p}(\tau) &\propto \vec{r}(\tau) , \\ \frac{d}{d\tau} \vec{r}(\tau) &\propto \vec{p}(\tau) , \end{aligned}$$

making the commutators vanish.

The fact that the bivector $L(\tau)$ is conserved has two important implications:

- the motion of the electron is confined to a plane that we could represent by some unit-bivector $\hat{J}$, $\hat{J}^2 = -1$;
- We can write $L = lJ$ for some constant l .

Now we start to work towards solving our rotor equation. First note that the unit-vector $\hat{\mathbf{r}}(\tau)$ can also be expressed as a function of an angle θ between $\hat{\mathbf{r}}(\tau)$ and, for example, the $\mathbf{e}_1$ -direction. Obviously if we write it like that θ will depend on τ . As $\hat{\mathbf{r}}(\theta)$ is a unit-vector it can be obtained from $\mathbf{e}_1$ by means of a rotation over the angle $\theta(\tau)$. So, there is a rotor $U(\theta)$

$$U(\theta(\tau)) = \exp\left(\frac{1}{2}\theta(\tau)\hat{J}\right) , \quad (6.2.27)$$

such that

$$\begin{aligned} \hat{\mathbf{r}}(\theta) &= U(\theta)\mathbf{e}_1\tilde{U}(\theta) , \\ &= U(\theta)^2\mathbf{e}_1 , \\ &= \exp(\theta(\tau)\hat{J}) , \end{aligned} \quad (6.2.28)$$

as the motion occurs in the plane specified by $\hat{J}$, a plane that contains the $\mathbf{e}_1$, this rotation is also generated by that bivector.

Notice how easy this argumentation is because of the geometric meaning of these bivectors!. Anyway, we proceed, using that

$$\mathbf{p}(\tau) = m\mathbf{u}(\tau) = m\frac{d}{d\tau}\mathbf{x}(\tau) , \quad (6.2.29)$$

we find that

$$\begin{aligned}\vec{p}(\tau) &= m\mathbf{e}_0 \wedge \frac{d}{d\tau}\mathbf{x}(\tau) , \\ &= m\frac{d}{d\tau}\vec{r}(\tau) .\end{aligned}$$

Now using the above we also have

$$\begin{aligned}\frac{d}{d\tau}\vec{r}(\tau) &= \frac{d}{d\tau}\{r(\tau)\mathbf{e}_0\hat{\mathbf{r}}(\tau)\} , \\ &= \left\{\frac{d}{d\tau}r(\tau)\right\}\mathbf{e}_0\hat{\mathbf{r}}(\tau) + r(\tau)\mathbf{e}_0\frac{d}{d\tau}\hat{\mathbf{r}}(\tau) , \\ &= \left\{\frac{d}{d\tau}r(\tau)\right\}\mathbf{e}_0\hat{\mathbf{r}}(\tau) + r(\tau)\mathbf{e}_0\hat{J}\hat{\mathbf{r}}(\tau)\frac{d}{d\tau}\theta(\tau) .\end{aligned}\quad (6.2.30)$$

If we now compute the angular momentum bivector by plugging these results in its definition we find,

$$L = l\hat{J} = mr(\tau)^2\frac{d}{d\tau}\theta(\tau)\hat{J} .\quad (6.2.31)$$

As a result we find that

$$mr(\theta)^2\frac{d}{d\tau}\theta(\tau) = l ,\quad (6.2.32)$$

and this is going to be so usefull in solving the rotor equation. Because, let us turn back to it now, it read

$$\begin{aligned}\frac{d}{d\tau}\Lambda(\tau) &= -\mathbf{e}_0\hat{\mathbf{r}}(\tau)\frac{ekQ}{2mr(\tau)^2}\Lambda(\tau) , \rightarrow \\ \left\{\frac{d}{d\tau}\theta(\tau)\right\}\frac{d}{d\theta}\Lambda(\theta) &= -\mathbf{e}_0\hat{\mathbf{r}}(\theta)\frac{ekQ}{2mr(\theta)^2}\Lambda(\theta) , \rightarrow \\ \frac{d}{d\theta}\Lambda(\theta) &= -\mathbf{e}_0\hat{\mathbf{r}}(\theta)\frac{ekQ}{2mr(\theta)^2\frac{d}{d\tau}\theta(\tau)}\Lambda(\theta) , \rightarrow \\ \frac{d}{d\theta}\Lambda(\theta) &= -\mathbf{e}_0\hat{\mathbf{r}}(\theta)\frac{ekQ}{2l}\Lambda(\theta) , \rightarrow \\ \frac{d}{d\theta}\Lambda(\theta) &= -\frac{ekQ}{2l}\mathbf{e}_0U(\theta)\mathbf{e}_1\tilde{U}(\theta)\Lambda(\theta) , \\ &= -\frac{ekQ}{2l}U(\theta)\mathbf{e}_0\mathbf{e}_1\tilde{U}(\theta)\Lambda(\theta) .\end{aligned}\quad (6.2.33)$$

Now this last equation looks a lot simpler. All the τ -dependence has been traded in for θ -dependence and, what's more, practically all reference to the trajectory of the electron has vanished from the rotor equation. One final step and we are done!

If we insert the *Ansatz* $\Lambda(\theta) = U(\theta)S(\theta)$ into the rotor equation we get an equation for $S(\theta)$ that reads

$$\frac{d}{d\theta}S(\theta) = \frac{1}{2}\{\kappa\mathbf{e}_0\mathbf{e}_1 - \hat{J}\}S(\theta) ,\quad (6.2.34)$$

$$\kappa = \frac{eQk}{l} .\quad (6.2.35)$$

This is a rotor equation with a constant bivector, hence this is something we can solve! The solution simply is,

$$S(\theta) = \exp\left(\frac{1}{2}(\kappa\mathbf{e}_0\mathbf{e}_1 - \hat{J})\theta\right), \quad (6.2.36)$$

and consequently the whole rotor becomes,

$$\Lambda(\theta) = \exp\left(\frac{1}{2}\theta(\tau)\hat{J}\right)\exp\left(\frac{1}{2}(\kappa K_1 - \hat{J})\theta\right). \quad (6.2.37)$$

Let us briefly discuss this result. Notice that the second rotor, $S(\theta)$, can be *boostlike* or rotation-like when

$$\begin{aligned} \left(\frac{1}{2}(\kappa\mathbf{e}_0\mathbf{e}_1 - \hat{J})\right)^2 &> 0, \text{ resp.} \\ \left(\frac{1}{2}(\kappa\mathbf{e}_0\mathbf{e}_1 - \hat{J})\right)^2 &< 0. \end{aligned}$$

If we work out this condition we find

$$\begin{aligned} \frac{eQk}{l} &> 1, \text{ resp.} \\ \frac{eQk}{l} &< 1. \end{aligned}$$

So the type of orbit is determined by the relationship between angular momentum and the strength of the interaction in a very direct manner.

We have obtained an expression for the eigen rotor of the electron as a function of the angle θ . As we do not know how θ depends on τ we have not completely solved the problem yet. Yet the proper time dependence of *theta* is restricted through the conservation of angular momentum, and the conservation of energy in the proton restframe. For the energy of the electron we have

$$E = mc^2\gamma(\theta(\tau)) - \frac{keQ}{r(\theta(\tau))}, \quad (6.2.38)$$

and this equation will allow one, by specifying the energy E and angular momentum l to solve for $r(\theta)$ which can then be used alongside with the rotor we just found to compute the worldline as well as $\theta(\tau)$. This process involves a few nasty integrals that require more elaborations that we can do here.

6.3. Fermi-Walker transport

The rotor equation allows us to calculate the eigen-rotor of a particle once we know the forces acting on the particle. We have also seen that the equations of motion from which we derived the rotor-equation can be interpreted as the result of a *extremization* principle: the principle of maximum action under end-point preserving variations of the worldline. In a couple of examples we have seen how the rotor-equation can be solved. But we have not fully revealed how the variational principle, which fixes the equation for the worldline, and the eigen-rotor, which

also determines the worldline but in a very different way, are tied together. The cause of this is our lack in properly understanding just how important the eigen-rotor is. In effect, it carries much more information about the particle than just what $\mathbf{u}(\tau)$ is in terms of $\mathbf{u}(0)$.

In this section we will see how we can use the eigen-rotor to specify not only $\mathbf{u}(\tau)$, but in fact a whole orthonormal frame $\mathbf{g}_\mu(\tau)$ carried along by the particle; its so-called *restframe*.

Subsequently we will study the inclusion of a new *dynamical* variable called *spin* in the following sections. Historically spin was not discovered until after the advent of quantum mechanics. In many cases one can view spin as an *angular momentum* representing some kind of internal rotational motion. However, to this day, no fully satisfactory picture of the spin degree of freedom has emerged. In a number of situations we used the term *restframe* of a particle. We know what the restframe looks like as seen from the restframe, but what does it look like from the outside? Let us call this rest-frame carried along by the moving particle $\mathbf{g}_\mu(\tau)$ and try to derive the equations that this frame satisfies.

Consider a particle moving with a proper velocity $\mathbf{u}(\tau)$ and recall that when

$$\mathbf{u}(\tau) = \Lambda(\tau)\mathbf{u}(0)\tilde{\Lambda}(\tau) ,$$

then

$$\mathbf{a}(\tau) = \frac{d}{d\tau}\mathbf{u}(\tau) = [\Omega(\tau), \mathbf{u}(\tau)] .$$

Now I would like to express $\Omega(\tau)$ directly in terms of $\mathbf{u}(\tau)$ and $\mathbf{a}(\tau)$ pretending I already know these. We get from the equations above

$$\begin{aligned} \mathbf{u}(\tau) \wedge \mathbf{a}(\tau) &= [\mathbf{u}(\tau), [\Omega(\tau), \mathbf{u}(\tau)]] \\ &= \frac{1}{2}(c^2\Omega(\tau) - \mathbf{u}(\tau)\Omega(\tau)\mathbf{u}(\tau)) . \end{aligned}$$

Using the fact that we know that $\mathbf{u}(\tau)$ and $\mathbf{a}(\tau)$ are perpendicular, and thus that $\mathbf{u}(\tau) \wedge \mathbf{a}(\tau) = \mathbf{u}(\tau)\mathbf{a}(\tau)$ it is easy to check that the following expression for $\Omega(\tau)$ would satisfy this equation

$$\Omega(\tau) = \mathbf{u}(\tau)\mathbf{a}(\tau) . \quad (6.3.1)$$

As $\mathbf{u}(\tau)$ is by definition proportional to the timelike unit-vector in the rest-frame $\mathbf{u}(\tau) = c\mathbf{g}_0(\tau)$ we find that this $\Omega(\tau)$ is a pure timelike bivector. As a result it generates a pure boost in the restframe of the moving particle. In other words: the bivector $\mathbf{u}(\tau)\mathbf{a}(\tau)$ representing the acceleration in the restframe of the moving particle represents a pure acceleration without adding any rest-frame rotation. If we define the restframe as the frame of the particle that is always boosted from one proper-time to another, i.e. does not contain any arbitrary restframe rotations, then these restframe vectors $\mathbf{g}_\mu(\tau)$ will satisfy

$$\frac{d}{d\tau}\mathbf{g}_\mu(\tau) = [\mathbf{u}(\tau)\mathbf{a}(\tau), \mathbf{g}_\mu(\tau)] . \quad (6.3.2)$$

We call this: the restframe is *Fermi-Walker transported* along the worldline. It is often useful to define the so-called *Fermi-derivative* along the worldline as

$$D_F\mathbf{b}(\tau) \equiv \frac{d}{d\tau}\mathbf{b}(\tau) - [\mathbf{u}(\tau)\mathbf{a}(\tau), \mathbf{b}(\tau)] . \quad (6.3.3)$$

The Fermi derivative of a vector represents the rate of change of the vector that remains after subtracting the change induced by the boost to velocity at $\tau + d\tau$. A vector $\mathbf{b}(\tau)$ that is Fermi-Walker transported along a worldline satisfies

$$D_F \mathbf{b}(\tau) = 0 . \quad (6.3.4)$$

If the particle carries additional *vector*-degrees of freedom, and if the dynamics of those degrees of freedom in the restframe is non-trivial then this should show up as deviations from the Fermi-Walker transport equation.

6.4. The Spin degree of freedom

An eigen-rotor expanded in terms of bivectors of an orthonormal basis has *six* independent components. We say it has six degrees of freedom. The time-evolution of $\mathbf{u}(\tau)$ requires three degrees of freedom due to the fact that $\mathbf{u}(\tau)^2 = c^2$. So this leaves three degrees of freedom in the eigen-rotor undetermined. By requiring Fermi-Walker transport along the worldline we, in a sense arbitrarily, fix the remaining three degrees of freedom by requiring that the acceleration bivector be a pure boost generator. Let's first see what this looks like for a charged particle in an electromagnetic field.

Fermi-Walker transport and charged particles

Let us consider a particle of charge q in an electromagnetic field described by the bivector field F . This particle follows a worldline described by an eigen-rotor $\Lambda(\tau)$. From the equations in the previous section we see that the proper acceleration of this particle is given by

$$\mathbf{a}(\tau) = \frac{q}{m} [F, \mathbf{u}(\tau)] , \quad (6.4.1)$$

We compute

$$\mathbf{u}(\tau) \mathbf{a}(\tau) = \frac{q}{m} E_u , \quad (6.4.2)$$

where we have introduced the electric E_u and B_u magnetic fields relative to the particle's restframe

$$E_u = \frac{1}{2} \{ F - \hat{\mathbf{u}}(\tau) F \hat{\mathbf{u}}(\tau) \} , \quad (6.4.3)$$

$$B_u = \frac{1}{2} \{ F + \hat{\mathbf{u}}(\tau) F \hat{\mathbf{u}}(\tau) \} . \quad (6.4.4)$$

So the equation for a vector that is Fermi-Walker transported along with this charged particle is

$$\frac{d}{d\tau} \mathbf{b}(\tau) - \frac{q}{m} [E_u, \mathbf{b}(\tau)] = 0 . \quad (6.4.5)$$

Spin, at last

Now elementary particles, such as electrons and quarks, carry another physically interesting quantity along with them, called their *spin*. Usually it appears in the equations of quantum mechanics as a form of *intrinsic angular momentum*. As angular momentum is a *spacelike* bivector quantity in the restframe, and so is spin S . However, such a spacelike bivector can be written in terms of a spacelike vector

$\mathbf{s}$ through

$$S = \frac{\hbar i}{2c} \mathbf{u}(\tau) \mathbf{s}(\tau) , \text{ where} \quad (6.4.6)$$

$$\mathbf{s}(\tau) \cdot \mathbf{u}(\tau) = 0 . \quad (6.4.7)$$

The $\hbar$ is just the quantum measure of its angular momentum, whereas $\mathbf{s}(\tau)$ is a spacelike unit-vector, $\mathbf{s}(\tau)^2 = -1$. In its restframe the spinning charged particle generates a magnetic moment that interacts with the external magnetic field B_u . This interaction typically is of the form

$$\frac{gq}{2m} [B_u, \mathbf{s}(\tau)] , \quad (6.4.8)$$

where the number g is called the *gyromagnetic ratio* and fixes the strength with which the spin produces a magnetic moment. For non-elementary particles the value of this g depends on the details of their internal structure.

Due to its interaction with the external magnetic field, the spinvector is deviates from the Fermi-Walker transport behaviour. It *does* rotate in particle's restframe. This rotation is described by

$$\begin{aligned} D_F \mathbf{s}(\tau) &= \frac{gq}{2m} [B_u, \mathbf{s}(\tau)] , \\ \frac{d}{d\tau} \mathbf{s}(\tau) + \frac{q}{m} [\mathbf{s}(\tau), E_u] &= \frac{gq}{2m} [B_u, \mathbf{s}(\tau)] , \\ \frac{d}{d\tau} \mathbf{s}(\tau) &= \frac{q}{m} [E_u, \mathbf{s}(\tau)] + \frac{gq}{2m} [B_u, \mathbf{s}(\tau)] , \\ &= \frac{q}{m} [F, \mathbf{s}(\tau)] + \frac{q}{2m} (g - 2) [B_u, \mathbf{s}(\tau)] . \end{aligned} \quad (6.4.9)$$

The last line in this equation is called the *Telegdi-Michel-Bargmann equation*. It describes the dynamics of the spin of a charged particle in an external electromagnetic field.

Note that for the very special choice of $g = 2$ this equation becomes equivalent to the one for $\mathbf{u}(\tau)$, i.e.

$$\frac{d}{d\tau} \mathbf{s}(\tau) = \frac{q}{m} [F, \mathbf{s}(\tau)] , \quad g = 2 . \quad (6.4.10)$$

This was long considered to be a quantum mechanical prediction. However, our *classical* equation also points out $g = 2$ as the most natural value. The values for g found experimentally for the electron and the muon are approximately $g = 2.0023$, though it is known up to a far greater accuracy. What this shows is that g is remarkably close to 2. The difference can be explained as a consequence of *quantum fluctuations* in the quantised electromagnetic field. Protons have a g which is very different (≈ 5.59) but that is due to their *finite-size* and their *internal structure*. The three charged quarks making up protons have $g \approx 2$ again. Similarly, the neutron has a $g \approx 3.83$ yet being a neutral particle it seems odd that the neutron interacts with a magnetic field at all! Again, the solution of this mystery is the internal structure of the neutron in terms of charged quarks with ... $g = 2$. Hence, $g = 2$ seems the appropriate classical value for elementary particles.

There is something wonderful about the above equation. If we would write the spin-vector $\mathbf{s}(\tau)$ as

$$\mathbf{s}(\tau) = \Psi(\tau)\mathbf{s}(0)\tilde{\Psi}(\tau) , \quad (6.4.11)$$

then this rotor $\Psi(\tau)$ satisfies the same rotor equation, as the one we introduced for the proper velocity, i.e.

$$\mathbf{u}(\tau) = \Lambda(\tau)\mathbf{u}(0)\tilde{\Lambda}(\tau) .$$

Because $\Psi(\tau)$ and $\Lambda(\tau)$ satisfy the same rotor-equation, and because $\mathbf{s}(\tau)$ is always perpendicular to $\mathbf{u}(\tau)$, we may in fact set $\Psi(\tau) = \Lambda(\tau)$. Solving the rotor equation for elementary charged particles amounts to finding the time-evolution of *both* spin and proper velocity in a single go. Two physical quantities for the prize of one! In the next chapter you can find two examples of how the use of eigen-rotors greatly facilitates the study of the dynamics of spinning particles in electromagnetic fields.

6.4.1. Spin precession

Our equations not only allow the solution of problems involving the motion of charged particles, but also those concerning the spin of $g = 2$ particles in external fields.

Consider a particle with spin that we fix at some position, immersed in a constant magnetic field. Although the particle cannot move its spin may interact with an external magnetic field $F = B$. The rotor $\Psi(t)$ determining the time-evolution of the spinvector $\mathbf{s}(t)$ thus satisfies

$$\frac{d}{dt}\Psi(t) = -\frac{q}{2m}B\Psi(t) , \quad (6.4.12)$$

and depends on time t as proper time coincides with coordinate time in this example. We know how to solve this equation, so we can immediately write down,

$$\Psi(t) = \exp\left(-\frac{q}{2m}Bt\right)\Psi(0) . \quad (6.4.13)$$

If we use that the initial spin is

$$\mathbf{s}(0) = \Psi(0)\mathbf{e}_3\tilde{\Psi}(0) , \quad (6.4.14)$$

we get

$$\mathbf{s}(t) = \exp\left(-\frac{q}{2m}Bt\right)\mathbf{s}(0)\exp\left(\frac{q}{2m}Bt\right) . \quad (6.4.15)$$

Now define

$$\hat{B} = \frac{B}{\sqrt{-B^2}} , \quad (6.4.16)$$

and note that B^2 is negative! Now we may split the initial spin into a part that is parallel with B and a part that is perpendicular to B ,

$$\begin{aligned} \mathbf{s}(0)_{//} &= \frac{1}{2}\{\mathbf{s}(0) + \hat{B}\mathbf{s}(0)\hat{B}\} , \\ \mathbf{s}(0)_{\perp} &= \frac{1}{2}\{\mathbf{s}(0) - \hat{B}\mathbf{s}(0)\hat{B}\} . \end{aligned}$$

Check explicitly whether these equations are correct! Now we can write down the final result,

$$\mathbf{s}(t) = \mathbf{s}(0)_{//} + \exp\left(-\frac{q}{m}Bt\right)\mathbf{s}(0)_{\perp} . \quad (6.4.17)$$

This equation clearly exhibits the fact that a spin will precess around the direction of the magnetic field with an angular frequency $\omega_B = \frac{q|B|}{m}$.

6.4.2. Magnetic Resonance of fixed Spins

The spin of a charged particle responds to the external magnetic field. Consequently if this field is time-dependent it may also be driven by this field. In this example we will study such a situation that is common in NMR imaging problems. Suppose we have a time-dependent magnetic field,

$$F(t) = B(t) = b_1\{\cos(\omega t)J_1 + \sin(\omega t)J_2\} + b_0J_3 . \quad (6.4.18)$$

This field precesses around the J_3 direction and thus will drive spins coupling to it to do the same. But let us see how that works exactly.

The rotor equation that we need to solve is

$$\frac{d}{dt}\Psi(t) = -\frac{q}{2m}B(t)\Psi(t) . \quad (6.4.19)$$

This is facilitated by rewriting the time-dependence of the magnetic field as follows

$$\begin{aligned} B(t) &= \exp\left(-\frac{\omega t}{2}J_3\right)\{b_1J_1 + b_0J_3\}\exp\left(\frac{\omega t}{2}J_3\right) , \\ &\equiv S(t)B(0)\tilde{S}(t) . \end{aligned} \quad (6.4.20)$$

This allows us to rewrite the rotor equation as

$$\frac{d}{dt}\tilde{S}(t)\Psi(t) = \left\{\frac{\omega}{2}J_3 - \frac{q}{2m}B(0)\right\}\tilde{S}(t)\Psi(t) . \quad (6.4.21)$$

The product $\tilde{S}(t)\Psi(t)$ satisfies a rotor equation with a time-independent bivector as a pre-factor, so it can be solved immediately as

$$\tilde{S}(t)\Psi(t) = \exp\left[-\frac{t}{2}\left(\frac{q}{m}B(0) - \omega J_3\right)\right]\Psi(0) , \quad (6.4.22)$$

and so we finally find

$$\Psi(t) = \exp\left(-\frac{\omega t}{2}J_3\right)\exp\left[-\frac{t}{2}\left(\frac{q}{m}B(0) - \omega J_3\right)\right]\Psi(0) . \quad (6.4.23)$$

Note that the motion of the spins contains three different frequencies. Next to the driving frequency ω we have

$$\begin{aligned} \omega_0 &= \frac{qb_0}{m} , \\ \omega_1 &= \frac{qb_1}{m} . \end{aligned}$$

The motion of the spin vector can be rather complicated despite the apparent simplicity of the rotor expression derived here.

Chapter 7

Spin, Symmetry and Extremal Worldlines

In the previous chapter we have seen how the worldlines of charged particles in electromagnetic fields can be found, even including the time-evolution of the particle's spin. The rotor-equation was derived from the spacetime generalization of Newton's second law yet it contained an unexpected additional degree of freedom: Spin. This is very suggestive of a point of view in which the rotor-equation is actually more fundamental than Newton's second law. In this chapter we will see that we can indeed consistently hold that view. As a result we will not only discover the foundation of Newton's laws but the expedition also takes us onto the brink of quantum mechanics.

The first thing we need to do is study the properties of integrals along lines, for example worldlines, in Spacetime. So we start with that.

7.1. Line-integrals

A line is a one-dimensional subset of Spacetime, so we need only one parameter to describe it. Let the parametrization of the points on a line l be given as $\mathbf{x}(z)$, where z is some parameter. For worldlines this parameter is most suitably chosen to be the proper time τ , but in principle the choice is arbitrary. We then define the tangent vector $g(z)$ as

$$\mathbf{g}(z) = \frac{d}{dz}\mathbf{x}(z) . \quad (7.1.1)$$

So an infinitesimally small piece of line $d\mathbf{x}$ can be written as

$$d\mathbf{x} = dz\mathbf{g}(z) . \quad (7.1.2)$$

Partial derivatives are not required here as the parametrization contains a single parameter. An integral of some function $f(\mathbf{x})$ *along the line* l can now be written as

$$\int_l d\mathbf{x} f(\mathbf{x}) \equiv \int dz \mathbf{g}(z) f(\mathbf{x}(z)) . \quad (7.1.3)$$

It is as simple as this. There are a few special cases that we can discuss.

7.1.1. Length of a line in Spacetime

Let us see how to find the *length* of the line. Obviously every increment dz of the parameter contributes a spacetime length $dz\sqrt{\mathbf{g}(z) \cdot \mathbf{g}(z)}$ to the total length of the line. Take care that in spacetime these tangents vectors can have negative squares and so the square root may become imaginary! Modulo these sorts of problems we would write

$$|l| \equiv \int dz\sqrt{\mathbf{g}(z) \cdot \mathbf{g}(z)}, \text{timelikecurves} , \quad (7.1.4)$$

$$= \int dz\sqrt{-\mathbf{g}(z) \cdot \mathbf{g}(z)}, \text{spacelikecurves} . \quad (7.1.5)$$

It is not possible to cover curves which have timelike and spacelike sections in a single definition.

7.1.2. Line-integral of a vectorfield

If we integrate a vectorfield $\mathbf{b}(\mathbf{x})$ along the line l we get an integral which quite naturally falls apart into two different portions

$$\begin{aligned} \int_l d\mathbf{x}\mathbf{b}(\mathbf{x}) &= \int dz\mathbf{g}(z)\mathbf{b}(\mathbf{x}(z)) , \\ &= \int dz\mathbf{g}(z) \cdot \mathbf{b}(\mathbf{x}(z)) + \int dz\mathbf{g}(z) \wedge \mathbf{b}(\mathbf{x}(z)). \end{aligned}$$

The scalar part of such a line-integral is encountered quite regularly in physics, the bivector-part is seen rarely. However, this is only partially reflecting its *applicability* as we are simply less intuitively attached to bivectors than we are to scalars.

7.1.3. Integral of a derivative

Suppose we have a line in spacetime $\mathbf{x}(z^1)$ with the parameter z^1 running from 0 to 1, and a function $f(\mathbf{x})$. There is a wellknown relation ship between the values of this function and the end-points of the fline, i.e. $f(\mathbf{x}(0))$ and $f(\mathbf{x}(1))$, and the behaviour of $f(\mathbf{x})$ along the line. It is easy to see that

$$\begin{aligned} f(\mathbf{x}(1)) - f(\mathbf{x}(0)) &= \int_0^1 dz^1 \frac{d}{dz^1} f(\mathbf{x}(z^1)) , \\ &= \int_0^1 \{dz^1 \mathbf{g}_1(\mathbf{x}(z)) \cdot \nabla\} f(\mathbf{x}(z^1)) , \\ &= \int_l \{d\mathbf{x} \cdot \nabla\} f(\mathbf{x}) . \end{aligned} \quad (7.1.6)$$

In this last statement any reference to the parametrization has been removed. This is the proper thing to do as the value of the integral does not depend on the parametrisation! An interesting question is however: how do the integrals depend on the lines along which the integrations are done? It may seem strange, but you can indeed say something sensible about that!

7.2. Variations of Line-integrals

Now suppose that we keep the end-points $\mathbf{x}(0)$ and $\mathbf{x}(1)$ of a line-integral fixed but we ask the question: *what is the extremal line connecting these end-points ?* With *extremal* we refer to the line being either the longest or the shortest. Due to the fact that length-squared in Spacetime can be negative, or even zero, it is not always easy to say a priori whether a certain line is long or short. But it will be no problem to think about *extremal* lines.

We would call a line $\mathbf{x}(z)$ *extremal* when a small variation on that line, leaving the end-points fixed, i.e.

$$\begin{aligned} \mathbf{x}(z) &\rightarrow \mathbf{x}(z) + \delta\mathbf{h}(z) , \\ \delta\mathbf{h}(z_0) &= \delta\mathbf{h}(z_1) = 0 , \end{aligned} \tag{7.2.1}$$

leaves the length of the worldline unaffected, i.e.

$$\delta|l| = 0 . \tag{7.2.2}$$

Such a line $\mathbf{x}(z)$ represents a line of maximum or minimum spacetime length. Can we actually compute such a line from this requirement? The answer is: yes! If we use the definition of the line with the small variation $\delta\mathbf{h}(z)$ to compute a new tangent vector we get

$$\mathbf{g}'_z(z) = \mathbf{g}_z(z) + \frac{d}{dz}\delta\mathbf{h}(z) .$$

As a result the integral becomes

$$\begin{aligned} |l'| &= \int_0^1 dz \sqrt{|\mathbf{g}'_z(z) \cdot \mathbf{g}'_z(z)|} , \\ &= \int_0^1 dz \sqrt{|\mathbf{g}_z(z)^2 + 2\left[\frac{d}{dz}\delta\mathbf{h}(z)\right] \cdot \mathbf{g}_z(z) + \left[\frac{d}{dz}\delta\mathbf{h}(z)\right]^2|} \end{aligned}$$

Now we assume that the variation is very small indeed, so we can apply a Taylor-approximation to the squareroot. If, for definiteness we assume that the line is a timelike line, i.e. a worldline of some particle, then we find from the Taylor-approximation

$$\sqrt{|\mathbf{g}_z(z)^2 + 2\left[\frac{d}{dz}\delta\mathbf{h}(z)\right] \cdot \mathbf{g}_z(z) + \left[\frac{d}{dz}\delta\mathbf{h}(z)\right]^2|} = \sqrt{|\mathbf{g}_z(z)^2|} + \frac{\left[\frac{d}{dz}\delta\mathbf{h}(z)\right] \cdot \mathbf{g}_z(z)}{\sqrt{|\mathbf{g}_z(z)^2|}} .$$

So the difference in the value between the two lines connecting $\mathbf{x}(z_0)$ and $\mathbf{x}(z_1)$ is

$$\begin{aligned} \delta|l| &= |l'| - |l| = \int_0^1 dz \frac{\left[\frac{d}{dz}\delta\mathbf{h}(z)\right] \cdot \mathbf{g}_z(z)}{\sqrt{|\mathbf{g}_z(z)^2|}} , \\ &= - \int_0^1 dz \delta\mathbf{h}(z) \cdot \left[\frac{d}{dz} \frac{\mathbf{g}_z(z)}{\sqrt{|\mathbf{g}_z(z)^2|}}\right] . \end{aligned}$$

For a (world)line of extremal length this integral would need to vanish for any arbitrary variation $\delta \mathbf{h}(z)$ that keeps the end-points fixed. Such a worldline thus should satisfy the equation

$$\frac{d}{dz} \frac{\mathbf{g}_z(z)}{\sqrt{\mathbf{g}_z(z)^2}} = 0 .$$

Suppose we are really dealing with the worldline of a physical particle. Then, as we saw earlier, we would parametrize the worldline by the proper-time τ and the proper-velocity $\mathbf{u}(\tau)$ defined as

$$\mathbf{u}(\tau) = \frac{d}{d\tau} \mathbf{x}(\tau) ,$$

is a tangent-vector along the worldline satisfying $\mathbf{u}(\tau)^2 = 1$. So for the worldline of classical particle we would find

$$\frac{d}{d\tau} \mathbf{u}(\tau) = 0 , \quad (7.2.3)$$

as the equation determining the *extremal* worldline. Clearly this is the equation for a straight line in spacetime, and the equation for a *free* particle moving without acceleration.

Problem 115: Differentiating with respect to vectors

Let $\mathbf{b}$ be some vector with the components b_μ in the orthonormal basis $\mathbf{e}_\mu$.

a: Show that

$$\nabla_{\mathbf{b}} = \sum_{\mu=0}^3 \mathbf{e}^\mu \frac{\partial}{\partial b_\mu} ,$$

is a derivative for which $\nabla_{\mathbf{b}} \mathbf{b} = 4$.

b: Show that if $L[\mathbf{b}]$ is a linear map with the components L_ν^μ then we have

$$\nabla_{\mathbf{b}} L[\mathbf{b}] = \sum_{\mu=0}^3 L_\mu^\mu .$$

c: Show that if the length of a line $\mathbf{x}(z)$ is given by $|l| = \int_{z_0}^{z_1} dz \sqrt{\mathbf{g}_z(z)^2}$, and a small variation $\delta \mathbf{h}(z)$ is applied, then the variation of its length can be written as

$$\delta |l| = \int_{z_0}^{z_1} dz \delta \mathbf{h}(z) \cdot \left[-\frac{d}{dz} \nabla_{\mathbf{g}_z(z)} \sqrt{\mathbf{g}_z(z)^2} \right] .$$

7.2.1. Line-integrals of Vectorfields

If we integrate a vectorfield $\mathbf{b}(\mathbf{x})$ along the line l we get an integral which quite naturally falls apart into two different portions

$$\begin{aligned} \int_l \mathbf{b}(\mathbf{x}) &= \int_{z_0}^{z_1} dz \mathbf{g}_z(z) \mathbf{b}(\mathbf{x}(z)) , \\ &= \int_{z_0}^{z_1} dz \mathbf{g}_z(z) \cdot \mathbf{b}(\mathbf{x}(z)) + \int_{z_0}^{z_1} dz \mathbf{g}_z(z) \wedge \mathbf{b}(\mathbf{x}(z)) . \end{aligned}$$

The physical or geometric interpretation of these integrals depends on the particular context in which they arise. Standard convention is however to call the scalar-part of the integral above *the* line-integral of $\mathbf{b}(\mathbf{x})$ along l .

Now suppose that we want to compute how the value of the line-integral of $\mathbf{b}(\mathbf{x})$ changes when we vary the line l by a very small amount but *without changing the end-points*. So suppose a second line l' is also described by the parameter z but through

$$\mathbf{x}(z) + \delta\mathbf{h}(z) ,$$

where the vector $\delta\mathbf{h}(z) \approx 0$ between the end-points and $\delta\mathbf{h}(z_1) = \delta\mathbf{h}(z_0) = 0$. The tangent-vector along l' becomes

$$\mathbf{g}'_z(z) = \mathbf{g}_z(z) + \frac{d}{dz}\delta\mathbf{h}(z) ,$$

and so we find for the line-integral of $\mathbf{b}(\mathbf{x})$ along the new line l'

$$\int_{l'} \cdot \mathbf{b}(\mathbf{x}) = \int_{z_0}^{z_1} dz \mathbf{g}'_z(z) \cdot \mathbf{b}(\mathbf{x}(z) + \delta\mathbf{h}(z)) .$$

Now we use that $\delta\mathbf{h}(z)$ is assumed to be very small indeed, so that we may approximate

$$\mathbf{b}(\mathbf{x}(z) + \delta\mathbf{h}(z)) \approx \mathbf{b}(\mathbf{x}(z)) + (\delta\mathbf{h}(z) \cdot \nabla_{\mathbf{x}(z)})\mathbf{b}(\mathbf{x}(z)) .$$

You may recognize this as a first-order Taylor approximation in $\delta\mathbf{h}(z)$. Inserting this into the integral yields

$$\int_{z_0}^{z_1} dz \mathbf{g}'_z(z) \cdot \mathbf{b}(\mathbf{x}(z)) + \int_{z_0}^{z_1} dz \mathbf{g}'_z(z) \cdot (\delta\mathbf{h}(z) \cdot \nabla_{\mathbf{x}(z)})\mathbf{b}(\mathbf{x}(z)) .$$

Now we insert the expression for $\mathbf{g}'_z(z)$ to obtain

$$= \int_{z_0}^{z_1} dz \mathbf{g}_z(z) \cdot \mathbf{b}(\mathbf{x}(z)) + \int_{z_0}^{z_1} dz \left[\frac{d}{dz} \delta\mathbf{h}(z) \right] \cdot \mathbf{b}(\mathbf{x}(z)) + \int_{z_0}^{z_1} dz (\delta\mathbf{h}(z) \cdot \nabla_{\mathbf{x}(z)}) [\mathbf{g}_z(z) \cdot \mathbf{b}(\mathbf{x}(z))] .$$

when we neglect terms that are quadratic in $\delta\mathbf{h}(z)$ and use that $\mathbf{g}_z(z)$ does not depend on $\mathbf{x}$. The variation of the line-integral under a small variation of the path is

$$\begin{aligned} \delta \int_l \cdot \mathbf{b}(\mathbf{x}) &= \int_{l'} \cdot \mathbf{b}(\mathbf{x}) - \int_l \cdot \mathbf{b}(\mathbf{x}) , \\ &= \int_{z_0}^{z_1} dz \left[\frac{d}{dz} \delta\mathbf{h}(z) \right] \cdot \mathbf{b}(\mathbf{x}(z)) + \int_{z_0}^{z_1} dz (\delta\mathbf{h}(z) \cdot \nabla_{\mathbf{x}(z)}) [\mathbf{g}_z(z) \cdot \mathbf{b}(\mathbf{x}(z))] , \\ &= \int_{z_0}^{z_1} dz \delta\mathbf{h}(z) \cdot \left[-\frac{d}{dz} \mathbf{b}(\mathbf{x}(z)) + \nabla_{\mathbf{x}(z)} \mathbf{g}_z(z) \cdot \mathbf{b}(\mathbf{x}(z)) \right] , \\ &= \int_{z_0}^{z_1} dz \delta\mathbf{h}(z) \cdot [-(\mathbf{g}_z(z) \cdot \nabla) \mathbf{b}(\mathbf{x}(z)) + \nabla_{\mathbf{x}(z)} (\mathbf{g}_z(z) \cdot \mathbf{b}(\mathbf{x}(z)))] . \end{aligned}$$

First of all note that what we have here is the commutator of a tangent-vector $\mathbf{g}_z(z)$ with a bivector-field B given by

$$B(\mathbf{x}) \equiv \nabla \wedge \mathbf{b}(\mathbf{x}) . \quad (7.2.4)$$

Problem 116: Check it

Show by doing the algebra that indeed

$$-[\mathbf{g}_z(z), B(\mathbf{x})] = -(\mathbf{g}_z(z) \cdot \nabla_{\mathbf{x}(z)})\mathbf{b}(\mathbf{x}(z)) + \nabla_{\mathbf{x}(z)}(\mathbf{g}_z(z) \cdot \mathbf{b}(\mathbf{x}(z))) . \quad (7.2.5)$$

So

the variation of the line-integral of $\mathbf{b}(\mathbf{x})$ along a line l under infinitesimal but arbitrary variations of this line vanishes, if

$$[\mathbf{g}_z(z), B(\mathbf{x})] = 0 .$$

This obviously gives a restriction on the way the line is allowed to go from starting-to end-point.

Problem 117: Partial integration and boundary-conditions

Show that a crucial step in rewriting the line-integrals above requires integration by parts and the use of the boundary conditions $\delta \mathbf{h}(z_1) = \delta \mathbf{h}(z_0) = 0$.

We can say that $\mathbf{x}(z)$, i.e. l , *extremizes* the value of $\int_l \cdot \mathbf{b}$. We just saw that *straight* lines in Spacetime are those which *maximize* the worldline length. In other relevant geometric or physical problems we may actually be on the look out for worldlines which extremize other line-integrals.

Problem 118: Linearity in Variations

Consider the line-integrals of two function $h(\mathbf{x})$ and $g(\mathbf{x})$ between two end-points $\mathbf{x}(z_0)$ and $\mathbf{x}(z_1)$.

- a: Why do we have $\int_l [g(\mathbf{x}) + h(\mathbf{x})] = \int_l g(\mathbf{x}) + \int_l h(\mathbf{x})$?
- b: Why will we have for the variations $\delta \int_l [g(\mathbf{x}) + h(\mathbf{x})] = \delta \int_l g(\mathbf{x}) + \delta \int_l h(\mathbf{x})$?
- c: Suppose a particle follows a worldline that extremises its length *and* extremises the value of $\int_l \mathbf{b}(\mathbf{x})$. Explain why the proper velocity of this particle must satisfy

$$\frac{d}{d\tau} \mathbf{u}(\tau) + [\mathbf{u}(\tau), B(\mathbf{x})] = 0 . \quad (7.2.6)$$

In

the next chapter we will see that this equation has *great physical significance*.

7.2.2. Line-integral of a gradient

Let us consider the integral of a derivative. In particular the derivative along the line. Suppose we have some function $f(\mathbf{x})$. Using the vector-derivative its gradient $\nabla f(\mathbf{x})$ can be calculated, i.e. the local rate of change of $f(\mathbf{x})$ when we make small changes in spacetime position. The total change measured along the line l is the integral

$$\begin{aligned} \int_l \cdot \nabla f(\mathbf{x}) &\equiv \int_{z_0}^{z_1} dz \mathbf{g}_z(z) \cdot \nabla f(\mathbf{x}(z)) , \\ &= f(\mathbf{x}(z_1)) - f(\mathbf{x}(z_0)) . \end{aligned} \quad (7.2.7)$$

The fact that we can immediately calculate this integral is because

$$\mathbf{g}_z(z) \cdot \nabla = \frac{d}{dz} ,$$

due to the chainrule. This result has two very important consequences:

- A line-integral of a gradient along a *closed* loop vanishes.
- The variation $\delta \int_l \cdot \nabla f$ of a line-integral of a gradient vanishes.

The first result is due to the fact that if there the starting- and end-point of a line are identical, i.e. the line is a closed loop, then $f(\mathbf{x}(x_1)) = f(\mathbf{x}(x_0))$, and so the integral vanishes. The second result follows from the fact that the variation $\delta \int_l \cdot \nabla f$ leaves the end-points of the line unaffected. As a result the value of the integral *does not vary when we vary the path*.

Now the nice thing is the first result is the same as the second result, but in different words. To see this, consider the following: suppose we have the line l and a second line l' that connects the same events $\mathbf{x}(z_1)$ and $\mathbf{x}(z_0)$. The difference between the two integrals can be interpreted written as a *third* line integral along the line l''

$$\int_{l''} \cdot \nabla f(\mathbf{x}) = \int_l \cdot \nabla f(\mathbf{x}) - \int_{l'} \cdot \nabla f(\mathbf{x}) .$$

The line l'' consists of going from $\mathbf{x}(z_0)$ to $\mathbf{x}(z_1)$ via l and then *going back* to $\mathbf{x}(z_0)$ via l' . The line l'' returns to its starting point and thus is a closed loop. The variation of the line-integral is identical to the line-integral along the infinitesimally narrow closed loop $l \odot l'$.

There is yet a third way of seeing this. Recall that we showed earlier that the variation of a line-integral of a vector field $\mathbf{b}(\mathbf{x})$ was proportional to $\nabla \wedge \mathbf{b}(\mathbf{x})$. We also saw earlier that $\nabla \wedge \nabla f(\mathbf{x}) = 0$.

7.3. Spin and Symmetry

Electrodynamics is a subject of great application, but also a window that offers a deep view of some of Nature's most inner workings. Although we will not omit the presentation of applications, we will first take a thorough excursion into important aspects of the origins of electromagnetic interactions. Having studied these will greatly facilitate the transition to quantum theory later on in the course.

We saw that a spinning, elementary, particle with charge q and gyromagnetic ratio $g = 2$, can be described on its motion through spacetime by a proper velocity $\mathbf{u}(\tau)$ and a spin-vector $\mathbf{s}(\tau)$ that satisfy

$$\begin{aligned} \mathbf{u}(\tau) \cdot \mathbf{s}(\tau) &= 0 , \\ m \frac{d}{d\tau} \mathbf{u}(\tau) &= q[F(\mathbf{x}), \mathbf{u}(\tau)] , \\ m \frac{d}{d\tau} \mathbf{s}(\tau) &= q[F(\mathbf{x}), \mathbf{s}(\tau)] . \end{aligned} \tag{7.3.1}$$

As a result there exists a single *eigen-rotor* $\Lambda(\tau)$ such that

$$\begin{aligned} \mathbf{u}(\tau) &= \Lambda(\tau) \mathbf{u}(0) \tilde{\Lambda}(\tau) , \\ \mathbf{s}(\tau) &= \Lambda(\tau) \mathbf{s}(0) \tilde{\Lambda}(\tau) . \end{aligned}$$

This rotor satisfies the rotor-equation

$$\frac{d}{d\tau} \Lambda(\tau) = \frac{q}{2m} F(\mathbf{x}(\tau)) \Lambda(\tau) . \tag{7.3.2}$$

The rotor $\Lambda(\tau)$ contains 6 degrees of freedom because it can be generated from 6 different bivectors. The equations written out above fix *only* 5 of these in terms of $\mathbf{u}(\tau)$ and $\mathbf{s}(\tau)$ from the knowledge of $\Lambda(\tau)$. So there is one degree of freedom left; the rotations that leave the $\mathbf{s}(0)\mathbf{u}(0)$ -plane invariant. These rotations are generated by a bivector called the *spin-plane* bivector

$$S(0) = -\mathbf{s}(0)\mathbf{u}(0)i . \quad (7.3.3)$$

Spin is generally interpreted a manifestation of *intrinsic* angular momentum of elementary particles (an odd concept for a point particle). So we would like to associate some kind of angular rotation with the presence of spin. Suppose we consider a spinning particle in a part of spacetime where there are no electromagnetic fields, i.e. $F(\mathbf{x}) = 0$. The rotor equation would read

$$\frac{d}{d\tau}\Lambda(\tau) = 0 .$$

However, if we take the spin-vector to be the *axis* around which the particle is spinning, then the bivector $S(0)$ is the generator of rotations around this axis in the restframe of the particle. Suppose such a particle rotates with an angular speed ω_0 , then its rotor equation becomes

$$\frac{d}{d\tau}\Lambda(\tau) = -\omega_0\Lambda(\tau)S(0) . \quad (7.3.4)$$

This equation can be easily solved, we find

$$\Lambda(\tau) = \Lambda(0) \exp(-\omega_0\tau S(0)) .$$

This is the eigen-rotor for a free spinning particle. If we use the fact that τ is the proper time, and thus the length of a particle's worldline l we can write this also as

$$\Lambda(\tau) = \Lambda(0) \exp\left(-\int_l d\tau \sqrt{\mathbf{u}^2(\tau)} \omega_0 S(0)\right) . \quad (7.3.5)$$

This integral counts the number of full rotations that the particle makes in the spin-plane as it travels along its worldline l .

7.3.1. A Gauge Symmetry

The rotor $\Lambda'(\tau)$ given by an expression such as

$$\Lambda'(\tau) = \Lambda(\tau) \exp(-\sigma[\phi(\mathbf{x}(\tau)) - \phi(\mathbf{x}(0))]S(0)) , \quad (7.3.6)$$

where σ is some number, gives the *same worldline* and the *same spin-evolution* as $\Lambda(\tau)$. The only difference is that there is an additional rotation in the spin-plane which, as far as we know now, is unobservable. This additional factor changes the rotor-equation satisfied by $\Lambda'(\tau)$ into

$$\frac{d}{d\tau}\Lambda'(\tau) - \sigma\mathbf{u}(\tau) \cdot \nabla\phi(\mathbf{x}(\tau))\Lambda'(\tau)S(0) = -\omega_0\Lambda'(\tau)S(0) .$$

Although the additional factor should not change the physics, it does change the equation. That is not how we want equations to be, so let us repair the rotor equation for a free, spinning particle.

To make the rotor-equation share this symmetry with the rotor, we need to introduce a new quantity called a *gauge field*. A vector-field $\mathbf{A}(\mathbf{x})$ is introduced into the rotor-equation for a spinning particle as follows

$$\frac{d}{d\tau}\Lambda(\tau) - q\mathbf{u}(\tau) \cdot \mathbf{A}(\mathbf{x}(\tau))\Lambda(\tau)S(0) = -\omega_0\Lambda(\tau)S(0) . \quad (7.3.7)$$

We now *require* that under the transformation

$$\Lambda(\tau) \rightarrow \Lambda'(\tau) = \Lambda(\tau) \exp -\phi(\mathbf{x}(\tau))S(0) ,$$

the gauge field $\mathbf{A}(\mathbf{x})$ transforms as

$$\mathbf{A}(\mathbf{x}) \rightarrow \mathbf{A}'(\mathbf{x}) = \mathbf{A}(\mathbf{x}) - \nabla\phi(\mathbf{x}) . \quad (7.3.8)$$

If the gauge fields $\mathbf{A}(\mathbf{x})$ and $\mathbf{A}'(\mathbf{x})$ are physically equivalent, then we can say that the two rotors satisfy the same physical equation! The rotor equation is, again, fairly easy to solve by means of direct integration. We find

$$\Lambda(\tau) = \Lambda(0) \exp \left(- \int_t d\tau [\sqrt{\mathbf{u}^2(\tau)}\omega_0 - \sigma\mathbf{u}(\tau) \cdot \mathbf{A}(\mathbf{x}(\tau))]S(0) \right) , \quad (7.3.9)$$

for any worldline $\mathbf{x}(\tau)$. Again the integral counts the number of rotations along the worldline, but now it also contains a contribution from this new gauge field. Now you may ask which worldline displays an extreme number of rotations in the spinplane along the worldline?

7.3.2. Extreme number of Spin-plane rotations

The answer to the question which path extremises the number of spin-plane rotations is: the worldline that satisfies

$$\omega_0 \frac{d}{d\tau} \mathbf{u}(\tau) = \sigma [\nabla \wedge \mathbf{A}(\mathbf{x}(\tau)), \mathbf{u}(\tau)] . \quad (7.3.10)$$

This is truly remarkable! The extremization argument yields $F(\mathbf{x}) = \nabla \wedge \mathbf{A}(\mathbf{x})$ as the physically relevant field. This field is indeed invariant under the gauge transformation,

$$\begin{aligned} \mathbf{A}(\mathbf{x}) &\rightarrow \mathbf{A}'(\mathbf{x}) = \mathbf{A}(\mathbf{x}) - \nabla\phi(\mathbf{x}) , \\ F(\mathbf{x}) &\rightarrow F'(\mathbf{x}) = F(\mathbf{x}) . \end{aligned}$$

Bivector forces can arise from underlying *local* symmetries. This is one of the deepest insights that modern physics has given us. Our current understanding of the world is that all *fundamental interactions* we know are *gauge interactions*.

7.3.3. *The Root of Quantum Mechanics?*

There is something even more spectacular in the above equation. Notice that the spin-plane angular speed ω_0 plays the role of a mass! Is there any way in which we can connect a frequency with a mass? Indeed there is; quantum physics associates frequencies with energies. In terms of the rest-mass m and Planck's constant $\hbar$ the rest-energy of a particle can be associated with a frequency according to

$$\omega_0 = \frac{mc^2}{\hbar} . \quad (7.3.11)$$

So to turn the equation for the worldline of a particle in the gauge field $\mathbf{A}(\mathbf{x})$ into that of a particle of mass m and charge q in an electromagnetic vector-potential $\mathbf{A}(\mathbf{x})$ we write

$$m \frac{d}{d\tau} \mathbf{u}(\tau) = q [\nabla \wedge \mathbf{A}(\mathbf{x}(\tau)), \mathbf{u}(\tau)] , \quad (7.3.12)$$

$$\sigma = \frac{qc^2}{\hbar} . \quad (7.3.13)$$

This would suggest that there must be some very direct connection between the eigen-rotor and quantum mechanics. We previously found that the rotor equation for a particle in the electromagnetic vector-potential $\mathbf{A}(\mathbf{x})$ can be solved for *arbitrary worldlines* between $\mathbf{x}(0)$ and $\mathbf{x}(\tau)$ with the result

$$\Lambda(\mathbf{x}(0), \mathbf{x}(\tau)) = \Lambda(0) \exp \left(-\frac{S(0)}{\hbar} \int_0^\tau d\tau' [mc^2 \sqrt{\mathbf{u}^2(\tau')} - q \mathbf{u}(\tau') \cdot \mathbf{A}(\mathbf{x}(\tau'))]) \right) . \quad (7.3.14)$$

Classical mechanics requires the extremization of the integral as the way to find the classical worldline. Quantum mechanics can be *derived* from the assumption that the particle may take any path with equal likelihood and that the *averaged* rotor represents the total quantum-amplitude to go from $\mathbf{x}(0)$ to $\mathbf{x}(\tau)$! This approach is known as the path-integral approach to quantum mechanics and what our "derivation" adds to this is that the origin of the complex numbers in quantum mechanics must be sought in the appearance of the bivector $S(0)$.

Is the root of quantum theory to be found here? This is at the moment of writing this still very unclear. There is mounting, but circumstantial evidence that indeed the origin of quantum mechanics may be related to the geometric structure of spacetime. It is still under active research and no definitive claims can and should be staked. We will return to this issue when we have come to a fuller understanding of how this gauge symmetry gives rise to electromagnetism and what its properties are.

7.4. **Action principles for point particles**

Where does the equation of motion for particles in spacetime really come from? First of all we saw that relativity dictates that forces reveals themselves as bivector fields. Then we saw that energy-momentum conservation, and the inclusion of potential energy-momentum, forces upon us that this bivector force is the curl of

this potential energy-momentum. In the last few sections we saw that the equation of motion can also be retrieved as the result of a extremisation procedure!

Suppose a classical particle with restmass m travels from $\mathbf{x}(\tau_0)$ to $\mathbf{x}(\tau_1)$ while passing through a potential energy-momentum vectorfield $\mathbf{V}(\mathbf{x})$. The classical worldline of this particle *extremizes* the integral

$$S[\mathbf{x}(\tau), \mathbf{u}(\tau)] \equiv \int_{\tau_0}^{\tau_1} d\tau [mc\sqrt{\mathbf{u}(\tau)^2} - \mathbf{u}(\tau) \cdot \mathbf{V}(\mathbf{x}(\tau))] , \quad (7.4.1)$$

if we consider variations of the worldline that leave the end-points unaffected. Then we have $\delta S[\mathbf{x}(\tau), \mathbf{u}(\tau)] = 0$ for the classical worldline $\mathbf{x}(\tau)$. This integral is called the *Action* of the particle, and the extremization principle together with the condition on the end-points is called the *action principle*.

Problem 119: Action in a Coulomb field

- a: Use the expressions given earlier to write down the action for a particle moving in a Coulomb field.
- b: Compute the variation of the action if we vary the worldline while keeping its end-points fixed.

Such action principles can be extremely useful tools in writing down new theories, or analysing systems of which we know very little. Usually a small number of reasonable assumptions already severely restricts the possible choices for the action.

Problem 120: General Actions

Consider an action that is given by an integral $S[\mathbf{x}(\tau), \mathbf{u}(\tau)] \equiv \int_{\tau_0}^{\tau_1} d\tau \mathcal{L}(\mathbf{x}(\tau), \mathbf{u}(\tau))$.

- a: Show that the variation, $\delta \mathbf{x}(\tau)$, of this action with respect to a small variation in the worldline, while keeping the end-points fixed reads

$$\delta S[\mathbf{x}(\tau), \mathbf{u}(\tau)] = \int_{\tau_0}^{\tau_1} d\tau \delta \mathbf{x}(\tau) \cdot [\nabla_{\mathbf{x}(\tau)} \mathcal{L}(\mathbf{x}(\tau), \mathbf{u}(\tau)) - \frac{d}{d\tau} \nabla_{\mathbf{u}(\tau)} \mathcal{L}(\mathbf{x}(\tau), \mathbf{u}(\tau))] .$$

- b: Show that if we seek a worldline that extremizes this action, then the worldline must satisfy the equation

$$\nabla_{\mathbf{x}(\tau)} \mathcal{L}(\mathbf{x}(\tau), \mathbf{u}(\tau)) - \frac{d}{d\tau} \nabla_{\mathbf{u}(\tau)} \mathcal{L}(\mathbf{x}(\tau), \mathbf{u}(\tau)) = 0 \quad (7.4.2)$$

These equations are called the *Euler-Lagrange equations*. The function $\mathcal{L}$ is usually called the *Lagrangian*.

The action-principles formulation of classical physics was discovered step-by-step in the eighteenth and nineteenth centuries by people like Hamilton, Lagrange and Fermat. The idea of an action principle is very attractive as it replaces Newton's second law with a principle that is much more general, while maintaining its predictive power for classical mechanics. Field theory and modern physics is formulated almost exclusively in terms of action principles, as are many important applications of mathematics. The question where the Action comes from, what it means is more elusive however. I shall be able to provide you with a par-

tial answer in the next chapter. Let us first study some of the merits of our new toy.

Problem 121: Conservation laws

Consider an arbitrary Lagrangian $\mathcal{L}(\mathbf{x}(\tau), \mathbf{u}(\tau))$ for some particle.

a: Write down the Euler-Lagrange equations in cylindrical coordinates.

b: Assume that due to some symmetry in the physical system the Lagrangian is *independent* of the angular variable θ of the cylindrical coordinates. Let $\dot{\theta}(\tau) = \frac{d}{d\tau}\theta(\tau)$ and show that this implies that the quantity

$$p_\theta \equiv \frac{\partial}{\partial \dot{\theta}(\tau)} \mathcal{L}(\mathbf{x}(\tau), \mathbf{u}(\tau)) .$$

is conserved.

c: Now consider the Euler-lagrange equations in a general parametrization $\mathbf{x}(z)$. Show that if the Lagrangian is effectively *independent* of one of the parameters, say z^3 , that there is a conserved quantity and give an expression for it.

7.4.1. Non-relativistic actions

The action principle is extremely usefull in physics. So let us analyze the non-relativistic content of it. We already know that non-relativistically we may approximate

$$\begin{aligned} \tau &\approx t , \\ mc^2 \frac{c}{\sqrt{\mathbf{u}(\tau)^2}} &\approx mc^2 + \frac{m(v_1(t)^2 + v_2(t)^2 + v_3(t)^2)}{2} , \\ \mathbf{u}(\tau) &\approx c\mathbf{e}_0 + v_1(t)\mathbf{e}_1 + v_2(t)\mathbf{e}_2 + v_3(t)\mathbf{e}_3 . \end{aligned}$$

If we substitute these into the relativistic action-integral we obtain

$$S_{NR}[\mathbf{x}(t)] = \int_{t_0}^{t_1} dt \left[\frac{1}{2} m \vec{v}(t)^2 - cV_0(\vec{x}(t)) + \vec{v}(t) \cdot \vec{V}(\vec{x}(t)) \right] , \quad (7.4.3)$$

where we ignored the constant path-independent contribution mc^2 and introduced the non-relativistic vector-notation for objects such as $\vec{v}(\tau)$ that really are bivectors representing relative vectors. By studying the variation of this action under small variations of the path, while keeping the end-points fixed we can derive the standard equations of motion.

Problem 122: Only potential Energy

Consider the situation where a particle moves in a potential energy-momentum field for which all the $V_j(\mathbf{x}(\tau)) = 0$ for $j = 1, 2, 3$.

a: Show that the action reduces to $S_{NR}[\mathbf{x}(t)] = \int_{t_0}^{t_1} dt \left[\frac{1}{2} m \vec{v}(t)^2 + cV_0(\vec{x}(t)) \right]$.

b: Show that a variation of this action leads to the equation of motion

$$m \frac{d}{dt} \vec{v}(t) = - \sum_{j=1}^3 \mathbf{e}_j \frac{\partial}{\partial x^j} cV_0(\vec{x}(t)) .$$

c: Why will we associate the non-relativistic potential $U(\vec{x})$ energy with $cV_0(\vec{x})$

7.5. Problems

This section contains a number of relevant problems.

Problem 123: Warming up

Given an orthonormal basis $\mathbf{e}_\mu$ and corresponding cartesian coordinates x_μ .

- a: Show that $\nabla \mathbf{x} = 4$.
- b: Show that $\nabla \mathbf{x}^2 = 2\mathbf{x}$.
- c: Compute $\mathbf{x}\nabla \mathbf{x}$ and compare with the previous result.
- d: Let $\mathbf{x}$ be a timelike vector and let $x = \sqrt{\mathbf{x}^2}$. Compute ∇x .

Problem 124: Some elementary algebra revisited

Consider two timelike unit vectors $\hat{\mathbf{u}}_1$ and $\hat{\mathbf{u}}_2$.

- a: Show that $(\hat{\mathbf{u}}_1 \hat{\mathbf{u}}_2)^{-1} = \hat{\mathbf{u}}_2 \hat{\mathbf{u}}_1$.
- b: Now check that $\hat{\mathbf{u}}_1 \hat{\mathbf{u}}_2 = \cosh(\alpha) + \sinh(\alpha)K$, for some α and some timelike bivector K with $K^2 = +1$.
- c: Determine two coefficients a and b such that $(a + bK)^2 = \cosh(\alpha) + \sinh(\alpha)K$. Why can we write $a + bK = \sqrt{(\hat{\mathbf{u}}_1 \hat{\mathbf{u}}_2)}$?
- d: Show that we *cannot* write $\sqrt{(\hat{\mathbf{u}}_1 \hat{\mathbf{u}}_2)} = \sqrt{\hat{\mathbf{u}}_1} \sqrt{\hat{\mathbf{u}}_2}$!

Problem 125: Parametrizing a Sphere

Consider the following parametrization of spacetime points $\mathbf{x}(\theta, \phi)$ in terms of two angular variables θ and ϕ ;

$$\mathbf{x}(\theta, \phi) = \cos(\theta) \cos(\phi) \mathbf{e}_1 + \cos(\theta) \sin(\phi) \mathbf{e}_2 + \sin(\theta) \mathbf{e}_3 . \quad (7.5.1)$$

- a:** Show that for all values of θ and ϕ the point $\mathbf{x}(\theta, \phi)$ is on a sphere with radius 1 around the origin.
- b:** Argue that we can consider θ and ϕ to be coordinates on the sphere with radius 1.
- c:** Calculate the basis vectors $\mathbf{g}_\theta$ and $\mathbf{g}_\phi$ and argue that they form a *local basis* in every point on the sphere.
- d:** Let $d^2 S(\theta, \phi)$ correspond to a small element of surface area on the sphere at $\mathbf{x}(\theta, \phi)$. Show that

$$d^2 S(\theta, \phi) = \pm d\theta d\phi \mathbf{g}_\theta \wedge \mathbf{g}_\phi . \quad (7.5.2)$$

- e:** Fix the sign by requiring that the vector $d\mathbf{n}(\theta, \phi)$ defined by

$$d\mathbf{n}(\theta, \phi) = \mathbf{e}_1 \mathbf{e}_2 \mathbf{e}_3 d^2 S(\theta, \phi) \quad (7.5.3)$$

is perpendicular to the surface element and points *outward*.

Problem 126: Bivector fields

Consider a bivector field

$$F = \sum_{j=1}^3 E_j K_j + B_j J_j . \quad (7.5.4)$$

- a:** Compute ∇F .
b: If we set $\nabla F = 0$. Extract four different equations for the e_j and b_j fields.
c: Compare these equations to the Maxwell equations without charges and currents.

Problem 127: Emission, Absorption and Resonance

Suppose A particle is in its first excited state with energy E_a before it decays to the ground state with energy E_g by emitting a photon with an energy $\hbar\omega_1$.

- a:** Why will the photon energy be smaller than $\Delta E = E_a - E_g$?
b: Show that

$$\hbar\omega_1 = \Delta E \left(1 - \frac{\Delta E}{2E_a}\right) .$$

- c:** Estimate $\frac{\Delta E}{2E_a}$ for the emission of a photon of visible light by a hydrogen atom.
 Now we consider the reverse process. A photon with an energy $\hbar\omega_2$ impacts on a particle at rest with a ground state energy E_g and excites it into the first excited state with energy E_a .
d: Show that

$$\hbar\omega_2 = \Delta E \left(1 + \frac{\Delta E}{2E_g}\right) .$$

Consider we have two particles A and B of the same species, thus with common energies for their ground states and their first excited state. Let particle A be in and excited state. It decays and emits a photon with energy $\hbar\omega_1$. Now particle B is hit by this photon.

- e:** What happens if particle B cannot absorb the photon?
f: How come such photons can be absorbed if the $\frac{\Delta E}{E_g}$ is sufficiently small?

Problem 128: Projectors

A projector P is defined as an operator for which the following algebraic rule holds $P^2 = P$. If we have two projects P_1 and P_2 that also satisfy $P_1 P_2 = P_2 P_1 = 0$, then we can use these to decompose expressions into separate parts by applying these projectors. We will study a few examples in this exercise.

- a:** Show that the following multivectors

$$P_{\pm} = \frac{1}{2}(1 \pm K_3), \quad (7.5.5)$$

are projectors and that they also satisfy $P_+ P_- = 0$.

- b:** Answer the same questions for

$$P_{\pm} = \frac{1}{2}(\mathbf{e}_0 \pm i). \quad (7.5.6)$$

Problem 129: Lightcones around Alice

Sometimes it is convenient to use a coordinate system that can be interpreted as a spacetime chart made by a special observer, i.e. around a specific worldline. We will discuss such a coordinate system in this and the next problem.

Consider a standard cartesian coordinate system with a corresponding orthonormal basis $\mathbf{e}_\mu$ on spacetime and let $\mathbf{x}(\tau)$ be the worldline of Alice. Let $\mathbf{X}$ be an arbitrary spacetime point.

a: Show that the cartesian time of $\mathbf{X}$ is given by

$$t = \mathbf{X} \cdot \mathbf{e}_0 . \quad (7.5.7)$$

b: Show that the cartesian radial distance r of $\mathbf{X}$ to the time-axis is given by

$$r = \sqrt{(\mathbf{e}_0 \wedge \mathbf{X})^2} . \quad (7.5.8)$$

Using t and r , supplemented by two angles gives us a spherical coordinate system on spacetime.

c: Write down the definitions of the distance $r(\tau)$ and the time $t(\tau)$ of Alice.

Introduce the lightcone coordinates

$$\begin{aligned} u(r, t) &= \frac{1}{2}(ct - r) , \\ v(r, t) &= \frac{1}{2}(ct + r) . \end{aligned}$$

d: Show that all the possible light signals received by Alice at proper time τ must have their sources in points $\mathbf{X}$ for which

$$v(r, t) = v(r(\tau), t(\tau)) . \quad (7.5.9)$$

e: Show that all the possible light signals emitted by Alice at proper time τ will only reach the points $\mathbf{X}$ for which

$$u(r, t) = u(r(\tau), t(\tau)) . \quad (7.5.10)$$

Now we will suppose that Alice *actively* observes the event $\mathbf{X}$ by receiving the *echo* of a light signal that she has emitted earlier. Said differently; she observes the spacetime surrounding her by means of an optical radar. Let $u_{\mathbf{X}}$ and $v_{\mathbf{X}}$ be the lightcone coordinates of the event $\mathbf{X}$.

f: Show that the proper times of emission τ_E and reception τ_R of the radar signal reflecting of $\mathbf{X}$ are given by the solutions of the two equations

$$\begin{aligned} u(\mathbf{X}) - u(\tau_E) &= 0 , \\ v(\mathbf{X}) - v(\tau_R) &= 0 . \end{aligned}$$

The solutions τ_E and τ_R are ofcourse functions of r and t , but we suppress that in our notation.

g: Argue that each of the two equations will have only one *physically admissible* solution. (hint: Can Alice ever intercept the same lightcone twice?)

Chapter 8

Strings, Membranes and Fields

All our discussions so far have been limited to point-particle living on one-dimensional timelike subsets of spacetime. These particles were interacting with external fields that are defined throughout spacetime. Extended objects such as strings or membranes are two respectively three-dimensional subsets of spacetime. So if we want to be able to describe these more general structures we need to enhance our ability in spacetime integration. So as we study what strings, membranes and fields in spacetime are, and how they may be fitted into action-principles, we will learn spacetime integration over surfaces, volumes and over spacetime itself.

8.1. Surface Integrals

Integrations over subsets of spacetime in more than a single dimension introduce lots of new features. We will devote quite some time on that now, and then leave away most of the details to exercises when we discuss three- and four-dimensional integrals.

Consider a surface Σ in spacetime. It is a two-dimensional object and thus needs *two* parameters $z = \{z^1, z^2\}$ instead of one. We will generically take it that both parameters range from 0 to 1. Given such a parametrisation, the points in the surface can be expressed as $\mathbf{x}(z^1, z^2)$. At every point on the surface we have two tangent vectors

$$\mathbf{g}_1(z) = \frac{\partial}{\partial z^1} \mathbf{x}(z) , \quad (8.1.1)$$

$$\mathbf{g}_2(z) = \frac{\partial}{\partial z^2} \mathbf{x}(z) . \quad (8.1.2)$$

In *parameter space* a small surface element has the area $dz^1 dz^2$. A surface element, including an orientation, in spacetime is

$$d^2 S(z) = dz^1 dz^2 \mathbf{g}_1(z) \wedge \mathbf{g}_2(z) . \quad (8.1.3)$$

An integral over the surface Σ can now be written as

$$\int_{\Sigma} d^2 S f(\mathbf{x}) = \int dz^1 \int dz^2 \mathbf{g}_1(z) \wedge \mathbf{g}_2(z) f(\mathbf{x}(z)) . \quad (8.1.4)$$

Depending on the geometric nature of f these integrals can have various grades or linear combinations thereof. But we can always sort the result of the integral as well as the integrals themselves into their bins of definite grade.

8.1.1. The Flux

One particular example that is regularly encountered in physics is that of a flux of a field $\mathbf{b}$ through a surface area Σ .

Flux of a vectorfield

Consider a vectorfield $\mathbf{J}(\mathbf{x})$ and a surface S . The local surface element d^2S is a bivector and as a result the quantity

$$d^S \wedge \mathbf{J}(\mathbf{x}) = \frac{1}{2}(d^S \mathbf{J}(\mathbf{x}) + \mathbf{J}(\mathbf{x})d^S)$$

is a trivector that represents the component of $\mathbf{J}(\mathbf{x})$ perpendicular to d^2S multiplied by the surface area of d^2S .

Problem 130: Perpendicular to surfaces

Let a spacelike surface be represented by a bivector B and let $\mathbf{a}$ be an arbitrary vector.

- a: Choose an orthonormal basis such that B takes on the simplest possible form.
- b: Show that $B \wedge \mathbf{a}$ indeed only contains the parts of $\mathbf{a}$ perpendicular to B .
- c: Does anything change when B is timelike?

Flux of a Bivectorfield

Because much of our intuition is deeply non-relativistic we are often not properly able to distinguish the flux of a vectorfield from the flux of a bivector field. Algebraically however they are deeply distinct. So there is a need to discuss them separately here.

8.1.2. Integral on the Boundary of a Surface

The following result is extremely important and usually goes under the name of *Stokes theorem*. But as a matter of fact it is only a particular example of a far more general theorem. However, fortunately the more general theorem is known as *generalized Stokes theorem*.

Consider a line ∂S which is parametrised by the parameter z^2 and which is the *boundary* at $z^1 = 1$ of a surface S with a parametrisation $\mathbf{x}(z^1, z^2)$. If you want to have a specific example in mind then just think of S as a disc around some axis with a radius equal to 1, let z^1 be the radial coordinate in a cylindrical coordinate system and z^2 being the angular variable $\frac{\varphi}{2\pi}$. We will also take the parametrisation to be such that the event $\mathbf{x}(0, z^2)$ is the same for all values of z^2 between 0 and 1. This leads to

$$\int_0^1 dz^2 \frac{\partial}{\partial z^2} f(\mathbf{x}(0, z^2)) = 0 ,$$

because the end-points of this integration are the identical events $\mathbf{x}(0, 0)$ and $\mathbf{x}(0, 1)$ in spacetime.

The integral of an arbitrary function $f(\mathbf{x})$ along the boundary of S can be written as an integral over the surface S itself. That goes as follows, starting out from the line-integral along the boundary ∂S

$$\begin{aligned} \int_{\partial S} d\mathbf{x} f(\beta x) &= \int_0^1 dz^2 \mathbf{g}_2(1, z^2) f(\mathbf{x}(1, z^2)) , \\ &= \int_0^1 dz^1 \int_0^1 dz^2 \frac{\partial}{\partial z^1} \mathbf{g}_2(z) f(\mathbf{x}(z)) , \end{aligned} \quad (8.1.5)$$

we have reinserted the full z^1 -integral using the properties of integrating a derivative from the previous subsection. Now we use that

$$\begin{aligned} \frac{\partial}{\partial z^1} \mathbf{g}_2(z) f(\mathbf{x}(z)) &= \mathbf{g}_2(z) \left[\frac{\partial}{\partial z^1} f(\mathbf{x}(z)) \right] + \left[\frac{\partial}{\partial z^1} \mathbf{g}_2(z) \right] f(\mathbf{x}(z)) , \\ &= \mathbf{g}_2(z) [\mathbf{g}_1(z) \cdot \nabla] f(\mathbf{x}) + \left[\frac{\partial}{\partial z^2} \mathbf{g}_1(z) \right] f(\mathbf{x}(z)) , \end{aligned}$$

where for the last term we used that parametrisation satisfy by definition

$$\frac{\partial}{\partial z^m} \mathbf{g}_k(z) = \frac{\partial}{\partial z^k} \mathbf{g}_m(z) . \quad (8.1.6)$$

Continuing the manipulations we try to get rid of the z^2 -derivative altogether

$$\begin{aligned} \left[\frac{\partial}{\partial z^2} \mathbf{g}_1(z) \right] f(\mathbf{x}(z)) &= \frac{\partial}{\partial z^2} [\mathbf{g}_1(z) f(\mathbf{x}(z))] - \mathbf{g}_1(z) \frac{\partial}{\partial z^2} f(\mathbf{x}(z)) , \\ &= \frac{\partial}{\partial z^2} [\mathbf{g}_1(z) f(\mathbf{x}(z))] - \mathbf{g}_1(z) [\mathbf{g}_2(z) \cdot] f(\mathbf{x}(z)) . \end{aligned}$$

Collecting these terms together gives

$$\int_{\partial S} d\mathbf{x} f(\beta x) = \int_0^1 dz^1 \int_0^1 dz^2 \frac{\partial}{\partial z^2} [\mathbf{g}_1(z) f(\mathbf{x}(z))] + \{ \mathbf{g}_2(z) [\mathbf{g}_1(z) \cdot \nabla] f(\mathbf{x}) - \mathbf{g}_1(z) [\mathbf{g}_2(z) \cdot] f(\mathbf{x}(z)) \} .$$

The first term gives an integral that vanishes. The two terms in curly brackets can be rewritten as

$$\mathbf{g}_2(z) [\mathbf{g}_1(z) \cdot \nabla] f(\mathbf{x}) - \mathbf{g}_1(z) [\mathbf{g}_2(z) \cdot] f(\mathbf{x}(z)) = -[\mathbf{g}_1(z) \wedge \mathbf{g}_2(z) \cdot \nabla] f(\mathbf{x}(z)) .$$

So we finally find the relationship

$$\int_{\partial S} d\mathbf{x} f(\mathbf{x}) = \int_S \{ d^2 S \cdot \nabla \} f(\mathbf{x}) . \quad (8.1.7)$$

This relates an integral along the boundary ∂S to the integral of a derivative over the surface S . The geometric interpretation of the righthandside is also straightforward. the expression $d^2 S \cdot \nabla$ contains only those differentiation in directions locally *parallel* to the surface S .

Problem 131: Over-dot notation

The over-dot notation is a usefull device to keep track in calculations about the functions that derivatives act on. The reason why we need it is because the vector-derivative ∇ is no longer a scalar and thus its position in an equation matters.

a: Consider the following two equations for arbitrary vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\begin{aligned}\frac{d}{d\tau}\mathbf{ab} &= \left[\frac{d}{d\tau}\mathbf{a}\right]\mathbf{b} + \mathbf{a}\left[\frac{d}{d\tau}\mathbf{b}\right], \\ \nabla\mathbf{ab} &= [\nabla\mathbf{a}]\mathbf{b} + \mathbf{a}[\nabla\mathbf{b}].\end{aligned}$$

Why is the second equation generally incorrect?

b: Now define the over-dot notation for the action of the derivative as

$$\dot{\nabla}\mathbf{ab} \equiv \sum_{\mu} \mathbf{e}^{\mu}\mathbf{a} \frac{\partial}{\partial x^{\mu}}\mathbf{b}. \quad (8.1.8)$$

Show that

$$\nabla\mathbf{ab} = \dot{\nabla}\mathbf{ab} + \dot{\nabla}\mathbf{ab}.$$

The integral identity we have just arrived is very usefull in physics, where we often have *physical* information about $f(\mathbf{x})$. To rewrite the integral identity in such a way that it contains $\nabla f(\mathbf{x})$ we need to do a few algebraic manipulations. In this context it is very usefull to introduce the so-called *anti-commutator* $\{A, B\}$ of two multivectors defined as

$$\{A, B\} \equiv \frac{1}{2}(AB + BA). \quad (8.1.9)$$

Problem 132: Commutator and Anticommutator

a: Look up the definition of the commutator $[A, B]$.

b: Show that for any two multivectors $AB = \{A, B\} + [A, B]$.

Now we are able to write out the products in the integration over S as follows

$$\begin{aligned}\int_S \{d^2S \cdot \nabla\} f(\mathbf{x}) &= \int_S \frac{1}{2}(d^2S \nabla f(\mathbf{x}) - \dot{\nabla} d^2S f(\mathbf{x})), \\ &= \int_S \frac{1}{2}(d^2S \nabla f(\mathbf{x}) + \dot{\nabla} f(\mathbf{x}) d^2S - \dot{\nabla} f(\mathbf{x}) d^2S - \dot{\nabla} d^2S f(\mathbf{x})), \\ &= \int_S \{d^2S, \nabla f(\mathbf{x})\} - \dot{\nabla} \{f(\mathbf{x}), d^2S\}.\end{aligned} \quad (8.1.10)$$

These integrals can now be split into terms of different grade and they contain a wealth of information. In many textbooks you will find several integral theorems scattered throughout man chapters and appendices that are all nicely wrapped together in the single-line expressions we've derived here.

8.2. Strings

A worldline of a particle is represented by a parametrisation $\mathbf{x}(\tau)$ with a single parameter τ , typically ranging from $\tau = -\infty$ to $\tau = \infty$. Physical particles are

always represented by timelike or lightlike worldlines, mathematically spacelike lines also exist in Spacetime but they do not correspond to the trajectory of a particle. Rather a spacelike line would represent an extended one-dimensional object that would only exist "instantaneously", there is no future or past copy of the object. As a result a physical *string* that propagates through time must be a two-dimensional subset of spacetime and is thus described by a parametrisation with two parameters $\mathbf{x}_S(\tau, \sigma)$. On the two-dimensional worldsheet surface we have two tangent-vectors

$$\begin{aligned}\mathbf{g}_\tau(\mathbf{x}(\tau, \sigma)) &= \frac{\partial}{\partial \tau} \mathbf{x}_S(\tau, \sigma) , \\ \mathbf{g}_\sigma(\mathbf{x}(\tau, \sigma)) &= \frac{\partial}{\partial \sigma} \mathbf{x}_S(\tau, \sigma) .\end{aligned}$$

For this to represent a truly propagating string the worldsheet surface needs to be timelike everywhere, i.e. the infinitesimal bivector

$$d^2 S_S \equiv d\tau d\sigma \mathbf{g}_\sigma(\mathbf{x}(\tau, \sigma)) \wedge \mathbf{g}_\tau(\mathbf{x}(\tau, \sigma)) ,$$

should satisfy

$$(d^2 S_S)^2 > 0 . \quad (8.2.1)$$

The total surface area of the worldsheet is represented by an integral that very much resembles the worldline length integral. The worldsheet area A is given by

$$A = \int d\tau d\sigma \sqrt{[\mathbf{g}_\sigma(\mathbf{x}(\tau, \sigma)) \wedge \mathbf{g}_\tau(\mathbf{x}(\tau, \sigma))]^2} . \quad (8.2.2)$$

8.3. Membranes

8.3.1. Volume Integrals

If V is a three-dimensional subset of spacetime parametrised by three parameters θ_j , then we define a volume integral of a function $f(\mathbf{x}(\theta))$ as

$$\int_V f(\mathbf{x}(\theta)) \equiv \int d\theta_1 d\theta_2 d\theta_3 \mathbf{g}_1(\theta) \wedge \mathbf{g}_2(\theta) \wedge \mathbf{g}_3(\theta) f(\mathbf{x}(\theta)) . \quad (8.3.1)$$

Like in the previous two cases the value of such an integral does not depend on the chosen parametrisation. Ofcourse there are parametrisation in which this integral maybe easier to compute than in others.

8.3.2. Charge Conservation and Volume-integrals

Conservation laws in nature typically take two forms. Either they specify a quantity which remains constant or a quantity whose distribution across spacetime satisfies the rule *what goes in has got to come out*, i.e. a quantity of which nothing is produced or destroyed. The first class usually represents some property of an individual particle remaining constant along the worldline. The second class is the one we consider in this section.

Let us start out with an example and then wonder whether it is general enough. Suppose we have a vector field $\mathbf{J}$ given in some basis as

$$\mathbf{J}(\mathbf{x}) = \rho c \mathbf{e}_0 + j_1 \mathbf{e}_1 + j_2 \mathbf{e}_2 + j_3 \mathbf{e}_3 . \quad (8.3.2)$$

Now the divergence of this current, i.e. $\nabla \cdot \mathbf{J}(\mathbf{x})$ is a measure for the strength of its sources. However, when electric charge is *conserved* there should not be any sources. So we must have

$$\nabla \cdot \mathbf{J}(\mathbf{x}) = 0 . \quad (8.3.3)$$

This is the *local* form of the charge-conservation law. When we write it out in cartesian coordinates pertaining to the basis $\mathbf{e}_\mu$ we get

$$\frac{\partial}{\partial t} \rho + \frac{\partial}{\partial x_1} j_1 + \frac{\partial}{\partial x_2} j_2 + \frac{\partial}{\partial x_3} j_3 = 0 . \quad (8.3.4)$$

In three-dimensional terminology this equation says: The rate at which the density ρ changes with time is compensated by the divergence of the current-vector $\vec{j}$. Hence, it are changes in the density of electric charge that form the sources for the electric current-density. The fact that *no other sources* are allowed is what leads to current conservation.

Integrals over Volumes and Boundaries of Volumes

It is also possible to take a global view of these matters. Consider a volume $\mathcal{V}$, i.e. a three-dimensional subset of spacetime. Given some parametrisation $\mathbf{x}(z)$ with $z = \{z^1, z^2, z^3\}$ of this part of spacetime. Let's assume for simplicity that all parameters run from 0 to 1. Locally, i.e. at every event in this volume, the infinitesimal volume-element corresponding to $dz^1 dz^2 dz^3$ is given by the trivector made from the usual induced basis-vectors,

$$d^3V = dz^1 dz^2 dz^3 \mathbf{g}_1(\mathbf{x}(z)) \wedge \mathbf{g}_2(\mathbf{x}(z)) \wedge \mathbf{g}_3(\mathbf{x}(z)) . \quad (8.3.5)$$

If we now want to integrate some quantity $\mathcal{Q}(\mathbf{x})$ over this volume $\mathcal{V}$ we need to compute

$$\int_{\mathcal{V}} d^3V \mathcal{Q}(\mathbf{x}) = \int_0^1 \int_0^1 \int_0^1 dz^1 dz^2 dz^3 \mathbf{g}_1(\mathbf{x}(z)) \wedge \mathbf{g}_2(\mathbf{x}(z)) \wedge \mathbf{g}_3(\mathbf{x}(z)) \mathcal{Q}(\mathbf{x}(z)) \quad (8.3.6)$$

Now assume that the parametrisation was chosen in such a way that the *boundary*, $\partial\mathcal{V}$, of the volume $\mathcal{V}$ is described by $z^3 = 1$. Then an integral over the two-dimensional boundary $\partial\mathcal{V}$ of this volume requires the evaluation of the integral

$$\int_{\partial\mathcal{V}} d^2S \mathcal{Q}(\mathbf{x}) = \int_0^1 \int_0^1 dz^1 dz^2 \mathbf{g}_1(\mathbf{x}(z)) \wedge \mathbf{g}_2(\mathbf{x}(z))(\mathbf{x}(z)) \mathcal{Q}(\mathbf{x}(z)) , \quad z^3 = 1 . \quad (8.3.7)$$

For a conserved current the net flow into the volume $\mathcal{V}$ is given by the part of the current $\mathbf{J}(\mathbf{x})$ that is perpendicular to the surface $\partial\mathcal{V}$. Locally the part of $\mathbf{J}(\mathbf{x})$ perpendicular to d^2S commutes with the bivector d^2S !

Problem 133: Perpendicularity to Surfaces and Flux

Let the bivector B represent a spacelike surface.

a: Choose an orthonormal basis such that B takes the simplest possible form.

b: Consider an arbitrary vector $\mathbf{a}$ and verify that the $\frac{1}{2}\{B\mathbf{a} + \mathbf{a}B\}$ is indeed proportional to the part of $\mathbf{a}$ that is perpendicular to B .

c: The *flux* $\Phi(\mathbf{a}, B)$ of a vectorfield $\mathbf{a}$ through a surface B is defined as the product of the magnitude of the perpendicular part of $\mathbf{a}$ with the surface area of B . Verify that

$$\Phi(\mathbf{a}, B) = \frac{1}{2}\{B\mathbf{a} + \mathbf{a}B\} . \quad (8.3.8)$$

So we need to compute the integral

$$\int_{\partial\mathcal{V}} \frac{1}{2}\{d^2S J(\mathbf{x}) + \mathbf{J}(\mathbf{x})d^2S\} ,$$

to determine the total flux going into or out of the volume $\mathcal{V}$ through the boundary $\partial\mathcal{V}$.

Problem 134: The integral over the Boundary

a: Argue that we are allowed to write

$$\int_{\partial\mathcal{V}} d^2S Q(\mathbf{x}) = \int_0^1 dz^3 \frac{\partial}{\partial z^3} \left[\int_{\partial\mathcal{V}} d^2S Q(\mathbf{x}) \right] .$$

b: Use that $\frac{\partial}{\partial z^3} = \mathbf{g}_3(\mathbf{x}(z)) \cdot \nabla$ and that $\frac{\partial}{\partial z^n} \mathbf{g}_m(\mathbf{x}(z)) = \frac{\partial}{\partial z^m} \mathbf{g}_n(\mathbf{x}(z))$ to show that

$$\int_{\partial\mathcal{V}} d^2S Q(\mathbf{x}) = \int_{\mathcal{V}} \{d^3V \cdot \nabla\} Q(\mathbf{x}) . \quad (8.3.9)$$

c: Explain that in this derivation we have not used anywhere that the volume is purely spacelike and that therefor this result is more general than the standard textbook versions of Gauss' theorem.

Now this integral over the boundary can be transformed into an integral over the interior volume $\mathcal{V}$. This is known as *Gauss' theorem* for the particular case of three-dimensional spacelike volumes. Applying it to our study of the influx of electric charge due to an electric charge current piercing the boundary of a volume $\mathcal{V}$ gives

$$\int_{\partial\mathcal{V}} \frac{1}{2}\{d^2S \mathbf{J}(\mathbf{x}) + \mathbf{J}(\mathbf{x})d^2S\} = \langle \int_{\mathcal{V}} \{d^3V \cdot \nabla\} \mathbf{J}(\mathbf{x}) \rangle_1 . \quad (8.3.10)$$

8.4. Actions for Electromagnetic Fields*8.4.1. Spacetime integrals**8.4.2. Field equations for Electrodynamics*

Maxwell's law, that describes the manner in which the electromagnetic field couple to their sources, is

$$\nabla F(\mathbf{x}) = 4\pi J(\mathbf{x}) .$$

In this section we will discuss some of its most fundamental aspects that are best treated in Spacetime, without making any arbitrary splits between space and time.

We have already seen that the electromagnetic interaction bivector $F(\mathbf{x})$ can be considered to be the result of extremising the worldline with respect to a Lagrangian that contains the vector-potential $\mathbf{A}(\mathbf{x})$. The connection between them is

$$F = \nabla \wedge \mathbf{A} \quad (8.4.1)$$

and because of fact that

$$\nabla \wedge (\nabla \wedge \mathbf{A}) = 0 ,$$

Maxwell's law can be rewritten in terms of $\mathbf{A}$ as,

$$\nabla \cdot (\nabla \wedge \mathbf{A}(\mathbf{x})) = \mathbf{J}_e(\mathbf{x}) . \quad (8.4.2)$$

Using that

$$\nabla \cdot (\nabla \wedge \mathbf{A}(\mathbf{x})) = \nabla^2 \mathbf{A}(\mathbf{x}) - \nabla(\nabla \cdot \mathbf{A}(\mathbf{x})) ,$$

we therefore find

$$\nabla^2 \mathbf{A}(\mathbf{x}) - \nabla(\nabla \cdot \mathbf{A}(\mathbf{x})) = \mathbf{J}_e(\mathbf{x}) . \quad (8.4.3)$$

The origin of this vector-potential is the *gauge*-symmetry satisfied by $\mathbf{A}(\mathbf{x})$ and the eigen-rotors $\Lambda(\tau)$ of charged particles

$$\begin{aligned} \Lambda(\tau) &\rightarrow \Lambda'(\tau) = \Lambda(\tau) \exp(-\phi(\mathbf{x}(\tau))S(0)) , \\ \mathbf{A}(\mathbf{x}) &\rightarrow \mathbf{A}'(\mathbf{x}) = \mathbf{A}(\mathbf{x}) - \nabla\phi(\mathbf{x}) . \end{aligned}$$

Maxwell's law is *gauge-invariant*, as is physics. At this point we have not seen yet what the origin is of Maxwell's law itself. We will look at that in a later section in this chapter. Gauge-invariance also means that instead of taking $\mathbf{A}$ we can always make a gauge transformation to some other $\mathbf{A}'$ to reduce the problem further. Let us see how this works.

8.4.3. Fixing the Gauge

The answer to the last questions is affirmative. For example we could wonder whether or not we could find a $\mathbf{A}'$ such that $\nabla \cdot \mathbf{A}' = 0$. If that were the case then one term could be dropped from the equation.

Now notice that if we start out from an arbitrary $\mathbf{A}$, the gauge transformation that brings us to this $\mathbf{A}'$ should satisfy,

$$\begin{aligned} \nabla \cdot \mathbf{A}' &= \nabla \cdot \mathbf{A} + \nabla^2 \alpha(\mathbf{x}) = 0, \rightarrow \\ \nabla^2 \alpha(\mathbf{x}) &= -\nabla \cdot \mathbf{A} . \end{aligned}$$

This last equation is just the wave-equation, with the divergence of $\mathbf{A}$ as its source-term, which admits infinitely many solutions depending on the boundary conditions. So, given some vector-potential $\mathbf{A}$ we will always be able to find a special potential $\mathbf{A}'$ that satisfies $\nabla \cdot \mathbf{A}' = 0$. If we express everything in terms of this vector-potential we say that *we have fixed the gauge*. For after we have made

this choice and simplified the equations we can no longer perform arbitrary gauge-transformations. The equations for $\mathbf{A}'$ read

$$\nabla^2 \mathbf{A}'(\mathbf{x}) = J_e(\mathbf{x}) , \quad (8.4.4)$$

$$\nabla \cdot \mathbf{A}'(\mathbf{x}) = 0 . \quad (8.4.5)$$

This choice of gauge is called the *Lorentz gauge*.

Is the gauge-symmetry completely fixed now? No, it isn't. If we would consider a function $\beta(\mathbf{x})$ satisfying $\nabla^2 \beta(\mathbf{x}) = 0$ then it is easy to prove that the transformation

$$\mathbf{A}'' = \mathbf{A}' + \nabla \beta ,$$

will preserve the Lorentz gauge. As a result we find that the Lorentz-gauge vector-potential still has at least one scalar function too many among its components. One remedy is to simply put one of the components to zero, for example

$$\mathbf{e}_0 \cdot \mathbf{A}'(\mathbf{x}) = 0 , \text{ Weyl gauge } ,$$

$$\mathbf{e}_j \cdot \mathbf{A}'(\mathbf{x}) = 0 , \text{ Axial gauge } .$$

Another option that is usefull when the electromagnetic fields do not depend on time is

$$(\mathbf{e}_0 \wedge \nabla) \cdot (\mathbf{e}_0 \wedge \mathbf{A}'(\mathbf{x})) = 0 , \quad (8.4.6)$$

called the *Coulomb gauge*. The Lorentz gauge comes in very handy when discussing time-dependent fields. The Coulomb gauge is very instrumental in studying electrostatics. The main physical lesson to be learned from the fact that we need *two* gauge conditions on the four components of the vector-potential is the following: the electromagnetic field contains only $4 - 2$ is *two* physical degrees of freedom!

8.5. Conservation laws

An important characteristic of the electromagnetic processes in Nature is the conservation of electromagnetic charge. It, and the conservation of energy-momentum give us clues as to why Maxwell's law takes the form that it does.

Conservation laws in nature identify quantities which take constant values, or quantities whose distribution across spacetime satisfies the rule *what goes in has got to come out*, i.e. a quantity of which nothing is produced or destroyed. The quantities that are conserved are usually found by integrating *densities*, such as the charge-density or energy-density, over a whole equal-time slice of Spacetime. So before we discuss the conservation laws, let us look at how this integration would work in a given parametrization spacetime.

Suppose we have introduced some parametrization $z = \{z^0, z^1, z^2, z^3\}$ in spacetime in which the induced basis-vector corresponding to the z^0 is a timelike vector, i.e.

$$\mathbf{g}_0(\mathbf{x}) = \frac{\partial}{\partial z^0} \mathbf{x}(z) \rightarrow \mathbf{g}_0(\mathbf{x})^2 > 0 . \quad (8.5.1)$$

The *integral* over an equal-time slice $z^0 = t$, in such a parametrization, of a function $f(\mathbf{x})$ is given by

$$\int_{z^0=t} f(\mathbf{x}(z)) \equiv \int_{z^0=t} dz^1 dz^2 dz^3 \mathbf{g}_1(\mathbf{x}) \wedge \mathbf{g}_2(\mathbf{x}) \wedge \mathbf{g}_3(\mathbf{x}) f(\mathbf{x}(z)) . \quad (8.5.2)$$

The trivector $\mathbf{g}_1(\mathbf{x}) \wedge \mathbf{g}_2(\mathbf{x}) \wedge \mathbf{g}_3(\mathbf{x})$ is the *volume-element* that translates the infinitesimal parameter-volume $dz^1 dz^2 dz^3$ into the actual infinitesimal spacetime volume-element. Now suppose we want to compare the value of this integral at $z^0 = t$ with the value at $z^0 = t'$. If both integrals are equal for any choice of t and t' , then we say that it represents a *conserved quantity*. So we compute

$$\begin{aligned} \int_{z^0=t} f(\mathbf{x}(z)) - \int_{z^0=t'} f(\mathbf{x}(z)) &= \int_{z^0=t'}^{z^0=t} dz^0 \frac{\partial}{\partial z^0} \int_{z^0=t} f(\mathbf{x}(z)) , \\ &= \int_{z^0=t}^{z^0=t'} dz^0 dz^1 dz^2 dz^3 \mathbf{g}_0(\mathbf{x}) \cdot \nabla [\mathbf{g}_1(\mathbf{x}) \wedge \mathbf{g}_2(\mathbf{x}) \wedge \mathbf{g}_3(\mathbf{x}) f(\mathbf{x}(z))] , \\ &= \int_{z^0=t}^{z^0=t'} dz^0 dz^1 dz^2 dz^3 \mathbf{g}_0(\mathbf{x}) \wedge \mathbf{g}_1(\mathbf{x}) \wedge \mathbf{g}_2(\mathbf{x}) \wedge \mathbf{g}_3(\mathbf{x}) \nabla f(\mathbf{x}(z)) . \end{aligned}$$

8.5.1. Current and Charge Conservation

Now also consider a subset of spacetime $\mathcal{M}$ such that its boundary $\partial\mathcal{M}$ can be represented by two spacelike subsets. Geometrically what this means is that we have a basically infinite subset of spacetime that has a *past* boundary and a *future* boundary. An integral over these boundaries is an integral over three-dimensional subspaces in spacetime. Following our earlier construction this implies a tri-vector valued integration measure $d\theta_1 d\theta_2 d\theta_3 \mathbf{g}_1(\theta) \wedge \mathbf{g}_2(\theta) \wedge \mathbf{g}_3(\theta)$ on the boundary. Such tri-vectors represent either volume-elements (the time-like trivector) or flux-elements (the spacelike trivectors). As a result matching the density ρ with a volume-element, and the current-densities j_i with a flux-element requires us to take the *pseudo-scalar part* of the integral of $\mathbf{J}$ across this boundary $\partial\mathcal{M}$ should vanish, i.e.

$$\int_{\partial\mathcal{M}} \wedge \mathbf{J}(\mathbf{x}) = 0 . \quad (8.5.3)$$

Now note that this integral gives a pseudo-scalar result which we can translate into a scalar when we pre-multiply $\int_{\partial\mathcal{M}}$ by $i^{-1} = -i$. But using that

$$i^{-1} \mathbf{a} \wedge \mathbf{b} = (i^{-1} \mathbf{a}) \cdot \mathbf{b} , \quad (8.5.4)$$

we can write

$$i^{-1} \int_{\partial\mathcal{M}} \cdot \mathbf{J}(\mathbf{x}) = 0 . \quad (8.5.5)$$

This form of a conservation law is called the *global* form. When we apply what we have learned above about the relationship between integrals across boundaries and integrals of derivatives we can conclude

$$\int_{\mathcal{M}} \nabla \cdot \mathbf{J}(\mathbf{x}) = 0 . \quad (8.5.6)$$

Now if $\mathbf{J}$ is a *conserved* quantity this integral should be true for every conceivable $\mathcal{M}$. Hence, we must have

$$\nabla \cdot \mathbf{J}(\mathbf{x}) = 0 . \quad (8.5.7)$$

This is the *local* form of a conservation law. When we write it out in cartesian coordinates pertaining to the basis $\mathbf{e}_\mu$ we get

$$\frac{\partial}{\partial t} \rho + \frac{\partial}{\partial x_1} j_1 + \frac{\partial}{\partial x_2} j_2 + \frac{\partial}{\partial x_3} j_3 = 0 . \quad (8.5.8)$$

In three-dimensional terminology this equation says: The rate at which the density ρ changes with time is compensated by the *divergence* of the current-vector $\vec{j}$. An important example of this type of conservation laws is the conservation of electric charge. In that case the vector $\mathbf{J}$ is called the *charge current-density*. A conservation law of the form

$$\nabla \cdot \mathbf{J}(\mathbf{x}) = 0 ,$$

is associated with a conserved quantity $Q(\mathcal{M})$ given by

$$Q(\mathcal{M}) \equiv \int_{\mathcal{M}} (i^{-1} d\theta_1 d\theta_2 d\theta_3 \mathbf{g}_1(\theta) \wedge \mathbf{g}_2(\theta) \wedge \mathbf{g}_3(\theta)) \cdot \mathbf{J}(\mathbf{x}) . \quad (8.5.9)$$

This quantity is called a conserved *charge* and it is a scalar quantity.

8.5.2. Energy-Momentum Conservation in Electromagnetic fields

We know that according to the equations governing the electromagnetic field, we have

$$\mathbf{J}(\mathbf{x}) = \nabla F(\mathbf{x}) ,$$

and so we can also write

$$\begin{aligned} \mathbf{J}(\mathbf{x}) = \tilde{\mathbf{J}}(\mathbf{x}) &= \tilde{F}(\mathbf{x}) \nabla , \\ &= -(F(\mathbf{x}) \nabla) , \end{aligned} \quad (8.5.10)$$

where the derivative acts on the bivectorfield from the right!. This relatively trivial observation comes in usefull when we consider what happens to an electric current J_e interacting with the electromagnetic field F . It evolves under the influence of a force exerted on the current-density by the electromagnetic field. From the previous chapters we know this force to be just

$$\begin{aligned} \mathbf{f}(\mathbf{x}) &= [F(\mathbf{x}), \mathbf{J}_e(\mathbf{x})] , \\ &= \frac{1}{2} \{ F(\mathbf{x}) \mathbf{J}_e(\mathbf{x}) - \mathbf{J}_e(\mathbf{x}) F(\mathbf{x}) \} \\ &= \frac{1}{2} \{ F(\mathbf{x}) (\nabla F(\mathbf{x})) - (\nabla F(\mathbf{x})) F(\mathbf{x}) \} \\ &= \frac{1}{2} \{ F(\mathbf{x}) (\nabla F(\mathbf{x})) + (F(\mathbf{x}) \nabla) F(\mathbf{x}) \} \\ &= \frac{1}{2} F(\mathbf{x}) \nabla F(\mathbf{x}) . \end{aligned} \quad (8.5.11)$$

Note that the derivate acts both to the left- as well as to the right in a *product-rule* fashion. This force acts on a *density* of charges and currents it is therefor a *force-density*. It represents the density of energy-momentum transferred, per unit of time, from the electromagnetic field to the current-density. Note that this has units like $m^{-3}s^{-1}$ and thus cries out to be integrated over some spacetime volume Ω . As the expression contains a derivative we may use Stokes theorem to obtain an integral over the boundary $\partial\Omega$ of the spacetime volume,

$$\begin{aligned} \int_{\Omega} d^4\Sigma \frac{1}{2} F(\mathbf{x}) \nabla F(\mathbf{x}) &= \int_{\partial\Omega} \frac{1}{2} F(\mathbf{x}) d^3\mathbf{V} F(\mathbf{x}) , \\ &= \int_{\partial\Omega} d^3V \frac{1}{2} F(\mathbf{x}) \mathbf{n}(\mathbf{x}) F(\mathbf{x}) . \end{aligned} \quad (8.5.12)$$

In this equation $\mathbf{n}$ represents the unit-vector normal to the boundary $\partial\Omega$. The boundary of something four-dimensional is three-dimensional and thus dV^3 can represent two things,

- The integral contains 3 space-directions: a volume integral.
- The integral contains 2 space-directions and time: a surface-integral integrated over time.

Physically the first case yields the total energy-momentum contained in the volume. The second case yields the time-integrated flow of energy-momentum through a spacelike surface. We define a very important quantity now, called the *energy-momentum tensor* of the electromagnetic field,

$$T[\mathbf{n}] = -\frac{1}{2} F(\mathbf{x}) \mathbf{n} F(\mathbf{x}) . \quad (8.5.13)$$

If $\mathbf{n}$ is time-like, it represents the density of energy-momentum flowing through the electromagnetic field for an observer whose time-like direction is $\mathbf{n}$. If $\mathbf{n}$ is spacelike, then $T[\mathbf{n}]$ represents the total flux of energy-momentum through a unit surface-area perpendicular to $\mathbf{n}$.

The General Case

Suppose that we assign to a system a *total energy-momentum* $\mathbf{p}(\mathcal{M})$ as a functions of the $3-d$ subspace of spacetime on which we integrate the densities of energy and momentum. How can we relate this vector to a set of densities and fluxes? The fluxes and densities are tri-vectors and we would expect that map that assigns a energy-momentum density to such a trivector is linear. So to this end we introduce a linear map $T[\mathbf{a}]$ that turns vectors into vectors and that allows us to write

$$\mathbf{p}(\mathcal{M}) \equiv \int_{\mathcal{M}} d\theta_1 d\theta_2 d\theta_3 T[i^{-1}\mathbf{g}_1(\theta) \wedge \mathbf{g}_2(\theta) \wedge \mathbf{g}_3(\theta)] . \quad (8.5.14)$$

Ofcourse the map T itself will depend on spacetime position. Now if $\mathcal{M}$ is a boundary $\mathcal{M} = \partial\mathcal{S}$ of something, then we expect this integral to vanish as *what goes in has got to come out*. Following the same line of reasoning as above this requires that the

$$\int_{\partial\mathcal{S}} d\theta_1 d\theta_2 d\theta_3 T[i^{-1}\mathbf{g}_1(\theta) \wedge \mathbf{g}_2(\theta) \wedge \mathbf{g}_3(\theta))]$$

$$\begin{aligned}
&= \int_S d\theta_1 d\theta_2 d\theta_3 d\theta_4 i^{-1} \mathbf{g}_1(\theta) \wedge \mathbf{g}_2(\theta) \wedge \mathbf{g}_3(\theta) \wedge \mathbf{g}_4(\theta) T[\nabla] , \\
&= 0 .
\end{aligned}$$

Hence we should have $T[\nabla] = 0$ where the derivatives in ∇ only act on the full position-dependence of the linear map T . The energy-momentum conservation law will take this form, and T is the energy-momentum tensor. In the chapter on Gravity we will see that there is a far more natural way of deriving energy-momentum conservation, and thus on deriving the exact form of the energy-momentum tensor: it results from extremising the action with respect to small variations in spacetime geometry!

8.6. The Connection

This section can be omitted on first reading. No new concepts are being introduced, but the expressions for curls and divergences in general non-cartesian coordinates are worked out. These are very important in a number of applications, but most physicists can live quite well with knowing where to find them whenever they need them. For the mathematically minded however, this section may contain a number of nice surprises. For those who continue towards curved spacetime this section will clarify quite a bit of terminology that you will also encounter there.

8.6.1. The connection $\Gamma_{\mu\nu}^\lambda$

In principle the calculation goes through in the same way for non-cartesian coordinates. Yet there is one step that needs extra attention. In the general case we have

$$\nabla \mathbf{b}(\mathbf{x}) = \sum_{\mu,\nu} \mathbf{g}^\mu(z) \frac{\partial}{\partial z^\mu} \mathbf{g}_\nu(z) b^\nu(z) ,$$

where now the differentiation $\frac{\partial}{\partial z^\mu} \mathbf{g}_\nu(z) b^\nu(z)$ produces an extra term because the induced basis-vectors themselves *also* depend on the spacetime position. We get

$$\frac{\partial}{\partial z^\mu} \mathbf{g}_\nu(z) b^\nu(z) = \mathbf{g}_\nu(z) \frac{\partial}{\partial z^\mu} b^\nu(z) + b^\nu(z) \frac{\partial}{\partial z^\mu} \mathbf{g}_\nu(z) .$$

The first term is nothing new and we had that one in the cartesian case as well. The second term contains the derivative of a vector $\mathbf{g}_\nu$ with respect to a parameter z^μ . The result is ofcourse another vector which we can expand in terms of the full local basis. So we define the short hand symbol $\Gamma_{\mu\nu}^\lambda(z)$

$$\frac{\partial}{\partial z^\mu} \mathbf{g}_\nu(z) \equiv \sum_\lambda \Gamma_{\mu\nu}^\lambda(z) \mathbf{g}_\lambda(z) , \quad (8.6.1)$$

called a *connection*. When we work with non-cartesian coordinates these connections show up as a result of the position-dependence of the locally induced basis vectors. They have the nice property of being *symmetric* in the lower indices

$$\Gamma_{\mu\nu}^\lambda(z) = \Gamma_{\nu\mu}^\lambda(z) .$$

Problem 135: Connection properties

- a: Recall the definition of the induced local basis vectors $\mathbf{g}_\mu(z)$ in terms of partial derivatives.
 b: Show that the statement $\Gamma_{\mu\nu}^\lambda(z) = \Gamma_{\nu\mu}^\lambda(z)$ is the same as stating that *partial derivatives commute*.
 c: Show that $\Gamma_{\mu\nu}^\sigma(z) = \mathbf{g}^\sigma(z) \cdot \frac{\partial}{\partial z^\mu} \mathbf{g}_\nu(z)$.

So in our expression for the divergence and the curl we should expect to find some extra terms due to the presence of the connection $\Gamma_{\nu\mu}^\lambda(z)$. Continuing our computation

$$\begin{aligned}
 \nabla \mathbf{b}(\mathbf{x}) &= \sum_{\mu,\nu} \mathbf{g}^\mu(z) \frac{\partial}{\partial z^\mu} \mathbf{g}_\nu(z) b^\nu(z) = \sum_{\mu,\nu} \mathbf{g}^\mu(z) \left[\mathbf{g}_\nu(z) \frac{\partial}{\partial z^\mu} b^\nu(z) + \sum_\lambda \Gamma_{\mu\nu}^\lambda(z) b^\nu(z) \mathbf{g}_\lambda(z) \right] , \\
 &= \sum_{\mu,\nu} [\mathbf{g}^\mu(z) \mathbf{g}_\nu(z)] \frac{\partial}{\partial z^\mu} b^\nu(z) + \sum_\lambda b^\nu(z) \Gamma_{\mu\nu}^\lambda(z) [\mathbf{g}^\mu(z) \mathbf{g}_\lambda(z)] , \\
 &= \sum_{\mu,\nu} \mathbf{g}^\mu(z) \cdot \mathbf{g}_\nu(z) \frac{\partial}{\partial z^\mu} b^\nu(z) + \mathbf{g}^\mu(z) \wedge \mathbf{g}_\nu(z) \frac{\partial}{\partial z^\mu} b^\nu(z) \\
 &+ \sum_\lambda \Gamma_{\mu\nu}^\lambda(z) b^\nu(z) \mathbf{g}^\mu(z) \cdot \mathbf{g}_\lambda(z) + \Gamma_{\mu\nu}^\lambda(z) b^\nu(z) \mathbf{g}^\mu(z) \wedge \mathbf{g}_\lambda(z) .
 \end{aligned}$$

This last expression is again clearly split into a scalar part, the divergence, and a bivector part, the curl. So we have

$$Div[\mathbf{b}(\mathbf{x})] = \sum_\mu \left(\frac{\partial}{\partial z^\mu} b^\mu(z) + \sum_\nu \Gamma_{\mu\nu}^\mu(z) b^\nu(z) \right) . \quad (8.6.2)$$

For the curl we find something remarkable. If you workout the part involving the connection it turns out that it vanishes. In other words, the curl in non-cartesian coordinates has no connection-dependence.

Problem 136: Connection gymnastics

- a: Use the algebraic definition of the reciprocal basis to show that

$$\left[\frac{\partial}{\partial z^\sigma} \mathbf{g}^\lambda(\mathbf{x}) \right] \cdot \mathbf{g}_\nu(\mathbf{x}) = -\Gamma_{\sigma\nu}^\lambda(\mathbf{x}) .$$

- b: Argue that this implies that the dervative of a reciprocal basis-vector is

$$\frac{\partial}{\partial z^\sigma} \mathbf{g}^\lambda(\mathbf{x}) = - \sum_\lambda \Gamma_{\sigma\mu}^\lambda(\mathbf{x}) \mathbf{g}^\mu(\mathbf{x}) .$$

8.6.2. Connections and Forces

Chapter 9

Stationary Electromagnetic Fields

9.1. Electrostatics and Magnetostatics

9.1.1. Integral Laws

9.1.2. Solving by educated guessing

9.2. Potentials, Poisson and Laplace

9.2.1. Separation of variables

9.2.2. Multipole Expansions

9.2.3. Green's functions

9.3. Stationary fields in a Medium

9.4. Problems

Chapter 10

Time-dependent Fields

10.1. Induction phenomena

10.2. Electromagnetic waves

10.2.1. Polarization

10.2.2. Reflection and Refraction

10.2.3. Harmonic fields in Matter

10.2.4. Waveguides

10.3. Problems

Chapter 11

Radiation

11.1. Retarded and Advanced Green's functions

11.2. Liénhard-Wiechert potentials

11.3. Radiation fields

11.4. Simple radiating systems

11.5. Problems

Chapter 12

Scalar and Rotorfields

12.1. Action principles for Scalar fields

12.2. Action principles for Rotor fields

12.3. The Dirac field

Suppose that the particles eigen-rotor is a function of spacetime, rather than of proper-time, i.e. suppose that $\Lambda(\tau) \rightarrow \Lambda(\mathbf{x}(\tau))$. It means that we are describing a *single particle* by a *field* $\Lambda(\mathbf{x})$ rather than by something which has a definite position! But let us see where this thought leads us. We can rewrite the rotor-equation to accomodate this conceptual shift as

$$\mathbf{u}(\tau) \cdot \nabla \Lambda(\mathbf{x}(\tau)) = -\omega_0(\mathbf{x}(\tau)) \Lambda(\mathbf{x}(\tau)) J_3 .$$

So far we haven't solved anything, because $\mathbf{u}(\tau)$ still figures explicitly in the equation. With the dot-product in it we can also not hope to somehow divide it out. Yet there is quite some arbitrariness in the role that $\mathbf{u}(\tau)$ plays in this equation. At every point on the trajectory $\mathbf{u}(\tau)$ could also be used as an initial velocity at that point and we would find the same trajectory. But every other initial velocity will also give a different but physically admissible trajectory. Furthermore, as $\Lambda(\mathbf{x}(\tau))$ depends on spacetime so does the proper velocity, i.e.

$$\mathbf{u}(\mathbf{x}(\tau)) = \Lambda(\mathbf{x}(\tau)) c \mathbf{e}_0 \tilde{\Lambda}(\mathbf{x}(\tau)) . \quad (12.3.1)$$

Our new rotor-equation should somehow be made to know about this! If we want to new rotor-equation to be true regardless of the specific direction of the initial velocity and regardless of the exact position of the particle, then we should write

$$\mathbf{u}(\mathbf{x}(\tau)) \nabla \Lambda(\mathbf{x}(\tau)) = -\omega_0(\mathbf{x}(\tau)) \Lambda(\mathbf{x}(\tau)) J_3 .$$

Left-multiplication of this equation by $\mathbf{u}(\mathbf{x}(\tau))$ we find

$$c^2 \nabla \Lambda(\mathbf{x}(\tau)) = -\omega_0(\mathbf{x}(\tau)) \Lambda(\mathbf{x}(\tau)) c \mathbf{e}_0 J_3 ,$$

where we used that

$$\mathbf{u}(\mathbf{x}(\tau))\Lambda(\mathbf{x}(\tau)) = \Lambda(\mathbf{x}(\tau))c\mathbf{e}_0 .$$

As we said we want this equation to be true everywhere in Spacetime, because a single worldline is no longer singled out.

So we come to propose the following equation for the *eigenrotor-field* of a free particle, for which in the simplest case we expect that $\omega_0(\mathbf{x})$ is a constant,

$$\begin{aligned} \nabla\Lambda(\mathbf{x}) &= -\mu\Lambda(\mathbf{x})\mathbf{e}_0J_3 , \\ \mu &= \frac{\omega_0}{c} . \end{aligned} \tag{12.3.2}$$

This equation is called the *free Dirac equation* and the current understanding in Physics is that this equation describes *all known forms of matter*. The quantity μ will turn out to be related to the restmass of the particle through

$$\omega_0 = \frac{mc^2}{\hbar} . \tag{12.3.3}$$

The Dirac equation is linear, which means that if $\Lambda_1(\mathbf{x})$ and $\Lambda_2(\mathbf{x})$ are both solutions to the Dirac equation then so is $\Lambda_1(\mathbf{x}) + \Lambda_2(\mathbf{x})$. Yet *if* for $j = 1, 2$ we have $\Lambda_j(\mathbf{x})\tilde{\Lambda}_j(\mathbf{x}) = 1$ then the sum of the two rotor-fields typically will **not** yield 1 when multiplied by its reverse. When we take the Dirac equation seriously this means that we can no longer require all its solutions $\Lambda_j(\mathbf{x})$ to do so. But whatever solution we have, if it can be considered to be a sum of rotor-fields than we do have the following

$$\Lambda(\mathbf{x})\tilde{\Lambda}(\mathbf{x}) = \rho(\mathbf{x}) \exp(i\beta(\mathbf{x})) , \tag{12.3.4}$$

where $\rho(\mathbf{x}) > 0$ and $\beta(\mathbf{x})$ are scalars. We call such rotors *weighed rotors*, they do not square to 1 but to a combination of scalars and pseudo-scalars. Note that any solution to the Dirac equation can thus be written as

$$\Lambda(\mathbf{x}) = \sqrt{\rho(\mathbf{x})} \exp\left(\frac{i}{2}\beta(\mathbf{x})\right) R(\mathbf{x}) , \tag{12.3.5}$$

where $R(\mathbf{x})$ is a rotor-field satisfying $R(\mathbf{x})\tilde{R}(\mathbf{x}) = 1$. The expectation values of functions position are straightforward to compute using ρ , say we would want to find the expectation value of $f(\mathbf{x})$ then we simply compute

$$\langle f \rangle \equiv \int d^3V \rho(\mathbf{x}) f(\mathbf{x}) . \tag{12.3.6}$$

Let us take another look at the Dirac equation with all this in mind, recall that

$$\begin{aligned} \nabla\Lambda(\mathbf{x}) &= -\frac{mc}{\hbar}\Lambda(\mathbf{x})\mathbf{e}_0J_3 , \\ &= -\frac{mc}{\hbar}\mathbf{u}(\mathbf{x})\Lambda(\mathbf{x})J_3 . \end{aligned}$$

We can now make the identification $\mathbf{p} = m\mathbf{u}$ for the local momentum of the particle to get

$$\frac{\hbar}{c} \nabla \Lambda(\mathbf{x}) J_3 = \mathbf{p}(\mathbf{x}) \Lambda(\mathbf{x}) . \quad (12.3.7)$$

This last equation tells us that the momentum of the particle is found by taking the vector-derivative, and multiplying by an appropriate factor and the spin-bivector J_3 . This is infact the standard prescription in quantum mechanics for associating momentum with a differential operator.

12.4. The Schrödinger Field

The Dirac equation is entirely relativistic and most of what we are familiar with in physics is non-relativistic. So if we can derive a sort of *non-relativistic* Dirac equation that possibly makes the physics easier to understand. Non-relativistic physics is hallmarked by the singling out of a particular time axis. So let us take $\mathbf{e}_0$ as such a time axis and see whether there are parts of $\Lambda(\mathbf{x})$ that have any special significance with respect to $\mathbf{e}_0$.

Suppose we would reflect all physical processes in the time-axis, i.e. in the straightline in the $\mathbf{e}_0$ -direction, then an arbitrary vector like the proper velocity of any particle transforms like

$$\mathbf{u}' = \mathbf{e}_0 \mathbf{u} \mathbf{e}_0 = \mathbf{e}_0 \Lambda c \mathbf{e}_0 \tilde{\Lambda} \mathbf{e}_0 . \quad (12.4.1)$$

This transformation is called *parity*. The rotorfield transforms as

$$\Lambda' = \mathbf{e}_0 \Lambda \mathbf{e}_0 . \quad (12.4.2)$$

It is easy to see that we can split up the rotor-field as

$$\Lambda_{P\pm} = \frac{1}{2} (\Lambda \pm \mathbf{e}_0 \Lambda \mathbf{e}_0) , \quad (12.4.3)$$

called the *parity eigenstates* of the fermion. Compare this to, for example, the split in electric and magnetic components of the electromagnetic bivectorfield F .

We can rewrite the Dirac equation in terms of these fields by applying as follows;

Writing out the Dirac equation, as well as sandwiching it between the two time-like basis vectors gives

$$\begin{aligned} \mathbf{e}_0 \frac{\partial}{c \partial t} \Lambda - \sum_{j=1}^3 \mathbf{e}_j \frac{\partial}{\partial x_j} \Lambda &= -\mu \Lambda \mathbf{e}_0 J_3 , \\ \mathbf{e}_0 \frac{\partial}{c \partial t} \mathbf{e}_0 \Lambda \mathbf{e}_0 + \sum_{j=1}^3 \mathbf{e}_j \mathbf{e}_0 \frac{\partial}{\partial x_j} \Lambda \mathbf{e}_0 &= -\mu \mathbf{e}_0 \Lambda \mathbf{e}_0 \mathbf{e}_0 J_3 . \end{aligned}$$

By adding and subtracting these equations we get equations for the fields $\Lambda_{P\pm}$ that read,

$$\mathbf{e}_0 \frac{\partial}{c \partial t} \Lambda_{P+} - \sum_{j=1}^3 \mathbf{e}_j \frac{\partial}{\partial x_j} \Lambda_{P-} = -\mu \Lambda_{P+} \mathbf{e}_0 J_3 ,$$

$$\mathbf{e}_0 \frac{\partial}{c \partial t} \Lambda_{P-} - \sum_{j=1}^3 \mathbf{e}_j \frac{\partial}{\partial x_j} \Lambda_{P+} = -\mu \Lambda_{P-} \mathbf{e}_0 J_3 . \quad (12.4.4)$$

Now we multiply everything from the left by $\mathbf{e}_0$ and use that

$$\mathbf{e}_0 \Lambda_{P\pm} \mathbf{e}_0 = \pm \Lambda_{P\pm} , \quad (12.4.5)$$

to get

$$\begin{aligned} \frac{\partial}{c \partial t} \Lambda_{P+} - \sum_{j=1}^3 K_j \frac{\partial}{\partial x_j} \Lambda_{P-} &= -\mu \Lambda_{P+} J_3 , \\ \frac{\partial}{c \partial t} \Lambda_{P-} - \sum_{j=1}^3 K_j \frac{\partial}{\partial x_j} \Lambda_{P+} &= +\mu \Lambda_{P-} J_3 . \end{aligned} \quad (12.4.6)$$

So far we have only rewritten the equation, but we haven't made any assumption. If we intend to find a non-relativistic equation we have to assume at some point that the *speeds* are low compared to c , or equivalently; that all energies are low compared to the rest-energy of the particle.

Consider the following *Ansatz* for the solution of these two equations,

$$\Lambda_{P\pm}(\mathbf{x}) = \Psi_{\pm}(\mathbf{x}) \exp(-\mu J_3 c t) . \quad (12.4.7)$$

The idea is that the frequency $c\mu$ is fast compared to the rate at which the functions $\Psi_{\pm}(\mathbf{x})$ change with time! Substituting this into the equations we get

$$\begin{aligned} \frac{\partial}{c \partial t} \Psi_+ - \sum_{j=1}^3 K_j \frac{\partial}{\partial x_j} \Psi_- &= 0 , \\ \frac{\partial}{c \partial t} \Psi_- - \sum_{j=1}^3 K_j \frac{\partial}{\partial x_j} \Psi_+ &= +2\mu \Psi_- J_3 . \end{aligned} \quad (12.4.8)$$

Now if assume that $c\mu$ is large compared to any other rate of change in time, so in particular the second equation becomes

$$-\sum_{j=1}^3 K_j \frac{\partial}{\partial x_j} \Psi_+ \approx 2\mu \Psi_- J_3 , \quad (12.4.9)$$

this is the key assumption to get a non-relativistic equation! For we can now easily solve the last equation to find Ψ_- as

$$\Psi_- \approx \frac{1}{2\mu} \sum_{j=1}^3 K_j \frac{\partial}{\partial x_j} \Psi_+ J_3 . \quad (12.4.10)$$

If we substitute this into the first equation for Ψ_+ we obtain

$$\frac{\partial}{\partial t}\Psi_+J_3 = -\frac{c}{2\mu}\sum_{j=1}^3\frac{\partial^2}{\partial x_j^2}\Psi_+ . \quad (12.4.11)$$

The multiplication from the right by J_3 can be removed from the equation as follows. Consider the multivectors

$$C_{\pm} = \frac{1}{2}\{1 \pm K_3\} , \quad (12.4.12)$$

$$D_{\pm} = \frac{1}{2}\{K_1 \mp J_2\} . \quad (12.4.13)$$

They satisfy

$$C_{\pm}J_3 = \pm iC_{\pm} , \quad D_{\pm}J_3 = \pm iD_{\pm} . \quad (12.4.14)$$

So if we expand

$$(12.4.15)$$

Compare this to the *Schrödinger equation* for the wavefunction ψ of a free particle with restmass m

$$i\hbar\frac{\partial}{\partial t}\psi = -\frac{\hbar^2}{2m}\sum_{j=1}^3\frac{\partial^2}{\partial x_j^2}\psi = 0 , \quad (12.4.16)$$

they are identical is we set

$$mc^2 = \hbar\mu c = \hbar\omega_0 , \quad (12.4.17)$$

$$i\psi \rightarrow \Psi_+J_3 . \quad (12.4.18)$$

We have identified the Dirac equation for Λ_P with the Schrödinger equation, in the non-relativistic limit. But why is Λ_{P-} different? Or isn't it? The answer is that it isn't that different at all. If we had chosen

$$\Lambda_{P\pm} = \Psi_{\pm}(\mathbf{x}) \exp(+\mu J_3 ct) ,$$

we would have found a free-particle Schrödinger equation for $\Psi_-(\mathbf{x})$ instead. The mass would have been the same as for $\Psi_+(\mathbf{x})$. This fact indicates that the Dirac equation described *more than one particle*. These partner-particles are called *anti-particles*! So we have seen that the parity-eigenstates of a rotorfield that solves the Dirac equation can, in the non-relativistic limit, be interpreted as quantum *wavefunctions* of a free particle.

12.5. Revision problems

Chapter 13

Quantum mechanics

13.1. The harmonic oscillator algebraically

13.2. The Pauli-Schrödinger Equation

13.3. Revision problems

Chapter 14

Quantum mechanics in Electromagnetic Fields

14.1. Constant magnetic fields

14.2. Stark effect in an oscillator

14.3. Zeeman effect in an oscillator

14.4. Revision problems

Chapter 15

Relativistic quantum mechanics

But what about the proper velocity of the particle? In principle it is easy to compute,

$$c\Lambda\mathbf{e}_0\tilde{\Lambda} = c\rho(\mathbf{x}) R\mathbf{e}_0\tilde{R} = \rho(\mathbf{x}) \mathbf{u}(\mathbf{x}), \quad (15.0.1)$$

where we have used that $i\mathbf{e}_0 = -\mathbf{e}_0i$. It is obvious here that ρ indeed can represent a probability density whereas $\mathbf{u}(\mathbf{x})$ represents a local flow. We could identify so-called *streamlines* $\mathbf{x}(\tau)$ in the flow by solving the equation,

$$\frac{d}{d\tau}\mathbf{x}(\tau) = \mathbf{u}(\mathbf{x}(\tau)) . \quad (15.0.2)$$

This idea will play a role when we try to understand the *classical* limit of the Dirac equation.

15.1. The classical limit

We have seen how the Dirac equation gives rise to a Schrödinger equation in the non-relativistic limit. But does it also give rise to a classical equation for $\mathbf{u}$ in some *classical limit*?

Consider what we have just derived, the equation

$$\frac{\hbar}{c}\nabla\Lambda(\mathbf{x})J_3 = \mathbf{p}(\mathbf{x})\Lambda(\mathbf{x}) .$$

It suggests that an approximate solution of the Dirac equation reads

$$\Lambda(\mathbf{x}) \approx \exp\left(\frac{-1}{\hbar}J_3\varphi(\mathbf{x})\right) , \quad (15.1.1)$$

for some unknown function $\varphi(\mathbf{x})$. This Ansatz is called the *Eikonal approximation*. If we insert it in the Dirac equation we find the expression

$$\frac{1}{c}\nabla\varphi(\mathbf{x}) = m\mathbf{u}(\mathbf{x}) .$$

Along a streamline for which

$$\frac{d}{d\tau}\mathbf{x}(\tau) = \mathbf{u}(\mathbf{x}(\tau)) ,$$

we have

$$\begin{aligned}\frac{d}{d\tau}\mathbf{u}(\mathbf{x}(\tau)) &= (\mathbf{u}(\mathbf{x}(\tau)) \cdot \nabla)\mathbf{u}(\mathbf{x}(\tau)) , \\ &= [\mathbf{u}(\mathbf{x}(\tau)), (\nabla \wedge \mathbf{u}(\mathbf{x}(\tau)))] ,\end{aligned}\quad (15.1.2)$$

where we use the fact that $\mathbf{u}^2 = c^2$. If we apply that $\nabla \wedge \nabla\varphi(\mathbf{x}) = 0$ for any $\varphi(\mathbf{x})$, we find that the free Dirac equation reduces to

$$\frac{d}{d\tau}\mathbf{u}(\mathbf{x}(\tau)) = 0 . \quad (15.1.3)$$

This is ofcourse what one would expect from a free particle!

15.2. Relativistic particle in a Box

15.3. Spin, Helicity and Anti-particles

15.4. Charged Electrons and gauging the Dirac equation

Let us recall that if eigen-rotor $\Lambda'(\tau)$ that is computed from $\Lambda(\tau)$ through

$$\Lambda'(\tau) = \Lambda(\tau) \exp(J_3\alpha(\tau)) ,$$

it is physically identical to $\Lambda(\tau)$ as it preserves both the spin as well as the proper velocity. We upgraded the eigen-rotors to rotorfields in this chapter without worrying about this issue. But now we need to resolve it for once and for all. As you will see, a big surprise is in store for us!

Consider the rotor-field $\Lambda(\mathbf{x})$ and suppose we study the change of Λ under a small change in one of the coordinates x_μ by computing

$$\frac{\partial}{\partial x_\mu}\Lambda(\mathbf{x}) .$$

Now this position dependence of Λ contains vital information on the momentum of the particle. So we don't want to spoil it by having useless information in there about rotations in the J_3 -plane. So let us subtract that part out of the derivative, i.e.

$$\frac{\partial}{\partial x_\mu}\Lambda(\mathbf{x}) - qA_\mu(\mathbf{x})\Lambda(\mathbf{x})J_3 , \quad (15.4.1)$$

where q is some arbitrary number and together with the function $A_\mu(\mathbf{x})$ it contains all the superfluous information. Multiplying by $\mathbf{e}_\mu$ and summing over all the $\mu = 0, 1, 2, 3$, gives us a *corrected* Dirac equation

$$\nabla\Lambda(\mathbf{x}) - q\mathbf{A}(\mathbf{x})\Lambda(\mathbf{x})J_3 = -\frac{mc}{\hbar}\Lambda(\mathbf{x})\mathbf{e}_0J_3 , \quad (15.4.2)$$

$$\mathbf{A}(\mathbf{x}) = \sum_{\mu=0}^3 A_\mu(\mathbf{x})\mathbf{e}_\mu^{-1} . \quad (15.4.3)$$

Ofcourse we haven't answered the question yet as to *how* we should get this superfluous information into the vectorfield $\mathbf{A}(\mathbf{x})$. But we can give it a name: it is called a *gauge field*. In a sense it *gauges* the meaning of these rotations in the spin-plane.

Before we ask ourselves how the information gets into $\mathbf{A}(\mathbf{x})$ let us see whether we can find out what the field does to the motion of the particle. We can try an *eikonal* approximation to get some feeling for the *classical* nature of $\mathbf{A}(\mathbf{x})$. So, inserting the eikonal approximation we find

$$\frac{1}{c}\nabla\varphi(\mathbf{x}) - q\mathbf{A}(\mathbf{x}) = m\mathbf{u}(\mathbf{x}) .$$

Now taking the curl on both sides gives,

$$-q\nabla \wedge \mathbf{A}(\mathbf{x}) = m\nabla \wedge \mathbf{u}(\mathbf{x}) .$$

In terms of streamlines this means

$$m\frac{d}{d\tau}\mathbf{u}(\mathbf{x}(\tau)) = qF(\mathbf{x}) \cdot \mathbf{u}(\mathbf{x}(\tau)) , \quad (15.4.4)$$

$$F(\mathbf{x}) = \nabla \wedge \mathbf{A}(\mathbf{x}) . \quad (15.4.5)$$

We have seen such an expression before! The gauge field can generate a bivector-field that acts as a force on the particle. We call such an interaction a *gauge interaction*. The number q is called the *charge* of the particle.

Electromagnetism is the simplest of these gauge theories, but all known forces of nature are known to be gauge interactions. Every of these comes with one or more charges. In the simple case of electromagnetism the charge is not fixed by the theory, yet in nature we only seem to find integer multiples of a fundamental charge unit. More involved gauge theories however also predict the quantisation of charge. The vectorfield $\mathbf{A}(\mathbf{x})$ evidently is a potential momentum, as we discussed in chapter 3. In the context of electromagnetism however, it is usuall referred to as a *vector-potential*.

15.5. Problems

Chapter 16

Solitons and Spontaneous Symmetrybreaking

In this chapter we return to the study of an elementary field theory model that displays features, such as solitons and instantons, that are very important in modern condensed matter physics and elementary particle physics. The model we use is not designed for its practical application to physical problems but due to its simplicity it allows certain issues to be resolved exactly, without making any approximation. It also provides a good starting point for your own independent work either on a project for this course, or as research project in a SCI301 research course.

16.0.1. A (quantum) mechanical model

16.0.2. A field theory

16.1. Groundstates and Symmetry breaking

16.1.1. Symmetry restoration in QM

16.1.2. Symmetry-breaking in field theory

16.2. Advanced topics

16.2.1. Fluctuations and elementary particles

16.2.2. Phasetransitions and Bounce solutions

16.2.3. Instantons

16.3. Revision problems

Chapter 17

Dislocation theory

The properties of materials are determined by a large number of factors. Among these a very important factor is the ordering of the atoms of which the material consists. Obviously fluids and gasses distinguish themselves from condensed matter due to a partial absence of such ordering. But even in solid matter the ordering in ideal crystals is a rare phenomenon. In this chapter we will consider a few simple crystal orderings and the description of defects in these crystals. The topic of dislocation theory is a subject which is generally taught in materials science curricula to 2nd or 3rd year undergraduates. It introduces very important geometric concepts such as torsion and curvature in a setting in which they have a very direct interpretation. Oddly enough, these same concepts are at the heart of Einstein's general theory of relativity describing the large scale structure of Spacetime.

17.1. Crystal lattices

17.1.1. Burger's vector

17.1.2. Gauge ideas

17.2. Dislocations and Torsion

17.2.1. Torsion

17.3. Disclinations and Curvature

17.4. Curvature

17.5. Revision problems

Chapter 18

The Standard model of Particle Physics

18.1. Electroweak interactions

18.2. The Higgs mechanism

18.3. Colour forces, Quarks and Confinement

18.4. Revision problems

Chapter 19

Gravity

In this chapter we will study the interaction between gravity and electromagnetism. Ofcourse this is a vast area and we cannot expect to even make a proper start covering it in a single chapter. Nevertheless it provides an opportunity to highlight some specific examples of new and surprising phenomena associated with the inclusion of gravity in the picture of classical electrodynamics. We will uncover gravity very much in the same way we uncovered electromagnetism. It all boils down to rethinking carefully what you assume to be obvious. So we start out by asking a few questions and providing tentative answers. In subsequent subsection we will show where these answers lead us.

- **Question 1:** What exactly do we mean when we say that Spacetime is four-dimensional?
- **Question 2:** What exactly do we mean when we say that Spacetime has one time- and three spacedirections?
- **Question 3:** What exactly do we mean when we say a spacetime vector at some point in spacetime has one or another direction?

It is tempting to think that these questions have all been answered before, which indeed is the case. But we implicitly assumed that that answer was *global*. With our discovery that electromagnetism arises from a local symmetry we should become more modest and provide *local* answers to these three questions.

I will take the point of view that spacetime is a *process* rather than an externally given fact. For such a process to be physically relevant its properties should, at least in principle, be measurable. So I will assume that the dimensionality of spacetime implies that this process allows us to experimentaly verify, at any event $\mathbf{x}$, the existence of four different directions in spacetime. In other words, spacetime should locally allow us to construct a, possibly largely arbitrary, set of basis-vectors $\mathbf{g}_\mu(\mathbf{x})$. Nothing more, and nothing less.

We have seen in earlier chapters that the main distinction between the time and the space-directions is the sign of the square of the corresponding unit-vectors. Now I already assumed that the process called spacetime allows the definition of a basis. So in addition I will now also assume that this basis at least allows us the distinction between one positively squared vector, $\mathbf{g}_0(\mathbf{x})$, and three others, $\mathbf{g}_j(\mathbf{x})$, which yield negative squares.

Finally we should worry about what it means when we assign a direction to anything in spacetime. Well, if we want to refrain from introducing even more

objects it seems logical to assume that any statement on direction is meant solely in reference to the local gravitational basis $\mathbf{g}_\mu(\mathbf{x})$. In other words, spacetime vectors are objects that are expressed locally in terms of the gravitational basis,

$$\mathbf{A}(\mathbf{x}) = \sum_{\mu=0}^3 A_\mu(\mathbf{x}) \mathbf{g}_\mu(\mathbf{x}) . \quad (19.0.1)$$

The dot-product of two of such vectors therefore can be written in terms of the local components as

$$\begin{aligned} \mathbf{A}(\mathbf{x}) \cdot \mathbf{B}(\mathbf{x}) &= \sum_{\mu=0}^3 \sum_{\nu=0}^3 A_\mu(\mathbf{x}) B_\nu(\mathbf{x}) \mathbf{g}_\mu(\mathbf{x}) \cdot \mathbf{g}_\nu(\mathbf{x}) , \\ &= \sum_{\mu=0}^3 \sum_{\nu=0}^3 A_\mu(\mathbf{x}) B_\nu(\mathbf{x}) g_{\mu\nu}(\mathbf{x}) . \end{aligned} \quad (19.0.2)$$

In the last line we introduced a matrix

$$g_{\mu\nu}(\mathbf{x}) = \mathbf{g}_\mu(\mathbf{x}) \cdot \mathbf{g}_\nu(\mathbf{x}) , \quad (19.0.3)$$

which is called the *spacetime metric* in the standard approach to spacetime geometry. It provides a local definition of length.

19.1. Local Special Relativity

We assumed the basis $\mathbf{g}_\nu(\mathbf{x})$ is the result of some experiment performed on the process called spacetime. As such, it cannot represent anything absolute but rather some observer-dependent set of directions and orientations. From our earlier considerations we know that any other inertial observer will observe a Lorentz transformed basis. So let us embed this principle locally into our theory: Any set of basis vectors $\mathbf{g}'_\nu(\mathbf{x})$ obtained from $\mathbf{g}_\nu(\mathbf{x})$ via a local Lorentzrotor $R(\mathbf{x})$ via

$$\mathbf{g}'_\nu(\mathbf{x}) = R(\mathbf{x}) \mathbf{g}_\nu(\mathbf{x}) \tilde{R}(\mathbf{x}) , \quad (19.1.1)$$

is considered to be *physically equivalent*. So whenever I speak of *the* gravitational basis I always mean this *modulo local lorentz transformations*. This is ofcourse a gauge-principle. And from our discussion of electroagnetism you will remember that this contains the seed of what we call interactions. That will happen here too! What we will find is *gravity*. But we are not there yet.

19.1.1. Coordinates and Differentiation

Spacetime has now become something more elusive on which we can locally define a gravitational basis on the basis of experiments. I will assume that generally we are able to distinguish different events from one another by the possibility of introducing four *coordinates*, i.e. scalar functions $x_\sigma(\mathbf{x})$, that take different values in different events. We will generally assume that we can always find *parametrisations* $\mathbf{x}(x_\mu)$ of parts of spacetime in terms of such coordinates. The rate at

which the event $\mathbf{x}(x)$ changes when one of the parameters is changed gives a local, coordinate-induced basis

$$\mathbf{e}_\mu(\mathbf{x}) = \frac{\partial}{\partial x_\mu} \mathbf{x}(x_\sigma). \quad (19.1.2)$$

Similarly we can say that coordinates change in value when we move through spacetime in the neighborhood of some event $\mathbf{x}$. This also gives us a sense of direction, but this time as a result of the *rate of change* of these coordinates $x_\sigma(\mathbf{x})$ as we change the chosen event $\mathbf{x}$. These are inversely related to the basis introduced earlier, and we have

$$\mathbf{e}_\mu^{-1}(\mathbf{x}) = \nabla x_\mu(\mathbf{x}), \quad (19.1.3)$$

where we used the vector-derivative, ∇ , which we know exists on such coordinatised patches of spacetime. All of these coordinates however, are physically completely meaningless. If we would like to study the rate of change of, say, scalar functions as a function of a direction we need to be able to assign such a rate of change to the directions $\mathbf{g}_\mu$ that form the gravitational basis. Ofcourse we may use the ∇ exists as a coordinate-free vector-derivative on scalars. Using the gravitational basis we may write down coordinate-free *scalar* derivatives

$$\partial_\mu = \mathbf{g}_\mu \cdot \nabla, \quad (19.1.4)$$

that will be useful in unravelling the algebra of differentiation that we want to merge with Spacetime algebra. This we will do in the next section.

19.1.2. The Algebra of Differentiation

Let us analyse the differentiation on scalars and see how it behaves under local Lorentz-rotations. Note that as a scalar-field ϕ itself does not transform at all, we have

$$\begin{aligned} (\nabla \phi)' &= R(\mathbf{x})(\nabla \phi) \tilde{R}(\mathbf{x}), \\ &= R(\mathbf{x}) \nabla \tilde{R}(\mathbf{x}) \phi = \nabla' \phi. \end{aligned}$$

This only works because in the expression $R(\mathbf{x}) \nabla \tilde{R}(\mathbf{x})$ the vector-derivative ∇ does not act on the reversed rotor $\tilde{R}$. So what we mean by the expression ∇' is

$$\nabla' = \sum_{\mu=0}^3 R \mathbf{g}_\mu^{-1} \tilde{R} \partial_\mu. \quad (19.1.5)$$

However, this causes a problem when we differentiate vectors. Because the expression $(\nabla \mathbf{v})'$ does *not* reduce to $\nabla' \mathbf{v}'$. To see this, just compute

$$\begin{aligned} \nabla' \mathbf{v}' &= [R \nabla \tilde{R}] [R \mathbf{v} \tilde{R}], \\ &= \left[\sum_{\mu=0}^3 R \mathbf{g}_\mu^{-1} \tilde{R} \partial_\mu \right] [R \mathbf{v} \tilde{R}], \\ &= \sum_{\mu=0}^3 R \mathbf{g}_\mu^{-1} \tilde{R} [(\partial_\mu R) \mathbf{v} \tilde{R} + R (\partial_\mu \mathbf{v}) \tilde{R} + R \mathbf{v} (\partial_\mu \tilde{R})], \end{aligned}$$

$$\begin{aligned}
&= (\nabla \mathbf{v})' + \sum_{\mu=0}^3 R \mathbf{g}_\mu^{-1} \tilde{R} [(\partial_\mu R) \mathbf{v} \tilde{R} + R \mathbf{v} (\partial_\mu \tilde{R})] , \\
&= (\nabla \mathbf{v})' + R \sum_{\mu=0}^3 \mathbf{g}_\mu^{-1} [\omega_\mu, \mathbf{v}]_- \tilde{R} ,
\end{aligned} \tag{19.1.6}$$

where we have defined the four bivectors

$$\omega_\mu = 2\tilde{R} \left(\frac{\partial}{\partial x_\mu} R \right) = -2 \left(\frac{\partial}{\partial x_\mu} \tilde{R} \right) R . \tag{19.1.7}$$

In other words, the standard vector-derivative breaks out of the nice spacetime algebra as soon as we make this algebra *local*. However, if we define a *new* vector derivative, $\mathcal{D}$, called the *covariant* vector-derivative given by

$$\mathcal{D} \mathbf{v} = \nabla \mathbf{v} - \sum_{\mu=0}^3 \mathbf{g}_\mu^{-1} [\Omega_\mu, \mathbf{v}]_- , \tag{19.1.8}$$

$$\tag{19.1.9}$$

where the bivector field Ω_μ is required to transform under local Lorentz-rotations as

$$\Omega'_\mu = R \Omega_\mu \tilde{R} + 2R (\mathbf{g}_\mu \cdot \nabla \tilde{R}) , \tag{19.1.10}$$

then we have a vector-derivative which satisfies two very convenient rules,

$$\mathcal{D} \phi = \nabla \phi , \tag{19.1.11}$$

$$(\mathcal{D} \mathbf{v})' = \mathcal{D}' \mathbf{v}' . \tag{19.1.12}$$

On ordinary scalar functions it reduces to the standard coordinate-free vector derivative and on vectors, and higher multivectors, it transforms neatly as a vector and thus perserving the algebra. The bivector fields that we have introduced transform oddly under Lorentz-rotations. But upon closer inspection you may recognise this as just a gauge-transformation of a form very similar to what we have already encountered. The covariant derivative can be expanded in terms of the gravitational basis

$$\begin{aligned}
\mathcal{D} &= \sum_{\nu=0}^3 \mathbf{g}_\nu^{-1}(\mathbf{x}) D_\nu , \\
\mathcal{D} \mathbf{v}(\mathbf{x}) &= \sum_{\mu=0}^3 \mathbf{g}_\mu^{-1}(\mathbf{x}) \partial_\mu \mathbf{v}(\mathbf{x}) + \sum_{\nu=0}^3 \mathbf{g}_\nu^{-1}(\mathbf{x}) [\Omega_\nu(\mathbf{x}), \mathbf{v}(\mathbf{x})]_- .
\end{aligned} \tag{19.1.13}$$

Yet in many books on general relativity, or differential geometry for physicists, you will find another covariant derivate. It identical to the one we present here, but it arises as follows. Consider the covariant scalar derivative of a gravitational basis-vector $D_\mu \mathbf{g}_\mu(\mathbf{x})$, it is a vector and hence it is expandable in terms of the $\mathbf{g}_\sigma(\mathbf{x})$ themselves. So there are scalars $\Gamma_{\mu\nu\lambda}$ such that

$$D_\mu \mathbf{g}_\nu(\mathbf{x}) = \sum_{\lambda=0}^3 \Gamma_{\mu\nu}^\lambda(\mathbf{x}) \mathbf{g}_\lambda(\mathbf{x}) . \tag{19.1.14}$$

As a result we can write for the components of the covariant derivative of a vector $\mathbf{v}(\mathbf{x})$

$$\begin{aligned} D_\nu \mathbf{v}(\mathbf{x}) &= \sum_{\mu=0}^3 (D_\nu v_\mu(\mathbf{x})) \mathbf{g}_\mu(\mathbf{x}) + v_\mu(\mathbf{x}) D_\nu \mathbf{g}_\mu(\mathbf{x}) , \\ &= \sum_{\mu=0}^3 (\partial_\nu v_\mu(\mathbf{x})) \mathbf{g}_\mu(\mathbf{x}) + \sum_{\mu=0}^3 \sum_{\lambda=0}^3 v_\mu(\mathbf{x}) \Gamma_{\nu\mu}^\lambda(\mathbf{x}) \mathbf{g}_\lambda(\mathbf{x}) , \end{aligned}$$

and this is the form usually found in these texts. We can give a formula relating the traditional $\Gamma_{\nu\mu}^\lambda(\mathbf{x})$ to our expressions, it is just a bit of algebra to check that

$$\Gamma_{\nu\mu}^\lambda(\mathbf{x}) = \mathbf{g}_\lambda^{-1}(\mathbf{x}) \cdot (\mathbf{g}_\nu(\mathbf{x}) \cdot \mathcal{D}(\mathbf{g}_\mu(\mathbf{x}))) . \quad (19.1.15)$$

Div and Curl

The covariant derivative gives rise to expressions similar to the *curl* and *divergence* that we introduced in chapter 4. From the above one can derive without much effort that

$$\begin{aligned} \mathcal{D} \cdot \mathbf{v}(\mathbf{x}) &= \sum_{\mu=0}^3 [\nabla v_\mu(\mathbf{x})] \cdot \mathbf{g}_\mu(\mathbf{x}) + \sum_{\nu=0}^3 \mathbf{g}_\nu^{-1}(\mathbf{x}) \cdot [\Omega_\nu(\mathbf{x}), \mathbf{v}(\mathbf{x})]_- , \\ \mathcal{D} \wedge \mathbf{v}(\mathbf{x}) &= \sum_{\mu=0}^3 [\nabla v_\mu(\mathbf{x})] \wedge \mathbf{g}_\mu(\mathbf{x}) + \sum_{\nu=0}^3 [\mathbf{g}_\nu^{-1}(\mathbf{x}) [\Omega_\nu(\mathbf{x}), \mathbf{v}(\mathbf{x})]_-]_- , \end{aligned}$$

These generalisations are usefull when discussing electromagnetic fields and their gauge-potentials in a context where gravity cannot be ignored. We will do that later. First we need to fully identify this emerging new gauge theory here as gravity.

19.2. Torsion

Now having a covariant derivative in our hand we should ask ourselves the question what happens when we take a covariant derivative of the gravitational basis-vectors? To answer this question let us first interpret what it would mean if the covariant derivative of the gravitational tetrad vanishes, $\mathcal{D}\mathbf{g}_\mu(\mathbf{x}) = 0$? First note that the statement is both locally Lorentz-invariant as well is coordinate-free. Hence, if correct it would be a truly physical statement.

Flat Spacetime

The covariant derivative of the tetrad compares the local changes in the tetrad, $\partial_\nu \mathbf{g}_\mu(\mathbf{x})$, to the action of an infinitesimal, local, Lorentz-rotation generated by bivector $\Omega_\nu(\mathbf{x})$. The vanishing of the derivative would imply that the tetrad varies across spacetime as the result of infinitesimal local lorentz-rotations. Then the remaining question would be whether these infinitesimal local transformations can be glued together to a globally defined rotor $R(\mathbf{x})$. If such a rotor would exist it means that we could express the gravitational tetrad $\mathbf{g}_\mu(\mathbf{y})$ *anywhere* in spacetime at the event $\mathbf{y}$ as the lorentz-rotated copy of the tetrad $\mathbf{g}_\mu(\mathbf{x})$ at $\mathbf{x}$. This would

make all the tetrads everywhere physically equivalent: this is what we call *flat spacetime*.

So we find that there are two conditions on the gravitational fields, the tetrad $\mathbf{g}_\mu$ and the gauge-field Ω_ν , that determine whether spacetime is flat or not. But we have not yet formulated them precisely. It is the purpose of the next two subsections to do just that. As a result we will have the concepts and equations in our hands to specify exactly how Einstein's theory of general relativity describes spacetime and gravity.

19.3. Curvature

19.4. Einstein's equation

19.5. Blackholes

19.6. Revision problems

Chapter 20

Appendix A: *Space, time and Observers*

This appendix reviews some basic elements of special relativity that are not covered in the main text, but that are assumed.

20.1. Observers, Standards and Units

Observers will want to discuss the state of the world (or the universe) with one another. In order to be able to do so they have to define a few *standards*. One such standard is, for example, the meaning of the length of one meter, or the duration of one second, etc. It is vital you realize that we obtain insight into the universe and its workings by performing *measurements*. Yet measurements are inherently *relational* in nature. We establish the properties of one object by making some kind of comparison to the properties of another. We call such standards of length, duration and for example temperature; *units*. This will include standard lengths and standard durations against which the results of measurements are to be compared.

Problem 137: Different Units for everything

This is the first "problem-box you encounter in these notes. You are advised to go through all these boxes during the course. Exam questions typically are variations on the problems in these boxes.

a: Find and write down the conversions between the units of energy: Joule, Calories and MeV.

b: Find and write down the conversions between the units of length: Fermi, Angstrom and Parsec.

A casual statement like *I am 6 feet tall*, actually is a comparison between the result of a measurement of my own length, and the measurement of the length of a standard called *foot*. If you measure things with your feet this may be a clever thing to do, but it is hardly a *standard* as everyone will have slightly different feet. So a lot of time has been invested in the past few centuries in establishing a few *generally accepted standards*. We call them *units* nowadays. But when we truly believe that measurements tell us something intrinsic about nature the choice of units should not matter fundamentally. Some units may be superior over others

depending on the context. For astronomers as well as for nuclear physicists the unit *meter* is not very practical. Light-years respectively Fermi is more appropriate in those contexts. The celebrated SI units are typically of "household" magnitude.

As the choice of units is, in a sense, immaterial units referring to the same quantity should be easily translatable into one another. Fahrenheit or Celsius, the choice is culturally determined and has nothing to do with the weather in your country. The amount of energy in 100 grams of chocolate remains the same irrespective of whether we express it in terms of Joules or Calories. The choice of units is usually made on the basis of what is *convenient*, and what is convenient for an astronomer probably will not be convenient for a pharmacist or an elementary-particle-physicist. As a result we never managed to agree on to a truly generally accepted set of units. But this is not a failure of communication skills in physics! It's just common sense. We'll use our communication skills not to negotiate about units, but to talk science. *

Sometimes we believe two quantities to be very different and introduce different units for them. For example you may express energies in Calories. But then Physics teaches us that there are relations between, say, the kinetic energy of an object and it's mass and speed. The well-known formula

$$E_k = \frac{1}{2} m \mathbf{v}^2 \quad .$$

This also induces a relation between the unit Calories and the unit "kilogram meter squared per second squared" or just "Joule" for short. So we don't need a separate unit for energy; it is a composite of units we already have. Although physically unnecessary, typographically it is convenient to introduce a separate unit. A choice of units is an example of choosing a *frame of reference*. The fact that different observers may choose different frames is at the heart of many of the *relativities* that you will encounter in physics.

The first thing any observer must do is establishing a frame of reference. For the sake of simplicity we will assume in this text that the observers we are going to talk about always use the same set of units. Unless explicitly mentioned we will use SI units. So our standard length will be the meter. However, the units are not the only things that require choices by an observer. The same thing holds true for *directions*. Many physically relevant quantities have more than just a *magnitude* that can be expressed in terms of some kind of unit; many things also have a direction. We call things with a magnitude *and* a direction by the name *vectors*. So velocity is a good example of a vector. In drawing vectors are always arrows with a certain magnitude and a direction. The nice thing about the English language is that the magnitude of the velocity goes by a different name: speed! Be aware that from this moment onwards I will expect you to make a conscious distinction between *speed* and *velocity*!

20.2. Units for Direction

In space, we know from experience, there are three directions. We can give them names, for example when specified with respect to your own human body

*This subject should be treated more carefully in technology and engineering. Where people use different units, people tend to make mistakes. Mistakes in technology are usually likely to be expensive and sometimes dangerous.

you might want to say:

- $\mathbf{e}_1$: Right;
- $\mathbf{e}_2$: Forward;
- $\mathbf{e}_3$: Up.

But here we will use the symbols $\mathbf{e}_j$ with $j = 1, 2, 3$ instead. All we really need

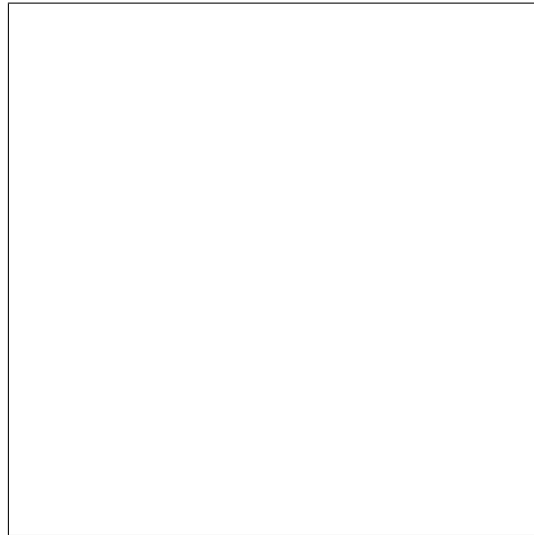


fig. 24: An Example of a basis set of directions

to know now is that we can always choose such a set of basic directions, and that we can set things up cleverly by choosing these words to mean *perpendicular* directions. As these three symbols represent units of mutually perpendicular directions we call them *orthogonal* unit-vectors, or *orthonormal* vectors.

There is a lot to be learned about vectors before you become fully proficient in using them. But once mastered they do provide a very powerful language to express ideas about geometry or physics. Your linear algebra course will have a lot more to say about these things.

Problem 138: Draw a Space diagram

It is important to learn how to make drawings that will help you understand problems and see solutions. So we start with a very simple one.

- a: Draw two perpendicular lines and denote a vector with a length of 1 cm on each of the lines, such that they represent $\mathbf{e}_1$ and $\mathbf{e}_3$ directions of some observer called Bob.
- b: Think of a method of depicting the direction $\mathbf{e}_2$ in your drawing without adding another axis.
- c: Suppose some other observer, Alice, uses different directions $\mathbf{e}'_1$ and $\mathbf{e}'_3$. Sketch an example of what they could look like in your drawing.
- d: Draw a perspective drawing of the three vectors $\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$ for Bob and a similar but different set for Alice.

From the above problem you see that it is not difficult to draw the directions chosen by different observers in one and the same diagram.

20.2.1. Time

Is time something that only has a magnitude, or does it also have a direction? As you and I can conceptually distinguish the Future from the Past, time seems to have a direction. Yet it does not seem to be related to any of the other three directions. So for *time* we have another symbol

- $\mathbf{e}_0$: into the Future.

In Newtonian physics the direction of time is completely disconnected from any of the other directions. As a result time *seems* to be merely a number. But of course Newton also understood the difference between future and past. Many people fail to see the difference between a number, and a vector in a one-dimensional space.

Problem 139: One-Dimensional Spaces

Consider an infinitely long, straight, line with one point singled out and called $\mathcal{O}$.

- Draw this line with $\mathcal{O}$.
- Draw a vector *in* the line, call it $\mathbf{e}_1$, starting from $\mathcal{O}$ with a length of 1 cm.
- Explain why any point $\mathbf{x}$ on the line can be identified by specifying the number of cm x that we need to walk, from $\mathcal{O}$ along $\mathbf{e}_1$ to get to $\mathbf{x}$.

The previous problem may have seemed very bureaucratic to you; distinguishing this x from $\mathbf{x}$ and introducing this $\mathbf{e}_1$. But it really isn't. Many things in spacetime physics become a lot more understandable when you make these distinctions carefully. Confusion often is the direct result of sloppiness in these *conceptual* matters! You should not underestimate the importance of the conceptual difference between x and $\mathbf{x}$ in the expression

$$\mathbf{x} = x\mathbf{e}_1 .$$

We saw earlier how different observers could decide to choose different basic directions. How is that with space and time?

Problem 140: Space-and-time diagrams

Consider a drawing in which we represent the $\mathbf{e}_0$ and the $\mathbf{e}_1$ directions for an observer called Bob.

- Make such a drawing and argue why it is reasonable that we can choose $\mathbf{e}_0$ (vertical) and $\mathbf{e}_1$ (horizontal) to be perpendicular.
- Now consider a second observer Alice. If you try to draw different vectors $\mathbf{e}'_0$ and $\mathbf{e}'_1$ in your drawing, do they make sense?
- Comment on how the disconnectedness of time and space limits the diagrams you can make.

From now on we will call diagrams in which space and time are drawn together, but conceptually disconnected, Space-and-time diagrams.

20.3. Standards for Position

To bring some order into the description of the universe we need some way of recording or writing down the position of the objects we observe. The necessary tool for this is the algebra of vectors and the notion of coordinates.

Problem 141: Choosing coordinates in Space

Take a flat piece of paper and look at it. It is a good example of a piece of flat, 2 dimensional, space.

a: Choose any arbitrary point; mark it with a dot and call it $\mathcal{O}$.

b: Choose two arbitrary, but perpendicular please, directions $\mathbf{e}_1$ and $\mathbf{e}_2$; represent them by two 1 cm long vectors drawn emanating from $\mathcal{O}$.

c: Explain why any point $\mathbf{x}$ on the piece of paper can be identified by specifying the two distances x and y that we need to walk, from $\mathcal{O}$ along $\mathbf{e}_1$ and $\mathbf{e}_2$ to get to $\mathbf{x}$.

These numbers x and y are called the *Cartesian coordinates* of the point $\mathbf{x}$.

d: Choose an arbitrary point, determine it's coordinates using your ruler and redraw the vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ but now emanating from the point you have chosen.

What you have done in the previous problem is called *choosing coordinates*. It means that you devise of a method to assign

- a set of numbers to each and every point in space;
- an *appropriate* set of basic directions in each and every point in space.

When the basic directions have been chosen to be *perpendicular* and *constant* throughout all of space, the coordinate system is called *Cartesian*.

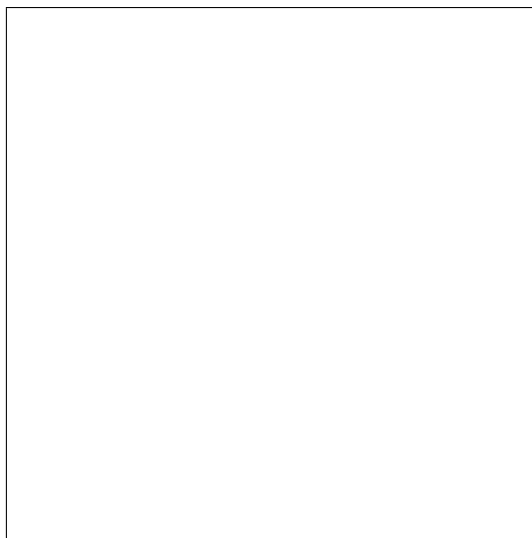


fig. 25: Skew coordinates

The Cartesian coordinates are simple, but they aren't always convenient.

Problem 142: Polar coordinates

Take a flat piece of paper and redraw the Cartesian coordinates, the corresponding basic directions, and the point $\mathbf{x}$ you have drawn in the previous problem.

a: Draw the line connecting the point $\mathbf{x}$ and the origin $\mathcal{O}$. Denote the angle that this line makes with the $\mathbf{e}_1$ -direction at $\mathcal{O}$ by the Greek letter θ .

b: Can we specify every point $\mathbf{x}$ in space not only by the Cartesian coordinates x and y , but also by the angle θ and the distance r between the point $\mathbf{x}$ and $\mathcal{O}$.

c: Show, by construction or by goniometric, that

$$\begin{aligned}x &= r \cos(\theta) , \\ y &= r \sin(\theta) .\end{aligned}$$

d: At each point $\mathbf{x}$ we can draw a basic direction $\hat{\mathbf{r}}$ which emanates from $\mathbf{x}$ and points in the direction of increasing r . Draw the 1 cm long vector representing $\hat{\mathbf{r}}$ in the special point $\mathbf{x}$ you had chosen earlier.

e: At each point $\mathbf{x}$ we can draw a basic direction $\hat{\theta}$ which emanates from $\mathbf{x}$ and points in the direction of *increasing* θ . Draw the 1 cm long vector representing $\hat{\theta}$ in the special point $\mathbf{x}$ you had chosen earlier.

The coordinates r and θ introduced in the previous problem, together with the basic directions $\hat{\mathbf{r}}$ and $\hat{\theta}$ are called a *polar* coordinate system. Note that these basic directions are different in different points. It is not a Cartesian system. We are setting up arbitrary standards so remain aware that the choice of coordinates is also arbitrary. In physics we choose coordinates that are most suitable and convenient for a particular situation. There is however one little pitfall here.

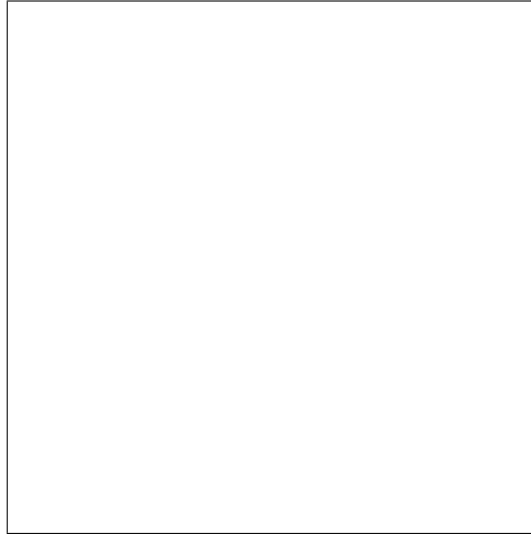


fig. 26: Spherical coordinates

20.3.1. Flat and curved space

You could wonder whether for *any* space it is *always* possible to choose a single set of Cartesian coordinates. The answer is a clear **No!** Spaces for which we can do so are called *flat*.

Problem 143: A cylinder without ends

Take a piece of flat paper and choose a Cartesian coordinate system on it. Then tape two opposite ends of the paper together carefully and without overlap. Of course this doesn't change the coordinates in any way. Explain why we can say that a cylinder is flat.

In this course we will only deal with flat spaces, so no need to worry. But there is one example of non-flatness that I cannot resist having you look at.

Problem 144: The Cone with a tip

Again pick a flat piece of paper, choose an origin $\mathcal{O}$ and draw a Cartesian coordinate system, on it.

a: Choose and draw two straight lines, both starting in $\mathcal{O}$ and making an angle θ_d (measure the angle with your ruler).

b: Cut away the part of the paper inside the lines with a scissor and reconnect the two cuts without overlap with tape. You should have obtained a cone.

c: Will there always be a little hole at the tip of the cone? Explain your answer.

The cone you have created is of course still flat, however the *true* cone without the hole in the tip would no longer be a flat space. There is no single set of Cartesian coordinates covering that entire cone *including* the tip. This phenomenon is called *curvature* and in two-dimensional space the curvature can be expressed by a *deficit angle* θ_d .

Problem 145: Circles on a Cone

Again pick a flat piece of paper, choose an origin $\mathcal{O}$ and draw two straight lines, both starting in $\mathcal{O}$ and making the angle θ_d (measure the angle with your ruler).

a: Draw two circles of a radius of 2 cm: one exactly around the origin $\mathcal{O}$, and one outside of the two lines and at a distance of, say, 5 cm from $\mathcal{O}$.

b: Now again cut away the part of the paper inside the lines with a scissor and reconnect the two cuts without overlap with tape to get a cone.

c: What is the difference in the circumference of the two circles after making the cone?

There is a hallmark of curvature: circles no longer have a circumference equal to 2π times their radius! In our example this was the piece of space that was cut-away in order to create a tipples version of the cone.

Problem 146: Where's the Curvature?

Consider the example of the cone again. Comment on the following statements;

a: Any circle that *does not* enclose the tipp of the cone has a circumference which is 2π times its radius.

b: The curvature of the cone is zero everywhere, except in the tipp.

In higher dimensions, such as in as in the real universe, curvature becomes a more complicated concept that requires many more numbers to be specified. In the context of relativity the notion of spaces that are not flat is very interesting

and relevant

- Special Relativity: the theory of the flat universe;
- General Relativity: the theory of *curved* universes.

In General Relativity you will get a full treatment of how to describe and work with curvature. But the problem of curved surfaces and spaces has an old and originally very practical history.

Problem 147: Cartography

Search the library or the web for resources on the main problem of cartography: how to map the entire Earth on a single sheet of paper.

a: Relate that problem to the notion of curvature.

b: What can you say about circles on the surface of the Earth?

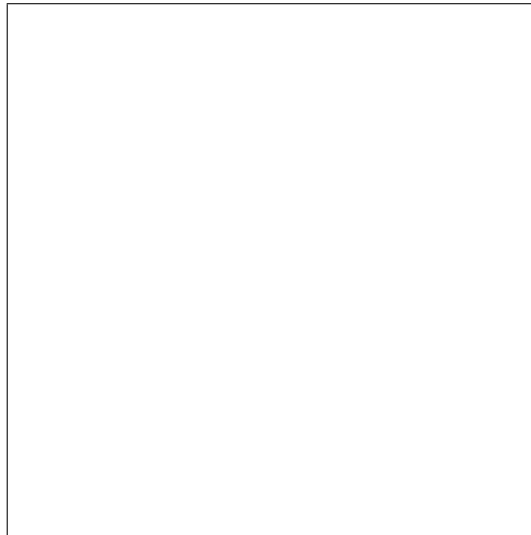


fig. 27: Cut Cone and Sphere

20.4. Relativity

Now let me introduce two observers that will guide you through this course: Alice and Bob.

Problem 148: Bob and Alice in Space-and-time

Consider a drawing in which we again represent the $\mathbf{e}_0$ and the $\mathbf{e}_1$ directions for Bob. Choose the origin $\mathcal{O}$ to be Bob's $t = 0$ and $x = 0$.

a: Draw where Bob is at different times (hint: Bob is egocentric and always at rest with respect to himself!).

b: Now consider Alice; she's moving with some velocity towards the positive $\mathbf{e}_1$ direction. You may assume Alice and Bob met at $t = 0$ on Bob's clocks. Draw where Alice is, relative to Bob, at different times.

Due to the fact that Alice is moving with respect to Bob she will perceive the world differently. We nevertheless expect them both to find the same laws of nature. This raises the core question of relativity: do all observers find the same laws of nature? The answer should of course be **yes**. What use would there otherwise be in doing physics? But the fact that the physical properties of objects are measured through comparison and setting standards should make you feel uncomfortable. What in our laws of physics is the result of choosing a frame of reference? Newton's laws give a partial answer to that. His first law says that whenever an object is moving at constant, possibly zero, velocity then there are no net forces acting on the object. This tells us that two observers that move at *constant relative velocity* should measure the same net forces acting on the other bodies in the universe. Such observers are called *inertial observers*. There is a widespread belief that

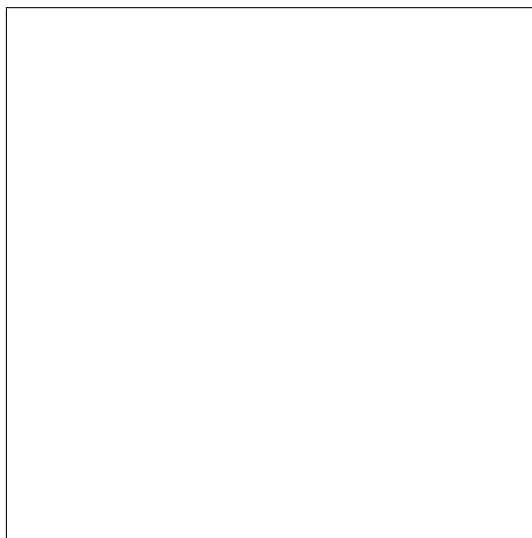


fig. 28: Bob and Alice in Space

the theory of special relativity deals with inertial observers only. This is not true: it deals with flat spaces only! Most books on special relativity deal with inertial observers only, and for a simple reason. Special relativity is easy in that case. But it is not restricted to that set of observers! Let us consider inertial observers first, and we'll have more say about non-inertial observers in later chapters.

20.4.1. Inertial Observers in Space-and-Time

Newton's first law suggests that observers at constant relative velocity should measure the same net forces acting on bodies. The other laws of Newton are laws about these forces; so inertial observers should observe the same laws of mechanics. We usually assume a number of things without much thought:

- All inertial observers agree on what Space is.
- All initial observers agree on what time is.

In fancier language we could say: Space and Time are absolute. When two inertial observers, Alice and Bob, move with respect to one another they drag along their reference frames. But due to the disconnectedness of time and space in Newtonian

physics, things are not really exciting. Let's draw some.

Problem 149: Moving Frames

Consider a diagram of space and time as seen by Bob.

- a: Draw a time-axis (vertically) and a space-axis (horizontally) and include Bob's unit-directions $\mathbf{e}_0$ and $\mathbf{e}_1$ in the origin $\mathcal{O}$.
- b: Why is Bob not staying in the origin $\mathcal{O}$, but rather everywhere along his time-axis?
- c: If we call Bob's vectors $\mathbf{e}_0$ and $\mathbf{e}_1$ his frame, then represent in your drawing Bob's frame at several different times.
- d: Alice moves in Bob's $\mathbf{e}_1$ -direction with a speed v . Draw the path she travels in the drawing and include the $\mathbf{e}'_0$ and $\mathbf{e}'_1$ of Alice's frame.

How do the observations of the *inertial couple* Alice and Bob on a moving body compare?

Problem 150: Galilean Relativity

Observer Bob has set up a frame of reference consisting of an origin $\mathcal{O}$ and a set of *space* basis vectors $\mathbf{e}_j$; According to himself Bob is at rest. Alice, is moving relative to Bob with a constant velocity $\mathbf{v}_A$. At time $t = 0$ the origin, $\mathcal{O}'$, of Alice's coordinates coincides with Bob's origin.

- a: If $\mathbf{x}_A(t)$ is the position of Alice's origin with respect to Bob's, then why do we know that $\mathbf{x}_A(t) = \mathbf{v}_A t$?
- Bob measures the position $\mathbf{y}(t)$ of some object at time t .
- b: Explain why the position of the object, as measured by Alice, $\mathbf{y}'(t)$, at time t is $\mathbf{y}'(t) = \mathbf{y}(t) - \mathbf{v}_A t$?
- c: Explain why the velocity of the object as measured by Bob, $\mathbf{u}(t)$ and the velocity as measured by Alice, $\mathbf{u}'(t)$ are related as $\mathbf{u}'(t) = \mathbf{u}(t) - \mathbf{v}_A$.
- d: Why is the acceleration of the object the same for Alice and Bob, and why does this mean that Alice and Bob will measure the *same force* acting on the body?

We can relate positions $\mathbf{x}$ and times t measured by Bob to those measured by Alice, $\mathbf{x}'$ and t' as

$$\mathbf{x}' = \mathbf{x} - \mathbf{v}_A t, \quad (20.4.1)$$

$$t' = t. \quad (20.4.2)$$

These equations express *Galilean relativity*; they give you a rule how to relate Cartesian positions measured by two observers moving at a constant relative velocity $\mathbf{v}_A$. Such observers, i.e. those that move at a constant relative velocity, are called *inertial observers*. It is noteworthy that time does not change from one observer to the other. This is the result of the disconnectedness discussed earlier. We could wonder whether this is really true? Yet common sense seems to say that it is.

Problem 151: A place is a *place*

When we use the word *place* we usually mean a location that can be identified at any given time (think about this for a moment).

- a: Why should we identify a *place* with a vertical line in a space-and-time diagram?
- b: Why do Alice and Bob always agree on vertical lines, and thus about the notion of a *place*?

Points in a space-time diagram do *not* denote positions or *places*. The things denoted by a point in a space-and-time diagram are called *events*.

Galilean relativity leaves the accelerations and thus the forces unaffected; inertial observers all observe the same forces. The Galilean transformation equations were considered to represent a *symmetry of nature*. This symmetry is called Galilean Invariance. As we shall see in this course this invariance is only an *approximate* symmetry.

Problem 152: Sonar Coordinates

The Cartesian coordinates that Bob assigns to Space-and-Time can be considered the result of a measurement. Suppose Bob is at rest with respect to some medium through which he can send *sound* signals. He sends out a sound-pulse in the $\mathbf{e}_1$ -direction at time t_e , the pulse reflects at some event $\mathbf{x}$ and Bob receives an echo back at t_a . Let c_s denote the speed of sound.

- a: Draw a frame for Bob (choose as units on the time-axis $c_s t$) and include the path of the sound pulse.
- b: Show, by drawing or algebra, that the coordinates of the event $\mathbf{x}$ are given by

$$\begin{aligned} t &= \frac{1}{2}(t_a + t_e) , \\ x &= \frac{c_s}{2}(t_a - t_e) . \end{aligned}$$

Problem 153: Alice's Sonar Coordinates

The coordinates that Alice assigns to Space-and-Time can be considered the result of the same measurement method. Copy the drawing you made in the previous problem. Alice moves with respect to Bob with a speed $v < c_s$ in the $\mathbf{e}_1$ -direction. She passes Bob exactly at the time t_e . For simplicity set $t_e = 0$.

- a: Draw Alice's path in Bob's Frame.
- b: Show that Alice receives the echo earlier than Bob at $t'_a = \frac{t_a}{(1 + \frac{v}{c_s})}$.
- c: Let x' be the spatial distance between Alice and the event of reflection, at the time of reflection t . Check that for Alice we indeed have $x' = x - vt$.

In the last two problems we have seen that Cartesian coordinates can be associated with a real measurement that both inertial observers can do. Indeed, when they do the same measurement their results are related by the Galilean transformation.

Note one important point about these sound signals. For Alice a sound-pulse she sends *forward* has a relative speed $c_s - v$, whereas the sound-pulse she receives

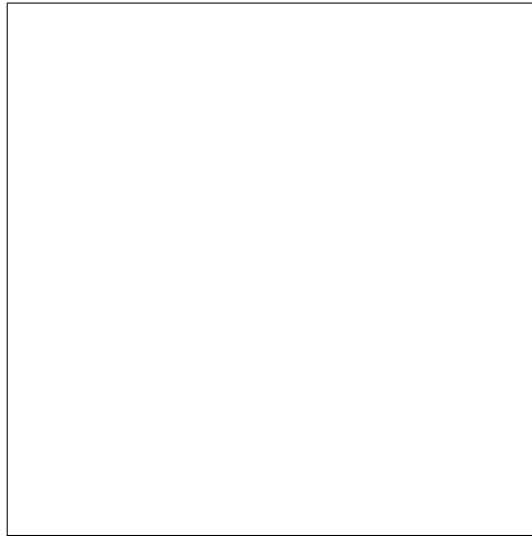


fig. 29: Sonar Coordinate sketch

back has a relative speed $c_s + v$. So by performing this measurement Alice can establish that she is moving. The presence of the medium gives rise to a *preferred frame*, namely the rest-frame of the medium. In a world mapped out with sonar-signals every observer can establish his or her motion with respect to this frame.

20.5. Reflections and double reflections

If you draw a point on a piece of paper and a line then it is not difficult to reflect the point in the line. It works as follows: measure the perpendicular distance between the point and the line, and then draw the reflected point at the same perpendicular distance on the opposite side of the line.

Problem 154: Reflections

Draw two perpendicular x - and y -axes on a piece of paper and draw the line $x = y$ in a different colour. Choose three different, but arbitrary points and reflect them in the line $x = y$.

When you do such reflections you'll see that different points are mapped by a varying degree depending on whether they are close to, or far away from, the line in which you reflect. This changes when you do a double reflection.

Problem 155: Double Reflections

Again draw two perpendicular x - and y -axes on a piece of paper and now draw the lines $x = \frac{1}{2}y$ and $x = 2y$ in different colours. Choose two different, but arbitrary points and reflect first them in the line $x = \frac{1}{2}y$ and then in the line $x = 2y$.

If you measure the angle by which each point has been displaced after a double reflection in lines through the origin, you will see all reflected points have been *rotated* around the origin by the same angle! In fact, if you compare that angle

to the angle between the lines you will see that the angle of rotation is twice the angle between the lines.

Problem 156: $\alpha \rightarrow 2\alpha$

Convince yourself of the point made in the text concerning the angle of rotation. Make several drawings, do reflections in different pairs of lines with varying angles between them and compare the results to the angles of rotation.

20.5.1. Reflections and double reflections in Spacetime

Now if you draw a point on a piece of paper and a line that represent spacetime objects it is not as easy to reflect the point in the line. However, if you remember that perpendicular lines always "sandwich" light signals!

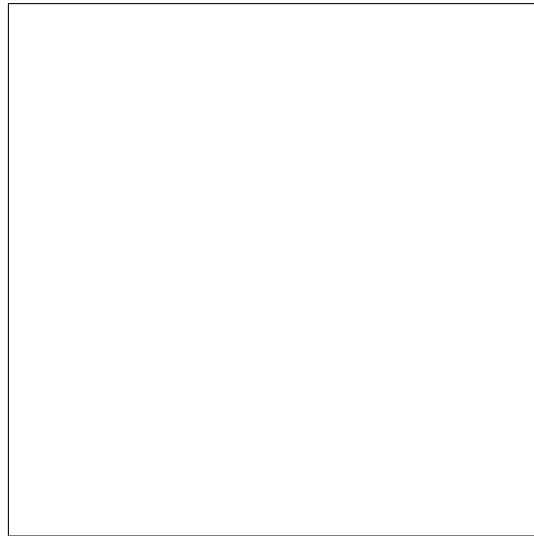


fig. 30: A double reflection in Spacetime

Problem 157: Perpendicular

Draw two perpendicular ct - and x -axes on a piece of paper and draw the lines, a , $ct = 2x$ and, b , $ct = -\frac{1}{4}x$ in different colours. Now construct the line through the origin perpendicular to a and the line perpendicular to b it by sandwiching a light signal through the origin.

With this "gut-feeling" for what perpendicular is in spacetime you can now also do reflections. For it now works the same way: measure the perpendicular distance between the point and the line, and then draw the reflected point at the same perpendicular distance on the opposite side of the line.

Problem 158: Spacetime reflections

Draw perpendicular ct - and x -axes on a piece of paper and draw the line $ct = 2x$ in a different colour. Choose three different, but arbitrary spacetime events and reflect them in the line.

When you do spacetime reflections this basically looks very much like space-reflections except for the odd skewness of what we call perpendicular. But there is a notable change when you do a double spacetime reflection.

Problem 159: Double Reflections

Again draw two perpendicular ct - and x -axes on a piece of paper and now draw the lines $ct = 2x$ and $ct = 4x$ in different colours. Choose two different, but arbitrary points and subsequently reflect them in the two lines.

You may not be extremely impressed by your result, and from the looks of it one doesn't easily recognise what this is; but it is a Lorentz transformation.

Problem 160: From Bob at rest to Alice at rest

- a: Draw a spacetime diagram of Bob at rest and Alice moving past Bob in their common origin $\mathcal{O}$. Draw both their worldlines and their space-axes.
- b: Now consider two lines, a , which is Bob's worldline and, b , the line that turns Alice's Worldline into Bob's when you do a reflection. Draw b .
- c: Check that the double reflection of Alice's worldline in first b then a gives Bob's "old" worldline, and that the double reflection of Alice's space-axis first in b and then in a indeed gives Bob's "old" space-axis.
- d: The double reflection $b; a$ transforms us to a frame in which Alice is at rest. Determine Bob's worldline in this frame.

What I want to do is use the little diagrams to derive a staggering result. There is no inertial observer for which $v > c$.

Problem 161: Drawing $v < c$

A Lorentz transformation carries you to the frame of a moving observer. So what happens if you repeat the same Lorentz transformation many times in succession? Do you get the frame of an observer moving at infinite speed? The answer is "no" as you'll see.

- a: Draw a spacetime diagram and add the lines $ct = 5x$ and $ct = 5.5x$. Choose the event with the coordinates $\{1, 0\}$ and do the double reflection.
- b: Now take the result of a: and do the double reflection again. c: Repeat this procedure a couple of times and comment on what you find. Where does the image of the point go towards after many successive double reflections?

This problem reveals two points. First of all, applying many Lorentz transformations in succession moves an event towards the lightcone, but it never pierces it! Secondly, if you apply a Lorentz transformation several times to a vector its images always lie on a single hyperbola.

The two lines making up a Lorentz transformation in our graphical method also define a plane in spacetime. As long as this plane contains a timelike direction the resulting transformation will be a Lorentz transformation. When the plane defined by the two lines contains only spacelike directions, then the transformation generated by the double reflections is a rotation. Lorentz transformations and rotations truly belong together in 4 dimensional Spacetime.

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